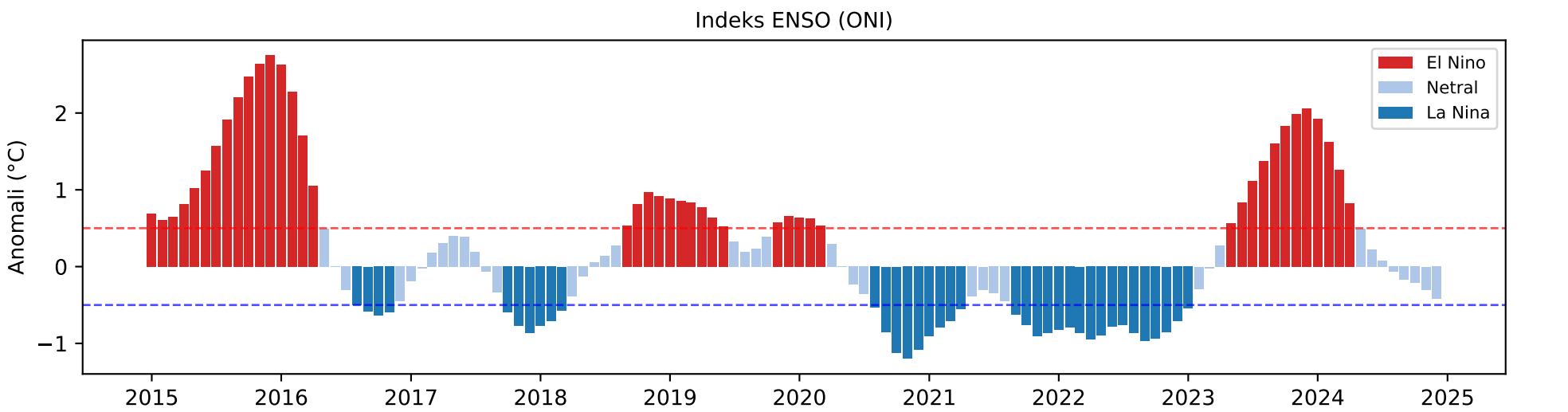
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN ROKAN HULU

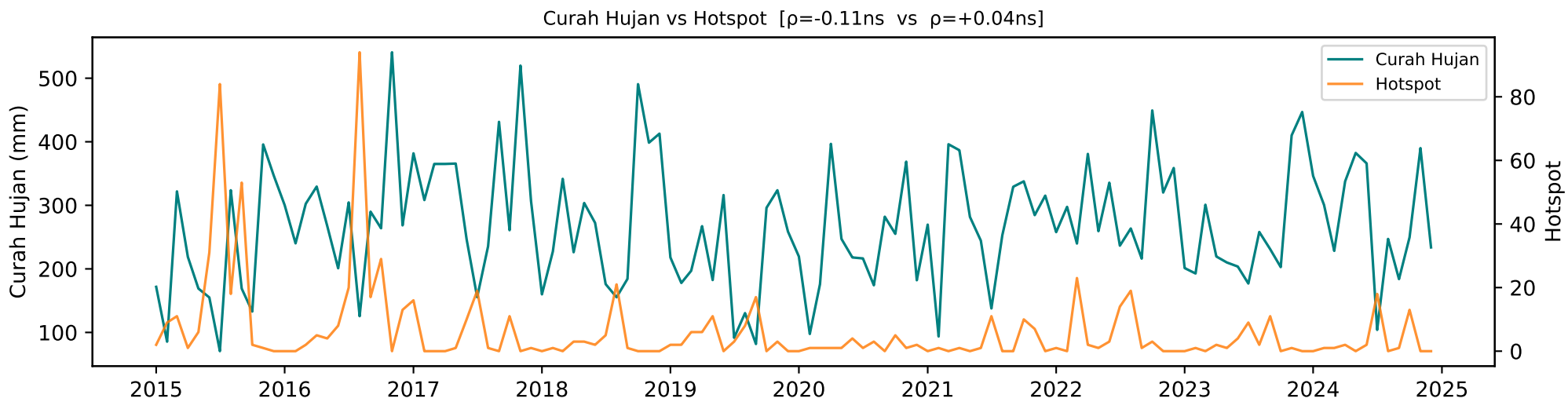
```
═══════════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.520 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                            : r=-0.108 ns    (p=0.239, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.108 ns    (p=0.239, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.040 ns    (p=0.661, Spearman ρ, efek=diabaikan, n=120)

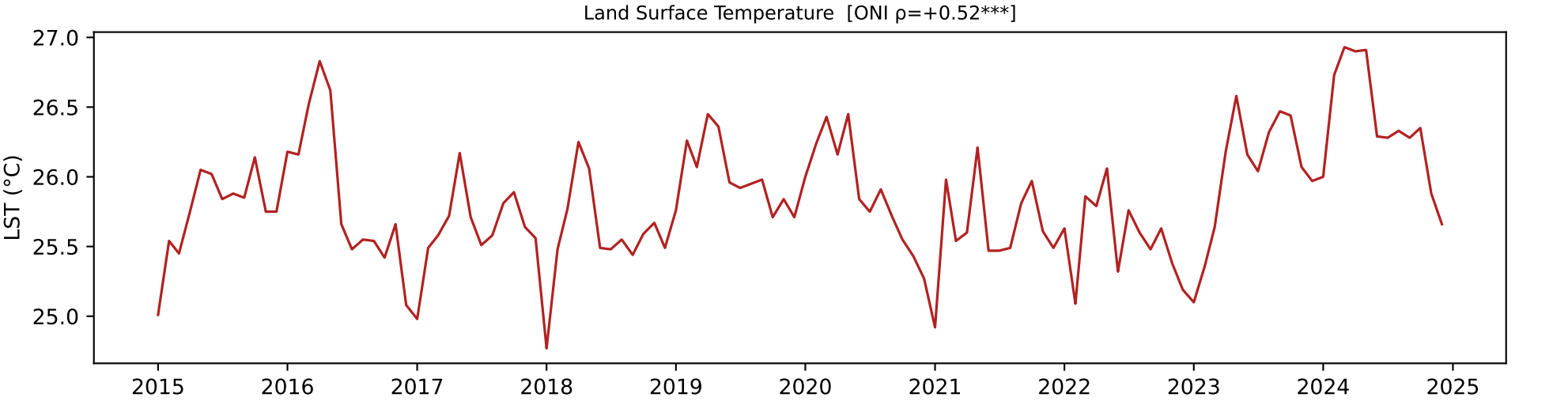
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────────
  Hotspot     : H=0.47, p=0.789ns, η²=-0.013
  LST         : H=27.21, p=0.000, η²=0.215
  Curah Hujan : H=3.46, p=0.177, η²=0.012

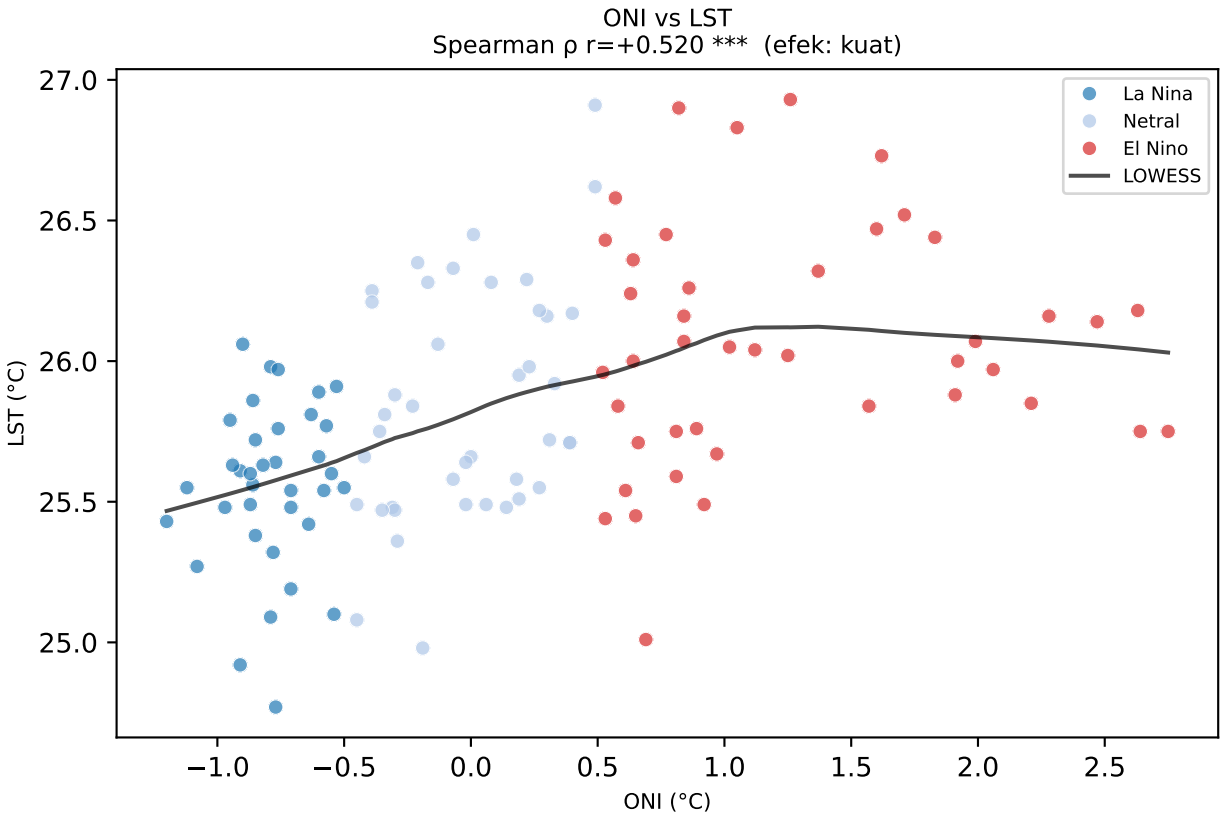
  INFO GAMBUT : total=89,147 ha | kelas dominan=100–<200 cm | median berbobot=200–<300 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

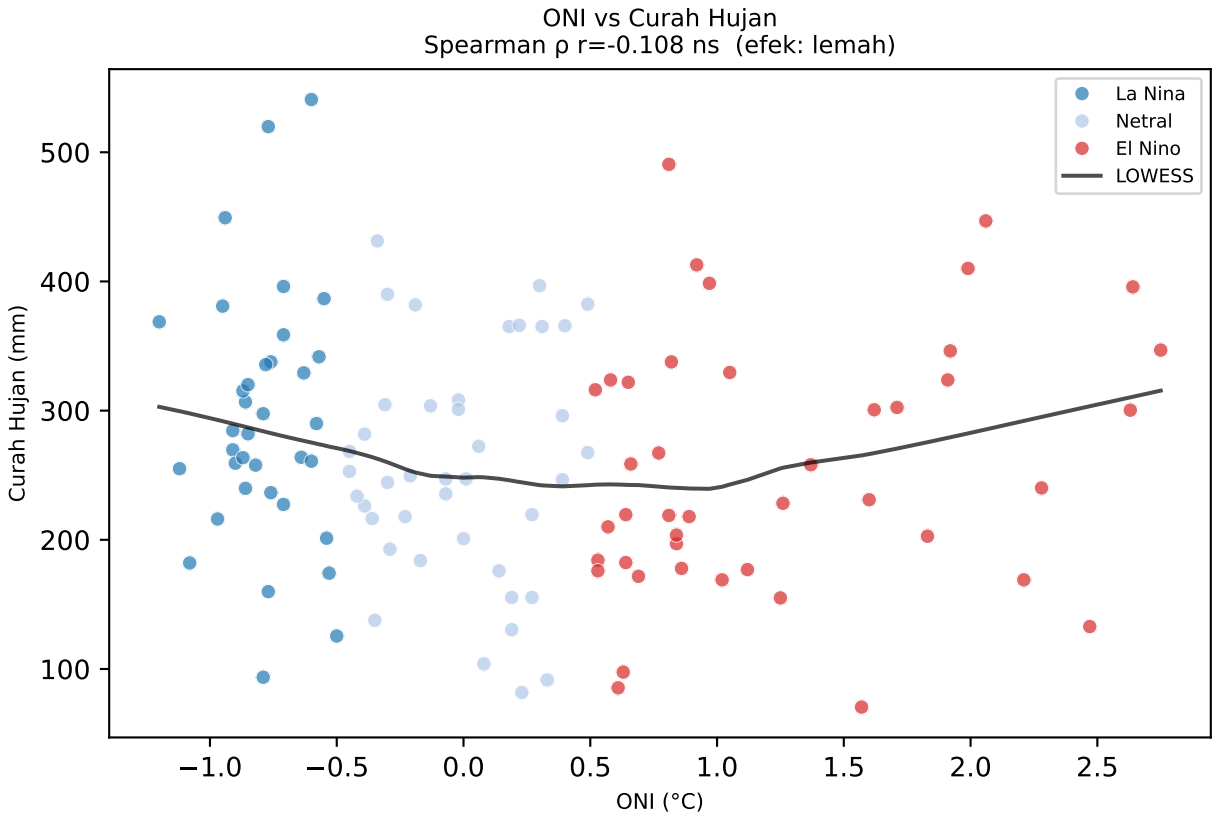


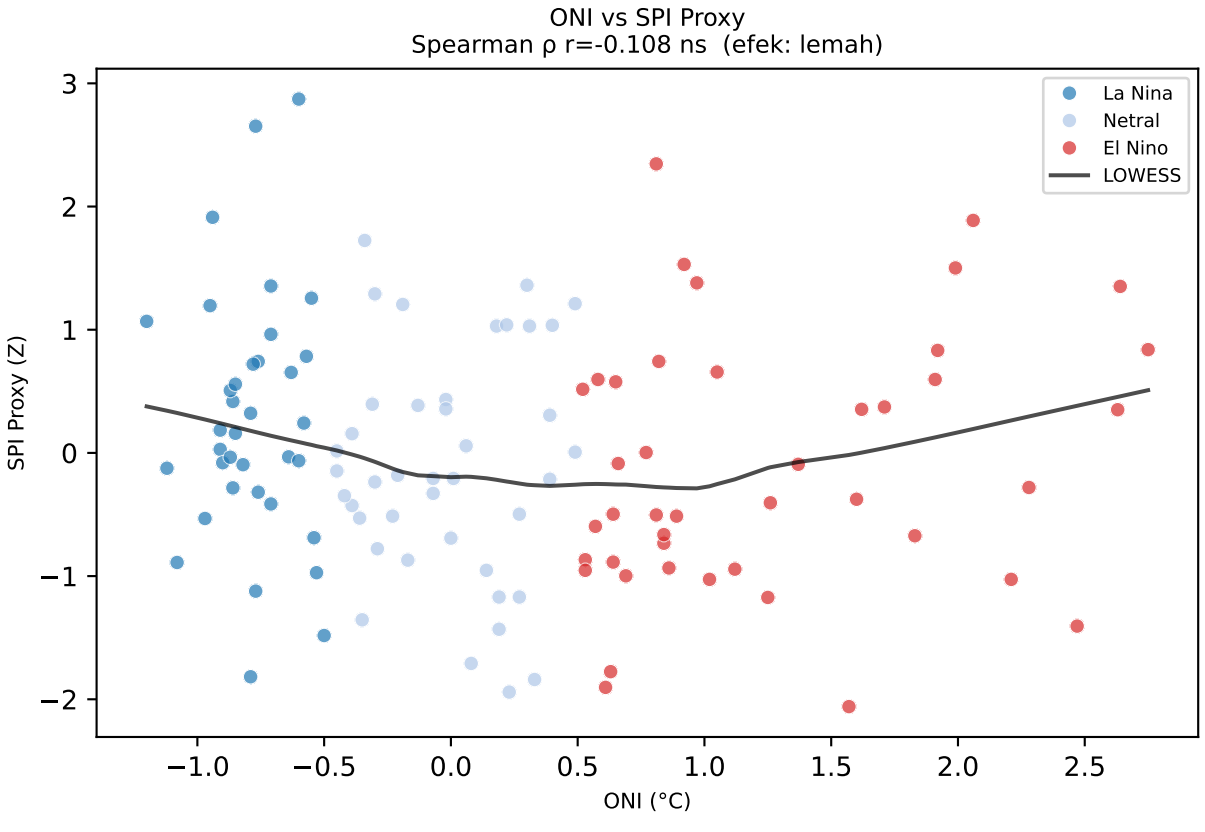


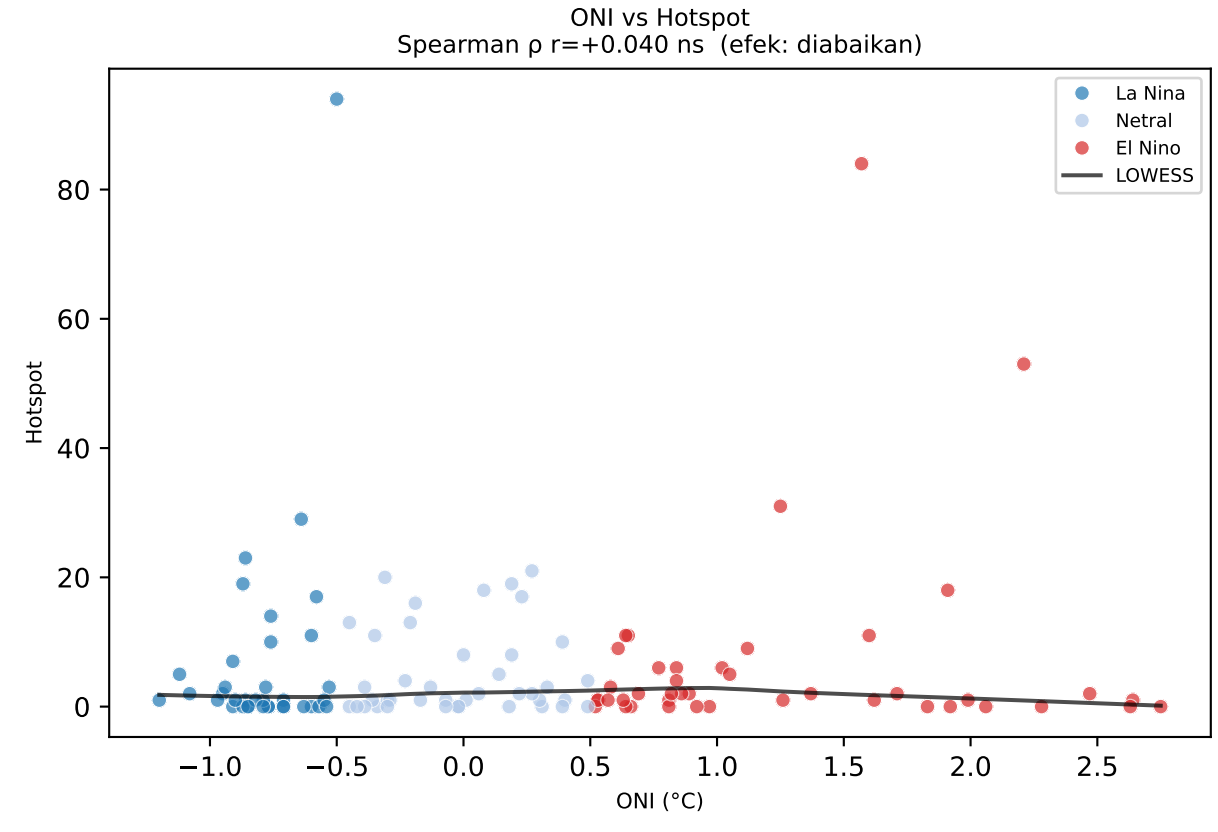


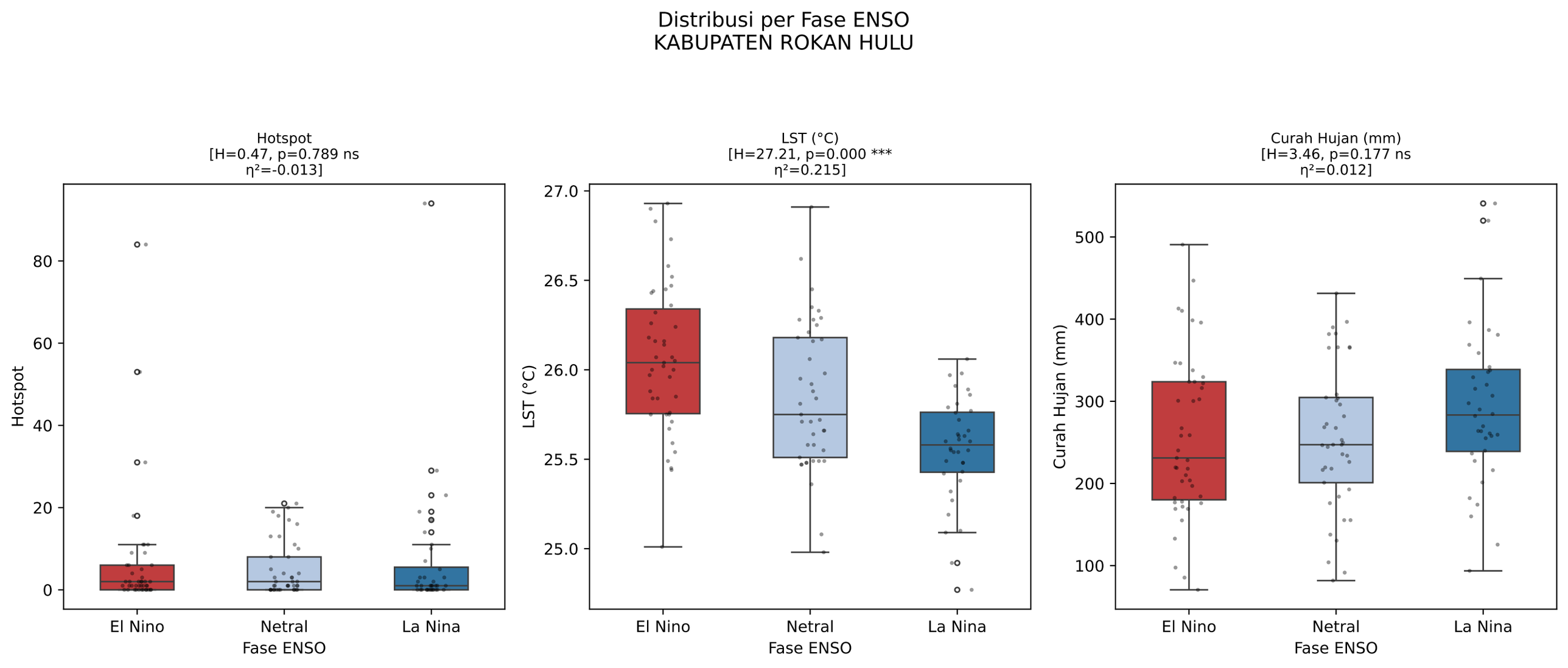
Scatter – ONI vs Variabel Iklim
KABUPATEN ROKAN HULU





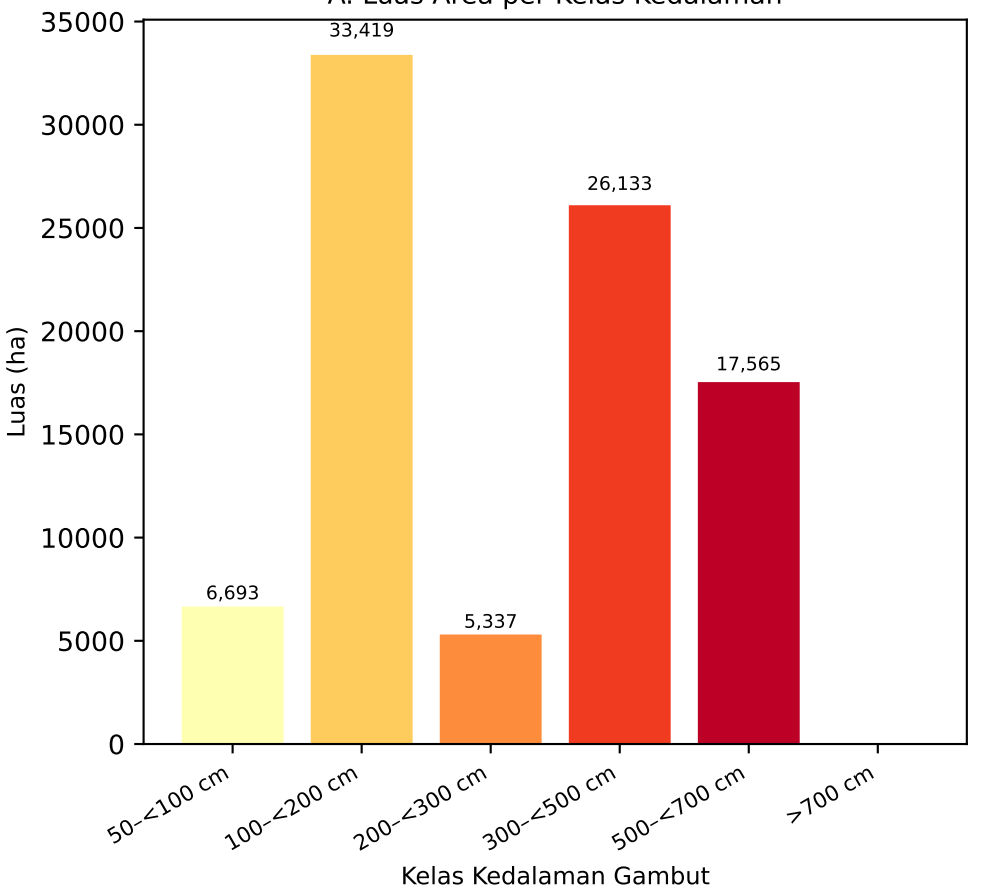




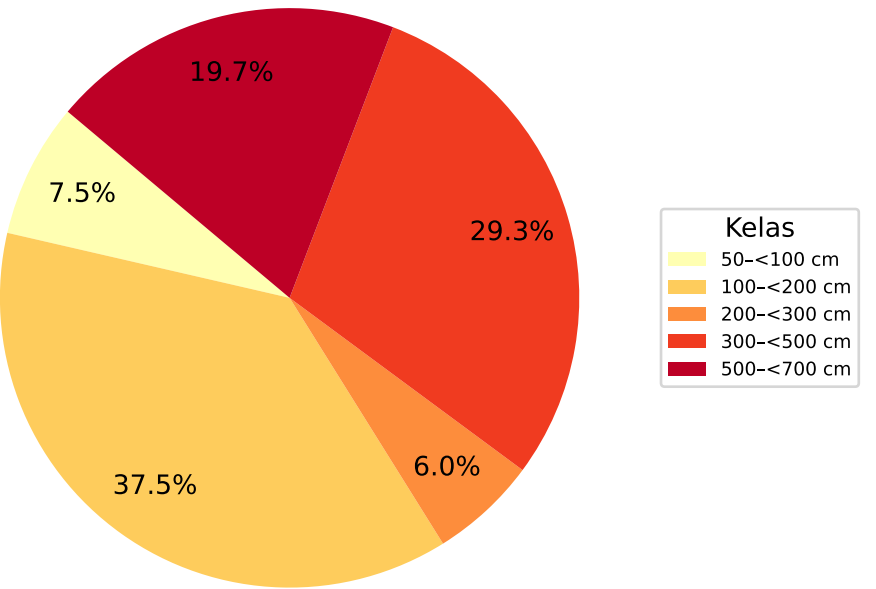

Distribusi per Fase ENSO
KABUPATEN ROKAN HULU
Hotspot
[H=0.47, p=0.789 ns
η²=-0.013]
Hotspot
0
20
40
60
80
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=27.21, p=0.000 ***
η²=0.215]
LST (°C)
25.0
25.5
26.0
26.5
27.0
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=3.46, p=0.177 ns
η²=0.012]
Curah Hujan (mm)
100
200
300
400
500
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN ROKAN HULU

A. Luas Area per Kelas Kedalaman



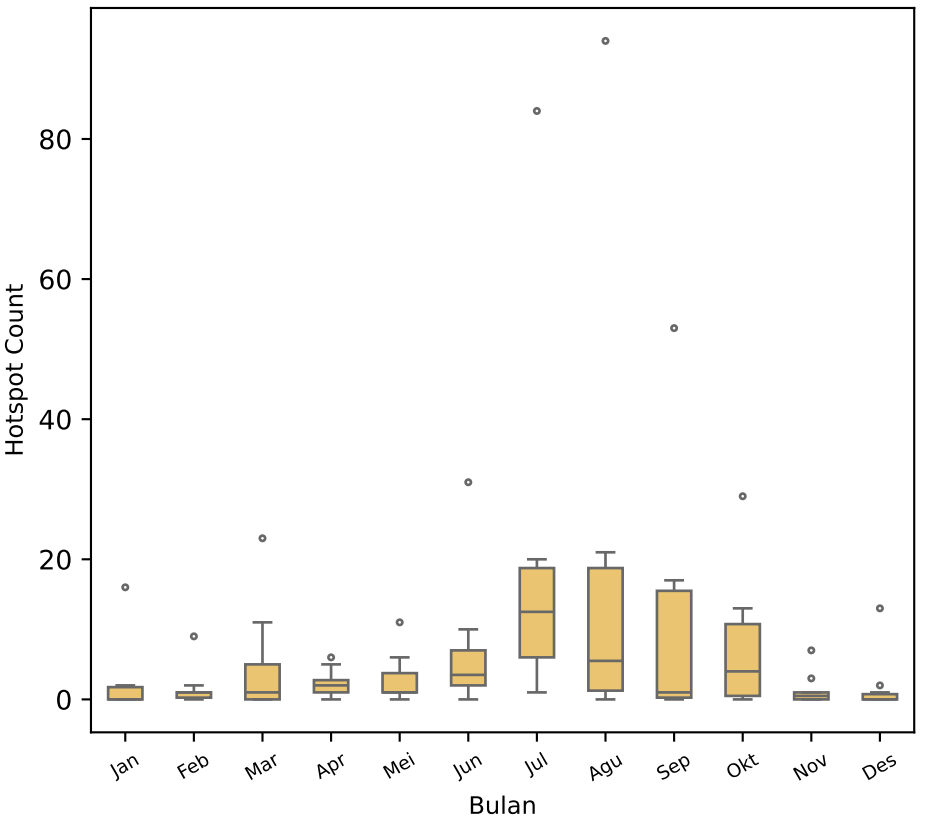
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
Statistik Gambut — Kabupaten Rokan Hulu
Total Luas          :       89,147 ha
Kelas 1    50–<100 cm:        6,693 ha  (  7.5%)
Kelas 2   100–<200 cm:       33,419 ha  ( 37.5%)
Kelas 3   200–<300 cm:        5,337 ha  (  6.0%)
Kelas 4   300–<500 cm:       26,133 ha  ( 29.3%)
Kelas 5   500–<700 cm:       17,565 ha  ( 19.7%)
Kelas 6       >700 cm:            0 ha  (  0.0%)
Kelas Dominan       : 100–<200 cm
Median Berbobot (σ) : 200–<300 cm
⚠  Kedalaman gambut = atribut statis
   Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 100–<200 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

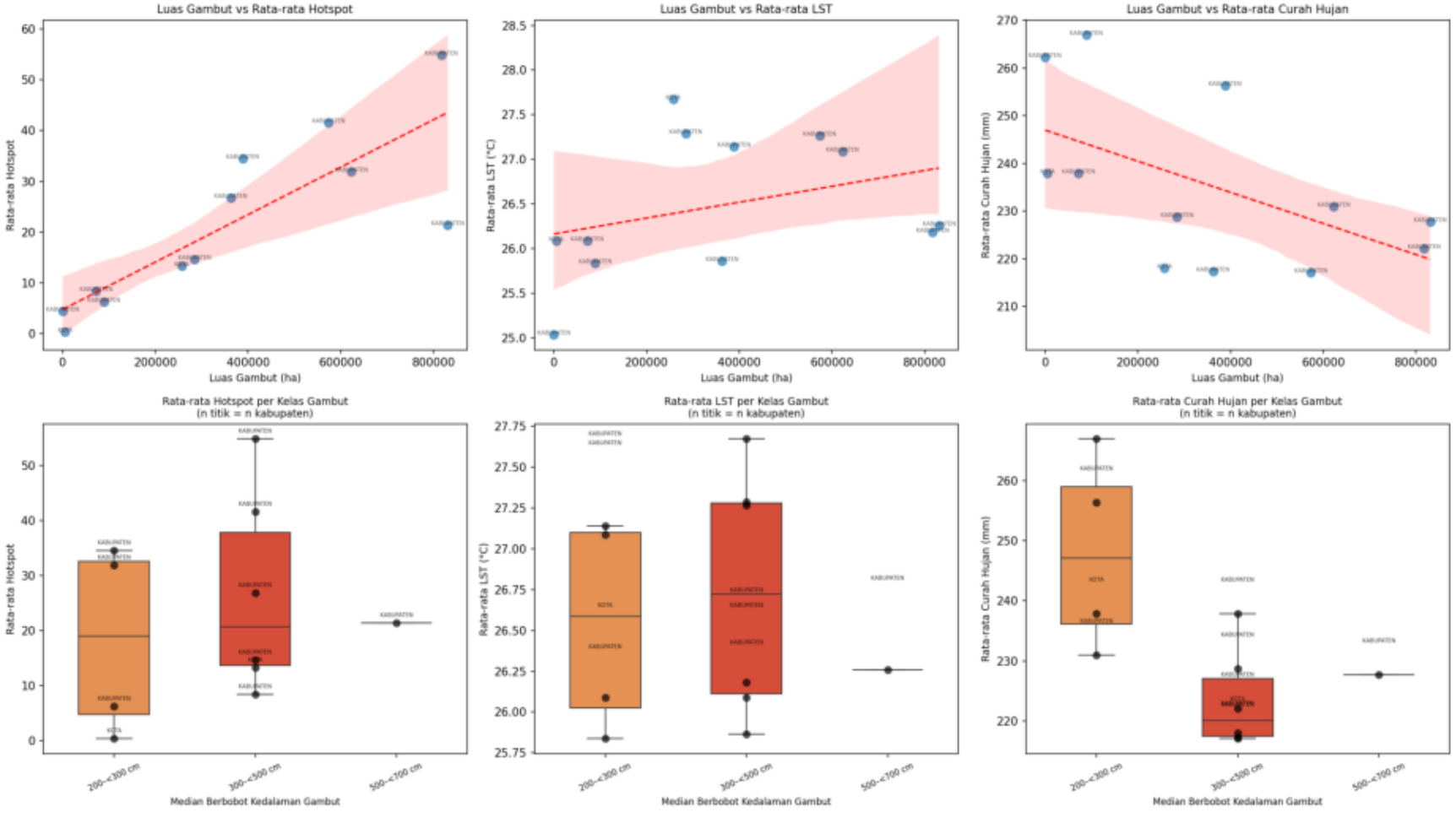
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



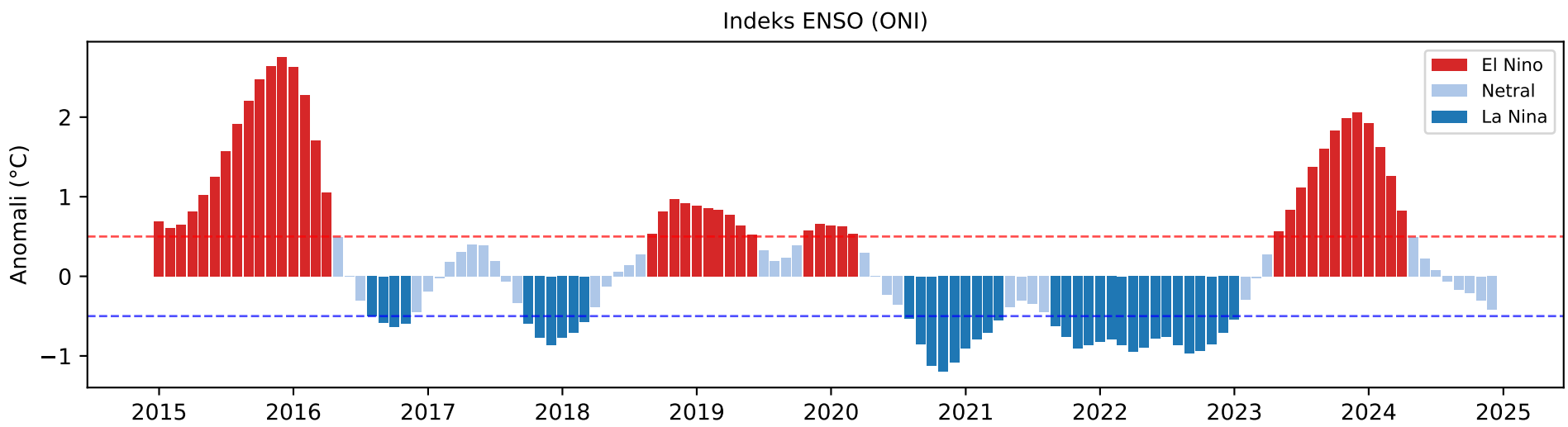
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN SIAK

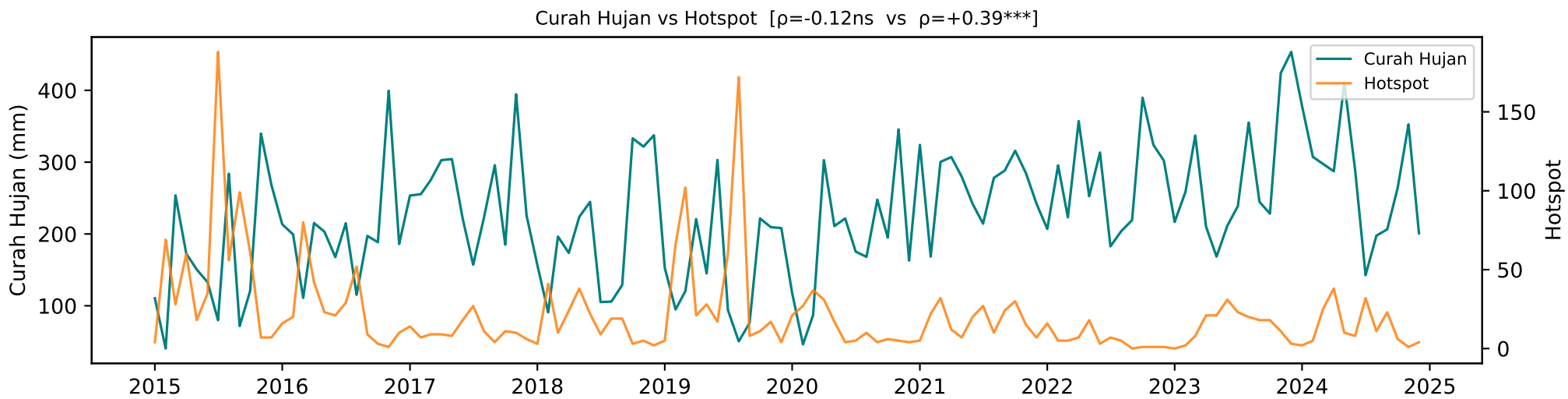
```
═══════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.542 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                            : r=-0.121 ns    (p=0.188, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.121 ns    (p=0.188, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.388 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

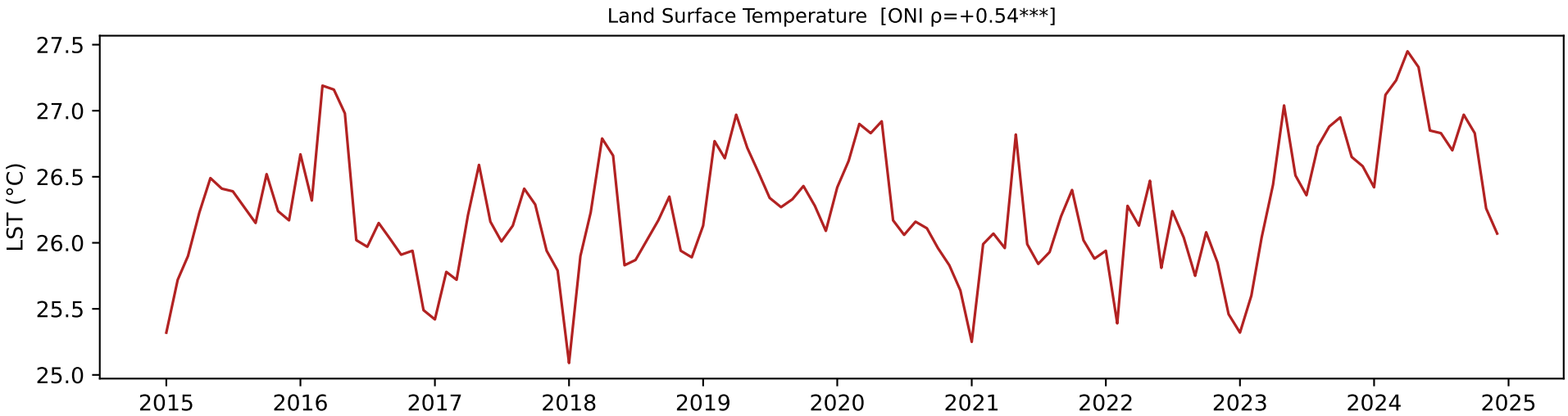
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────
  Hotspot     : H=16.62, p=0.000***, η²=0.125
  LST         : H=31.13, p=0.000, η²=0.249
  Curah Hujan : H=2.94, p=0.230, η²=0.008

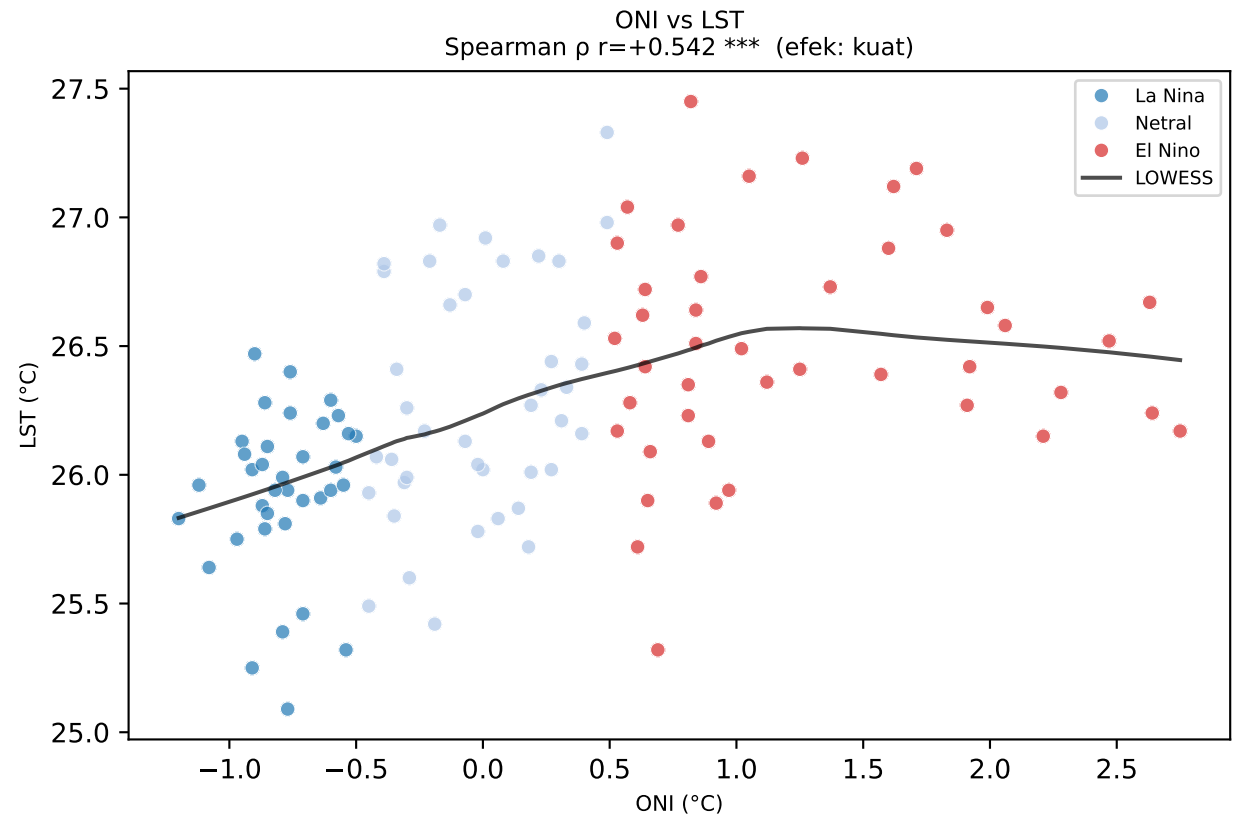
  INFO GAMBUT : total=830,788 ha | kelas dominan=500–<700 cm | median berbobot=500–<700 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

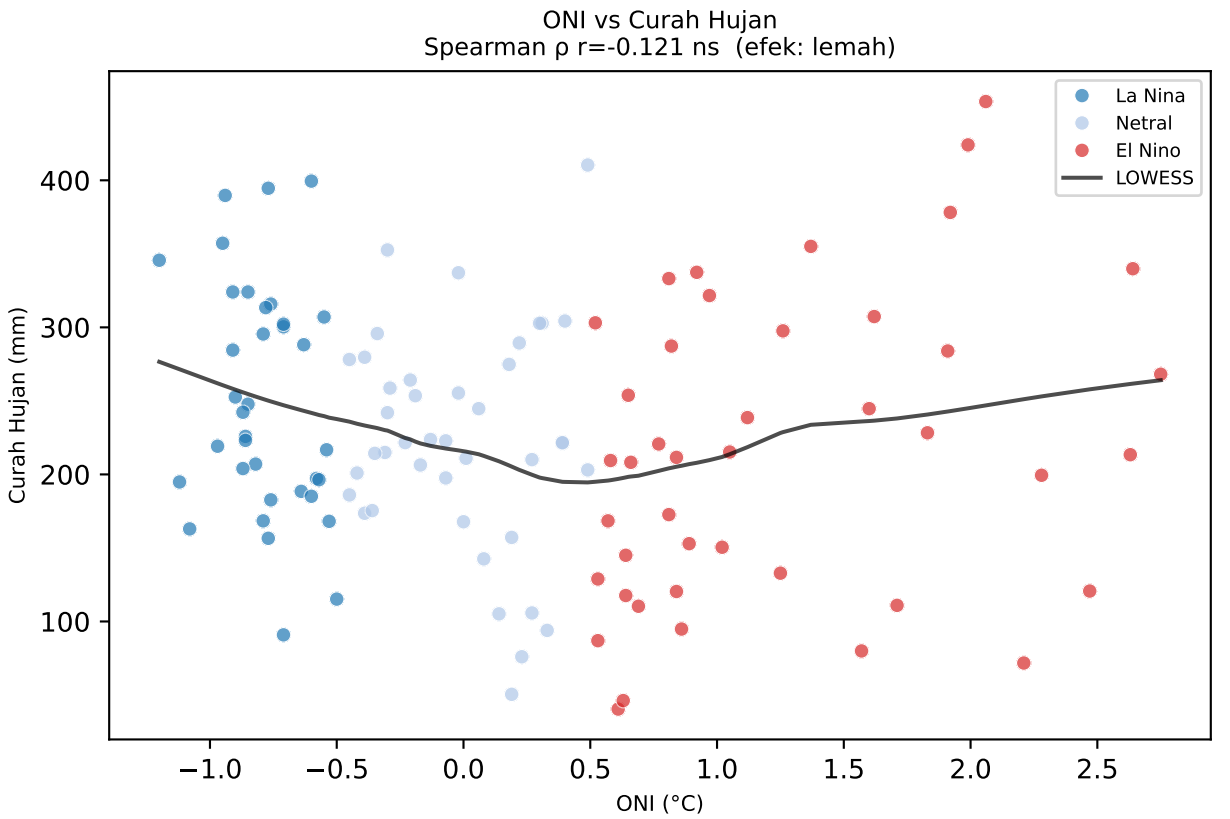


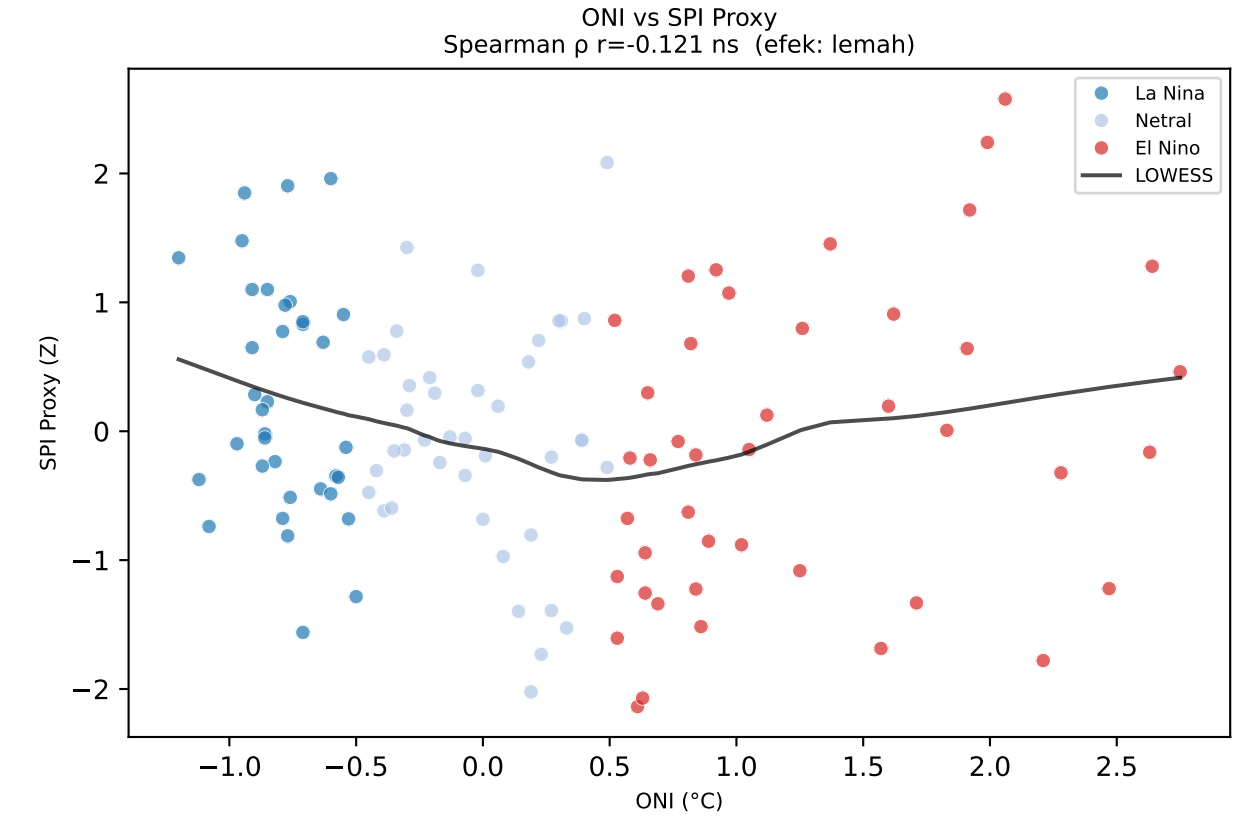


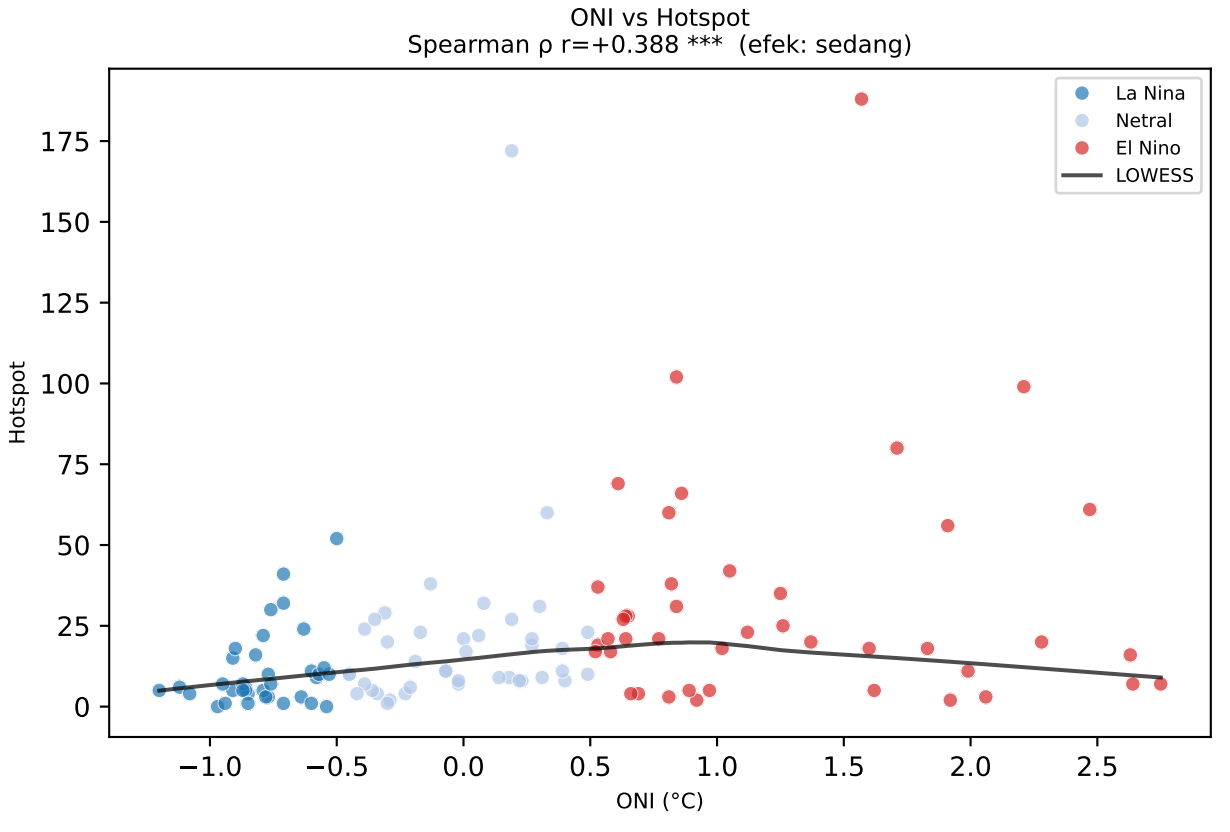


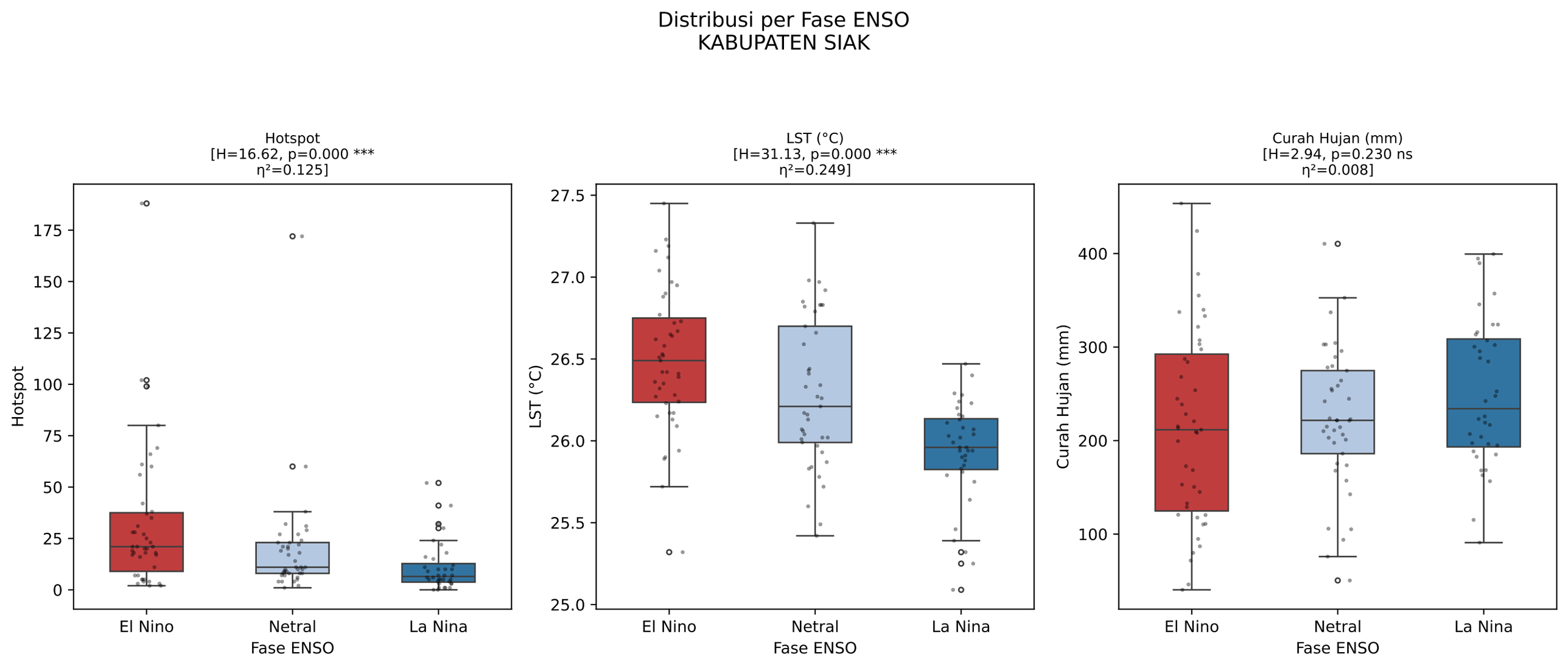
Scatter – ONI vs Variabel Iklim
KABUPATEN SIAK





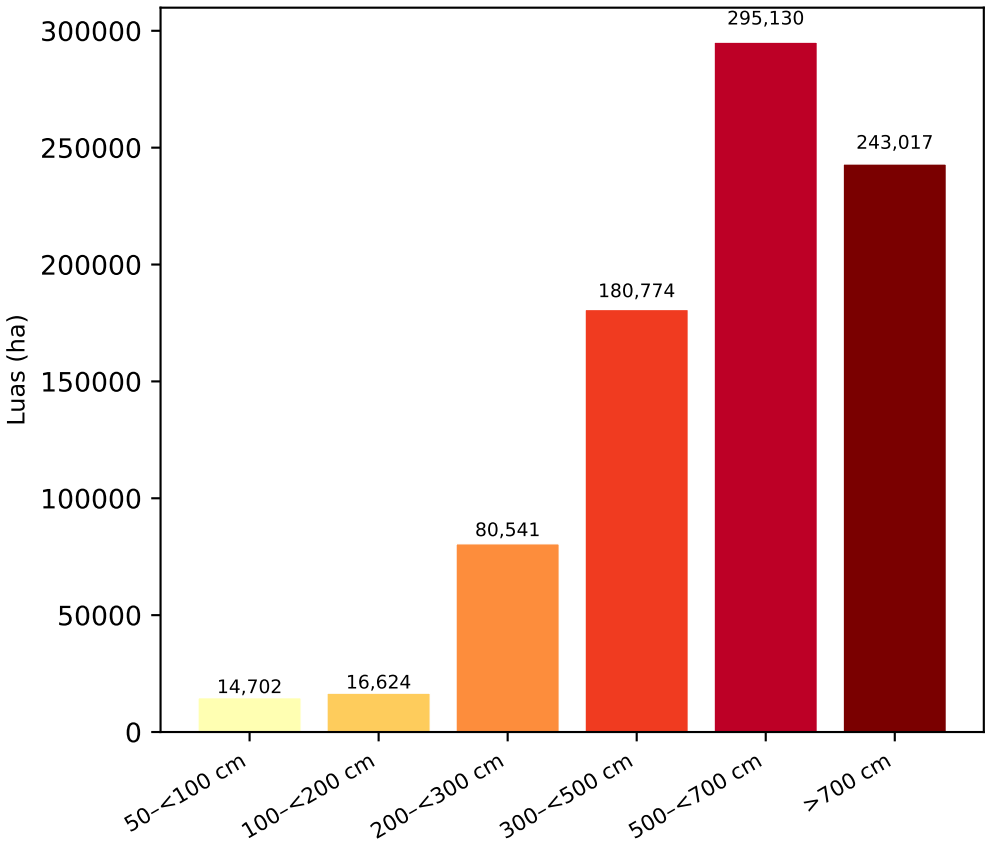




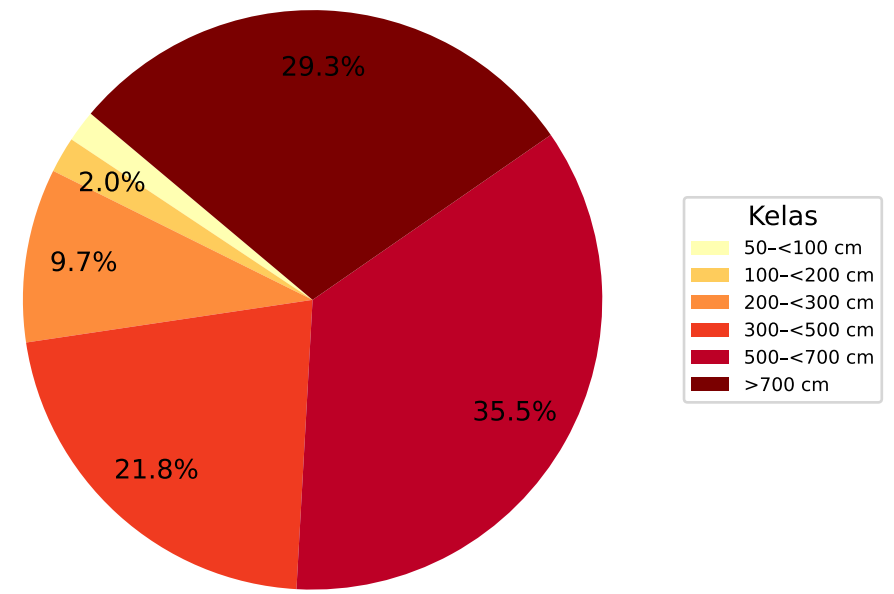

Distribusi per Fase ENSO
KABUPATEN SIAK
Hotspot
[H=16.62, p=0.000 ***
η²=0.125]
Hotspot
Fase ENSO
El Nino
Netral
La Nina
LST (°C)
[H=31.13, p=0.000 ***
η²=0.249]
LST (°C)
Curah Hujan (mm)
[H=2.94, p=0.230 ns
η²=0.008]
Curah Hujan (mm)

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN SIAK

A. Luas Area per Kelas Kedalaman



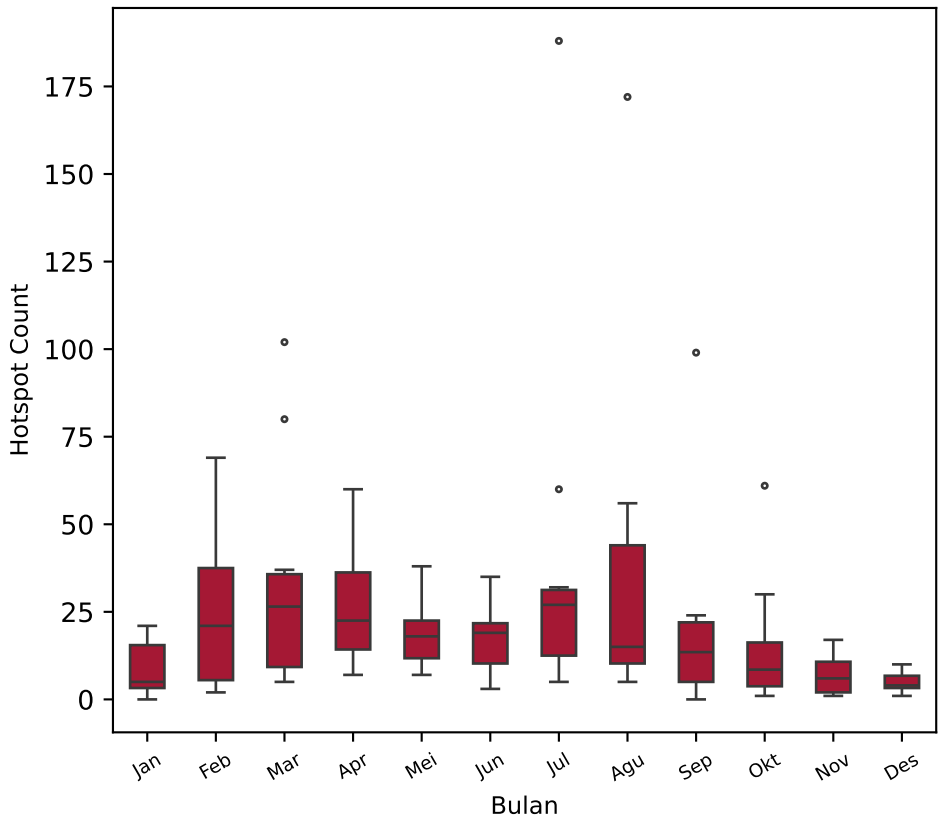
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════
  Statistik Gambut — Kabupaten Siak
───────────────────────────────────────────────
  Total Luas          :      830,788 ha
───────────────────────────────────────────────
  Kelas 1    50–<100 cm:     14,702 ha  (  1.8%)
  Kelas 2   100–<200 cm:     16,624 ha  (  2.0%)
  Kelas 3   200–<300 cm:     80,541 ha  (  9.7%)
  Kelas 4   300–<500 cm:    180,774 ha  ( 21.8%)
  Kelas 5   500–<700 cm:    295,130 ha  ( 35.5%)
  Kelas 6       >700 cm:    243,017 ha  ( 29.3%)
───────────────────────────────────────────────
  Kelas Dominan       : 500–<700 cm
  Median Berbobot (σ) : 500–<700 cm
───────────────────────────────────────────────
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 500–<700 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

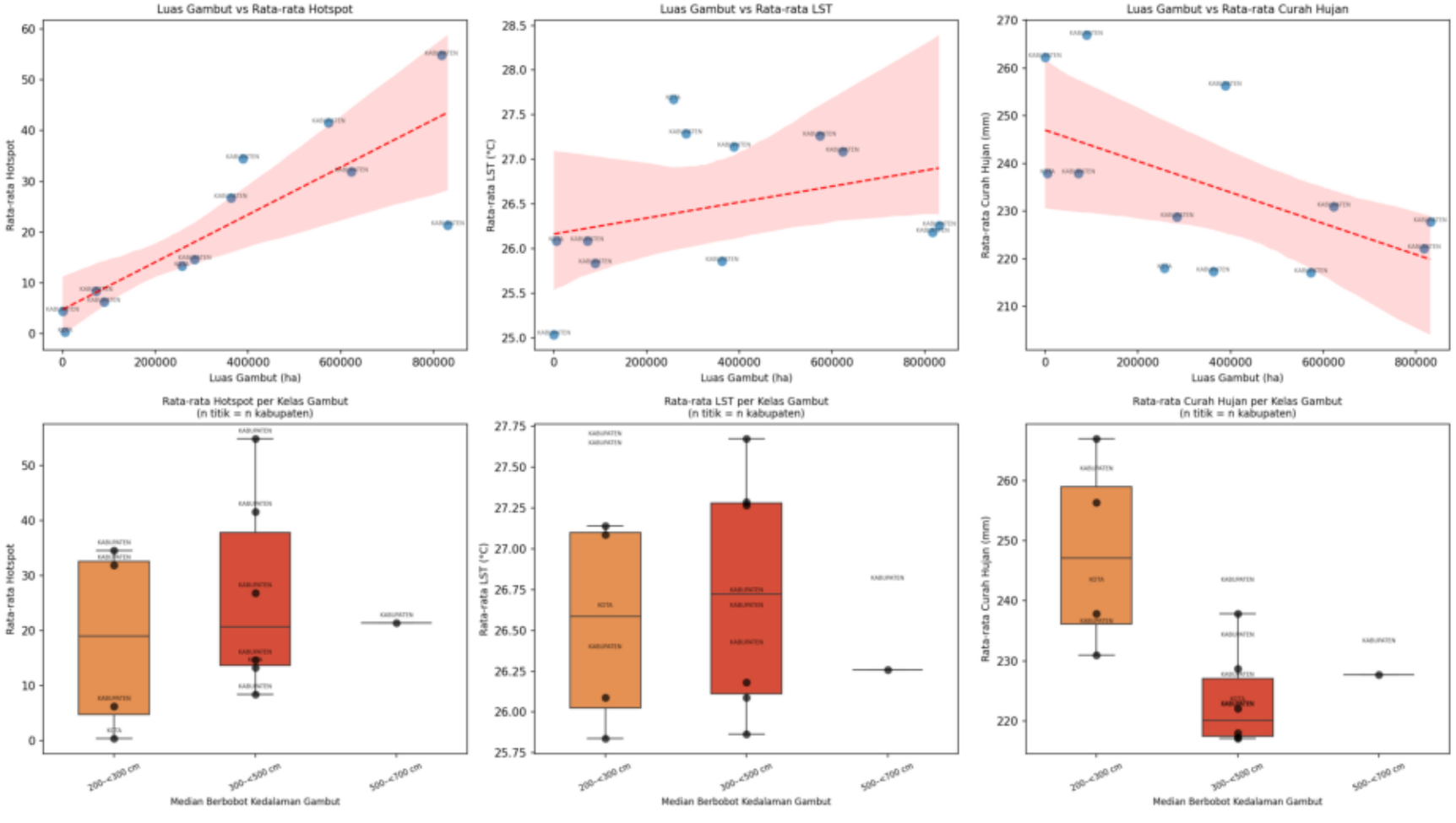
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



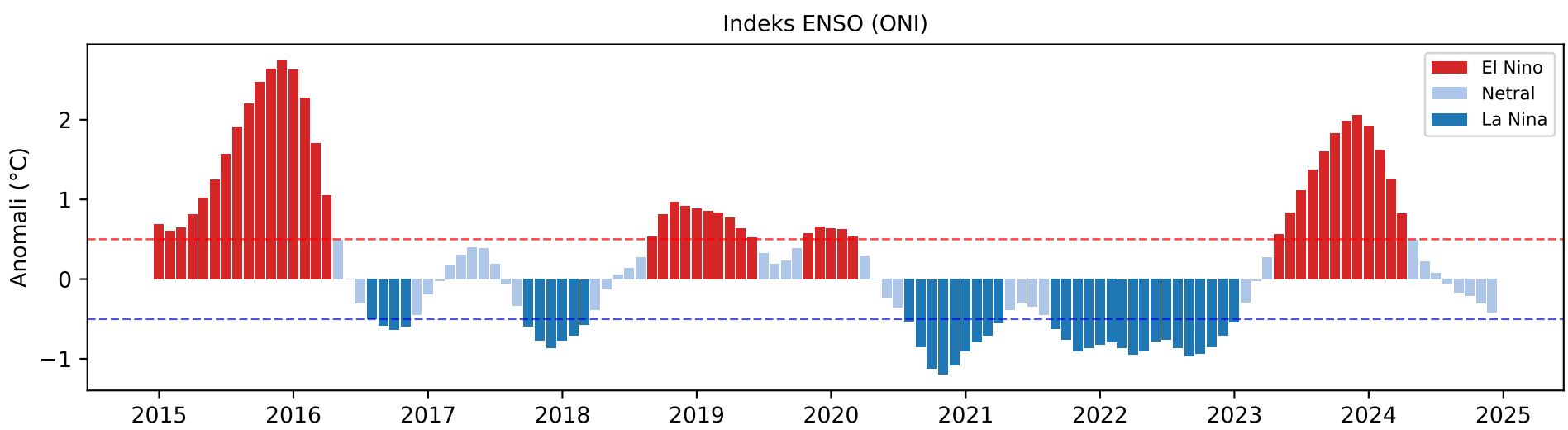
Laporan Korelasi Iklim, Gambut & Hotspot
KOTA DUMAI

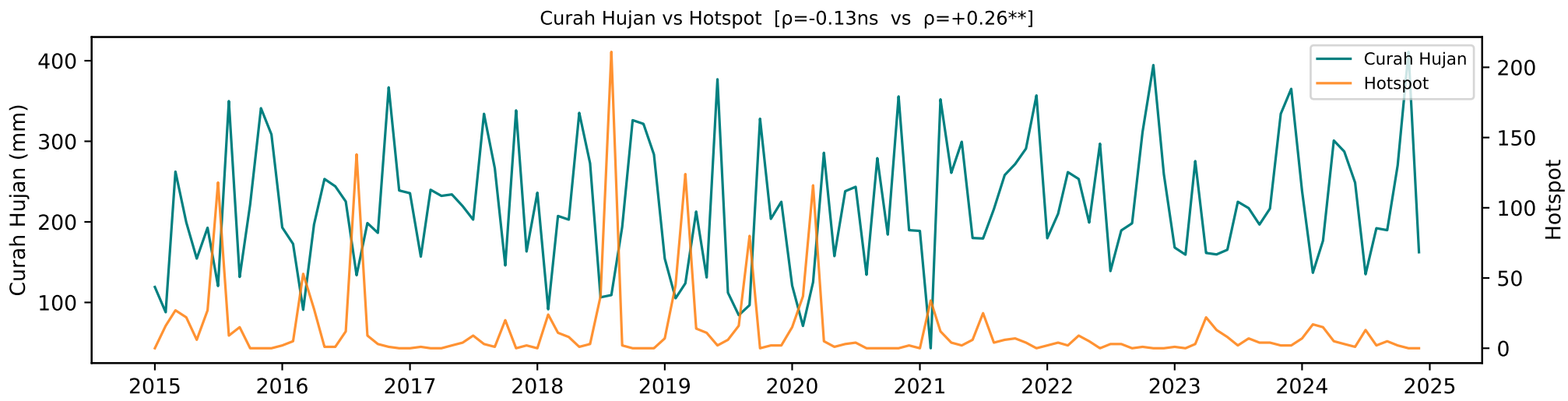
```
═══════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.431 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.126 ns    (p=0.170, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.126 ns    (p=0.170, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.262 **    (p=0.004, Spearman ρ, efek=lemah, n=120)

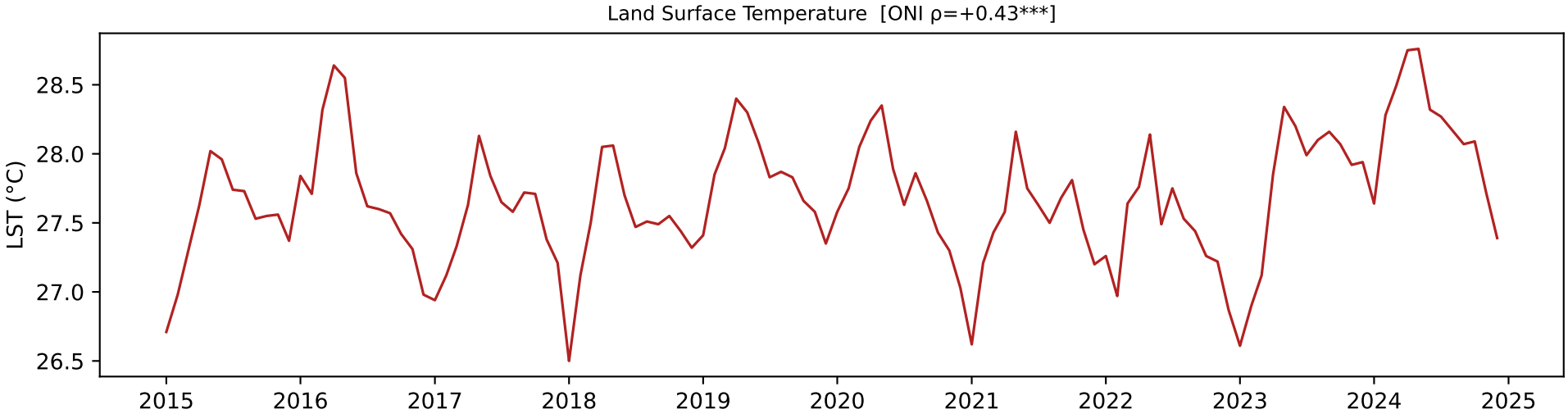
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────
  Hotspot     : H=8.38, p=0.015*, η²=0.055
  LST         : H=24.10, p=0.000, η²=0.189
  Curah Hujan : H=2.40, p=0.302, η²=0.003

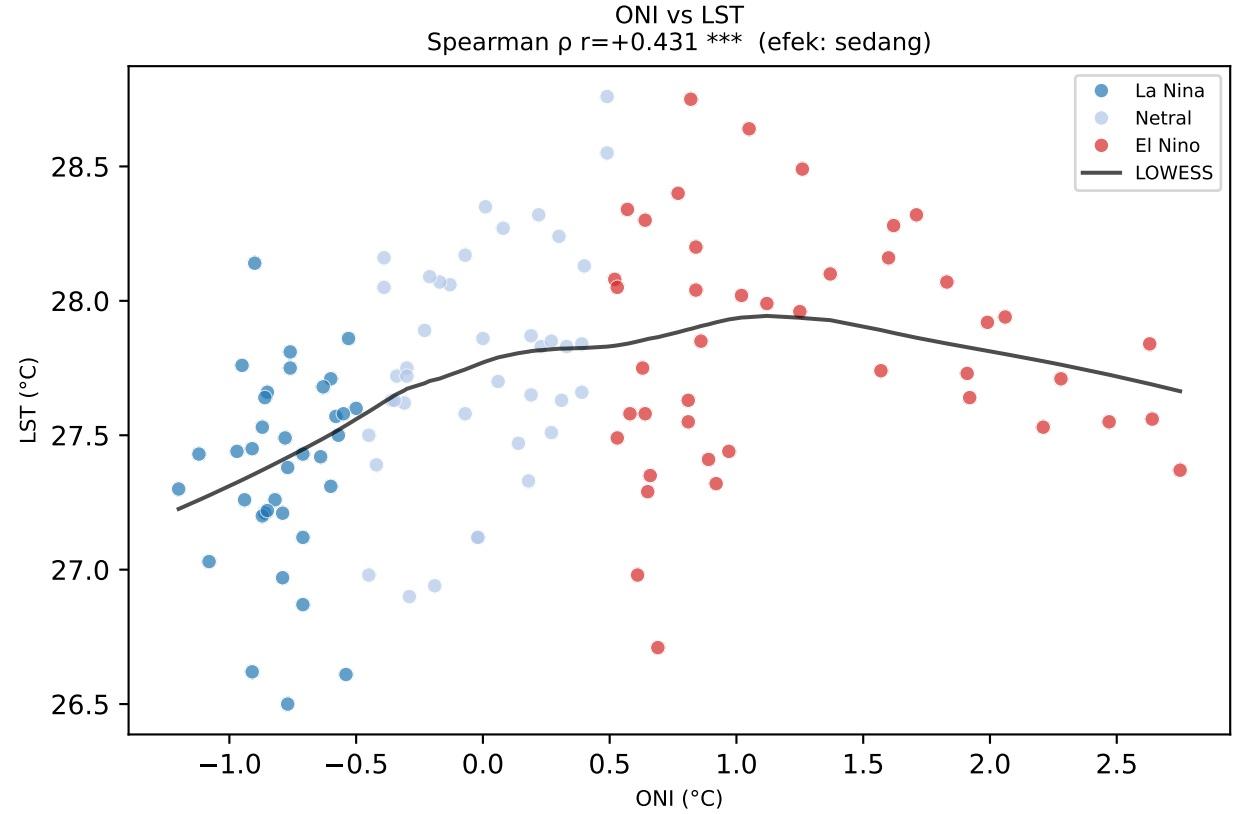
  INFO GAMBUT : total=257,356 ha | kelas dominan=200–<300 cm | median berbobot=300–<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

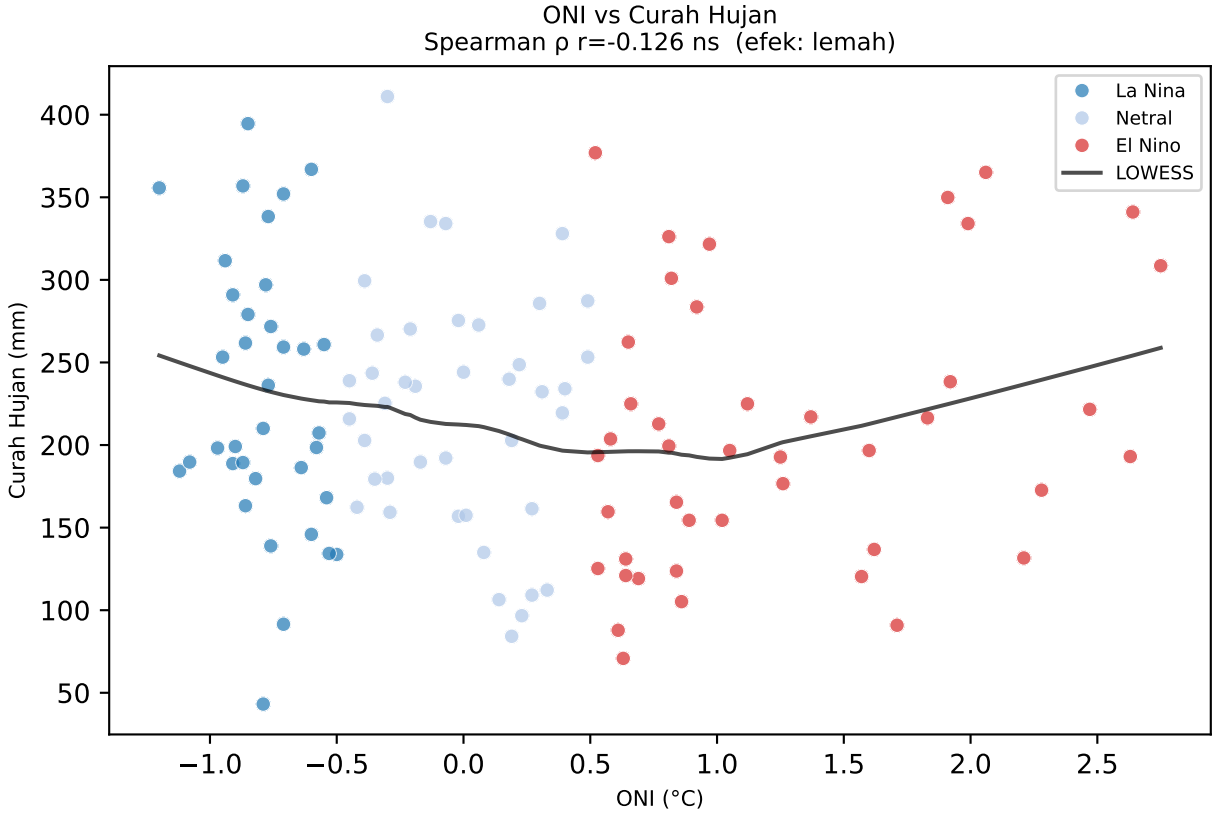


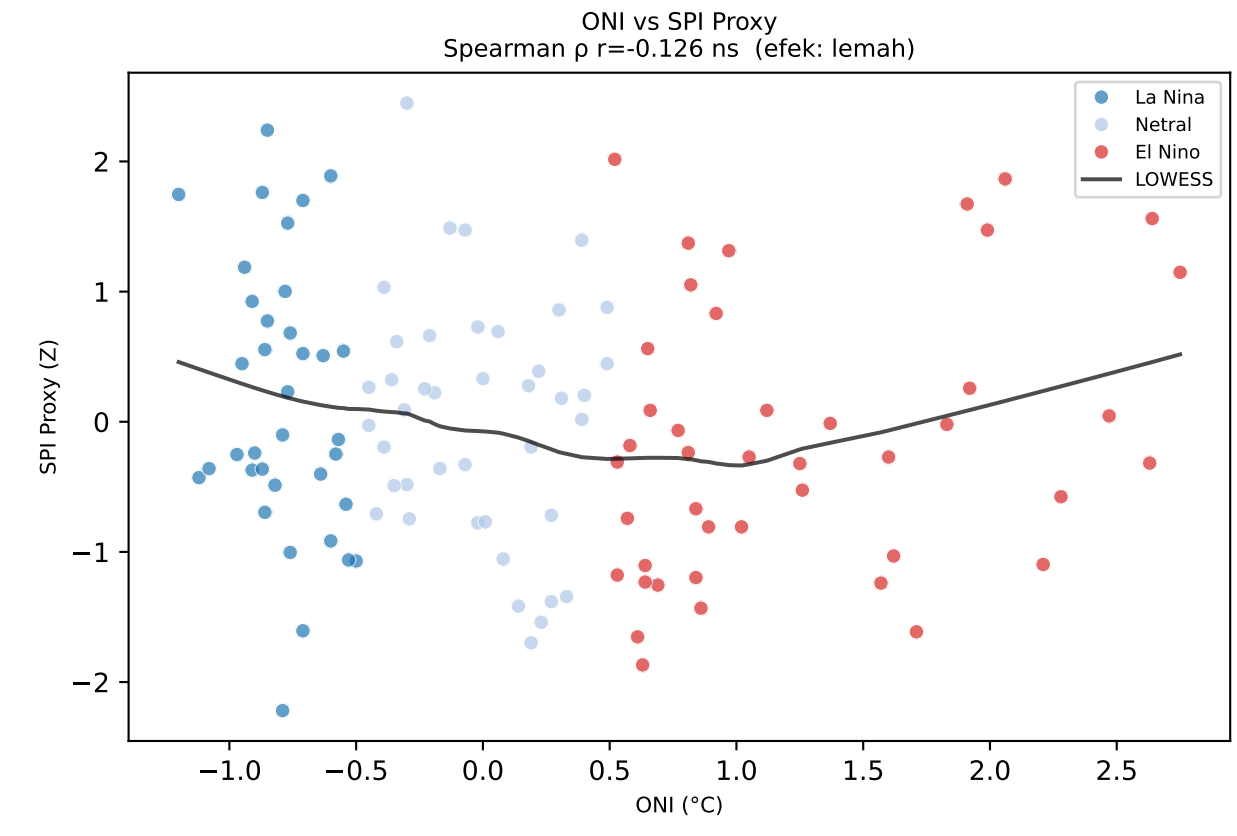


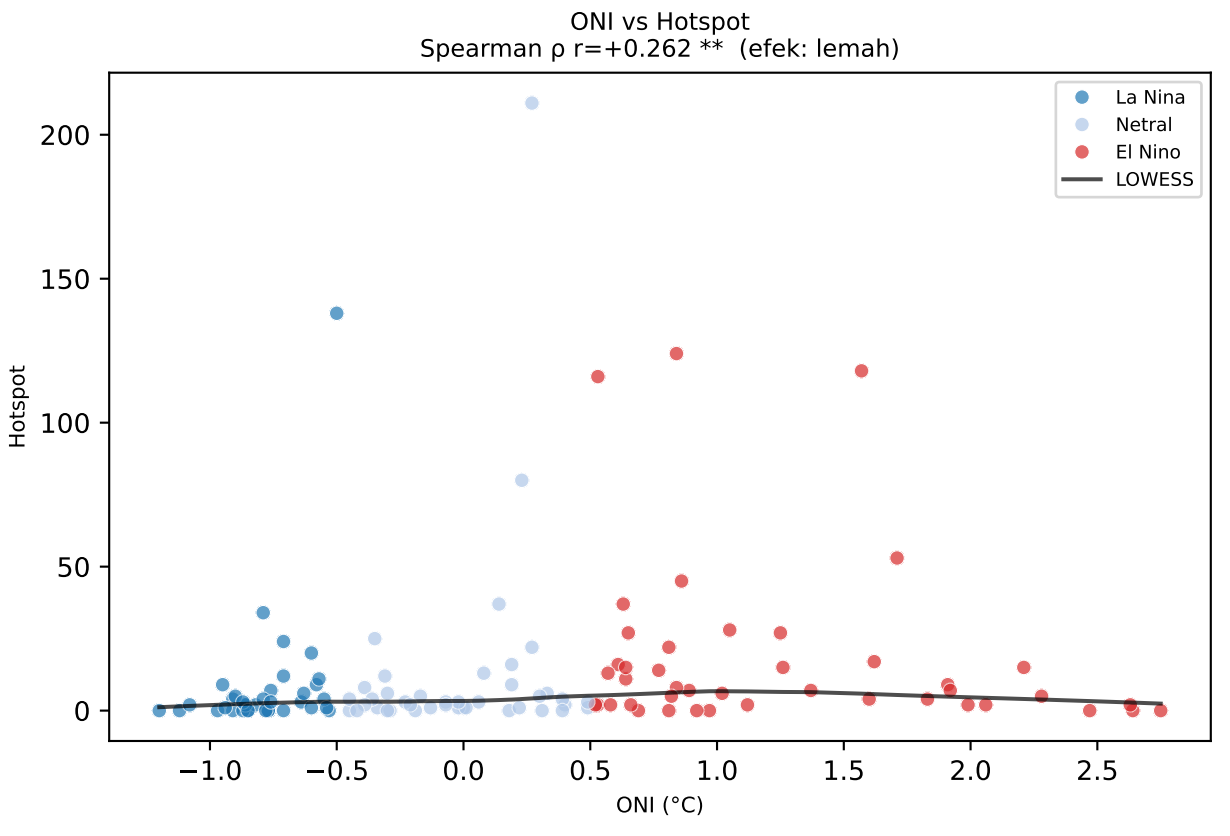


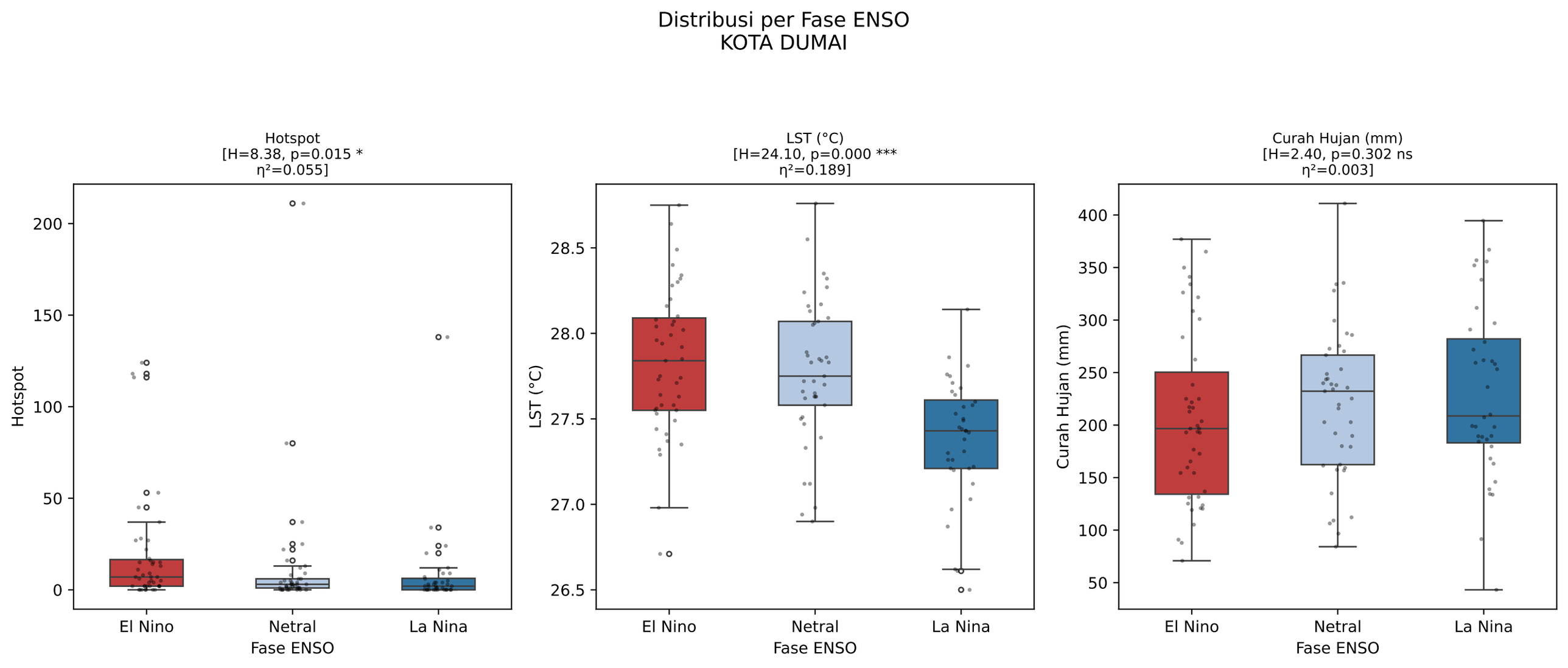
Scatter – ONI vs Variabel Iklim
KOTA DUMAI





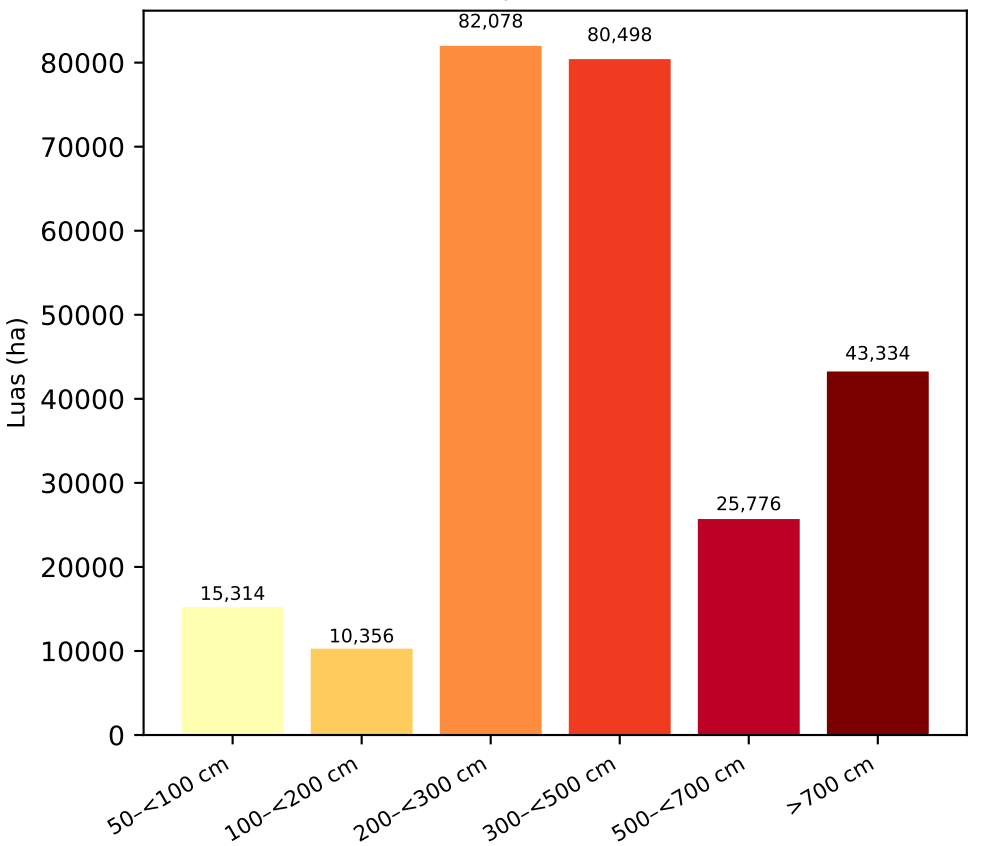




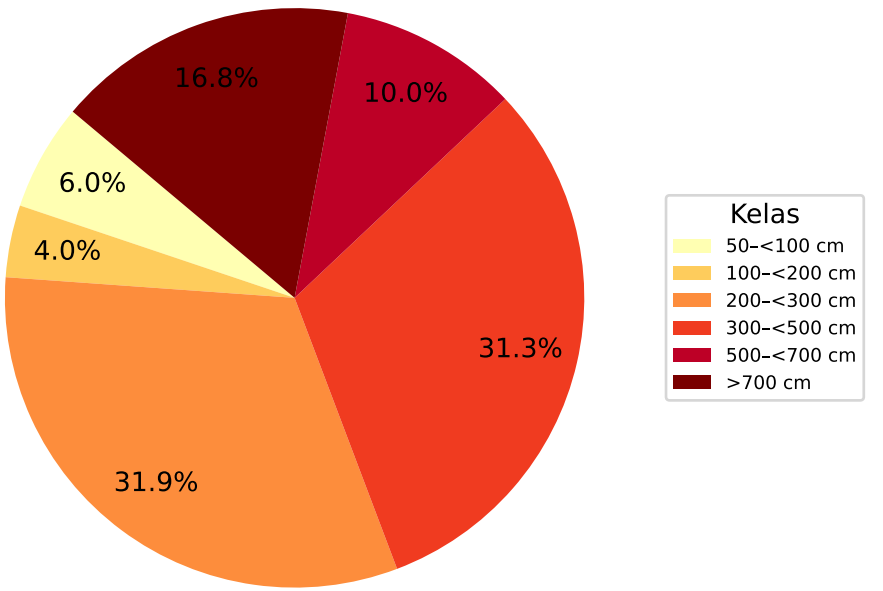

Distribusi per Fase ENSO
KOTA DUMAI
Hotspot
[H=8.38, p=0.015 *
η²=0.055]
Hotspot
0
50
100
150
200
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=24.10, p=0.000 ***
η²=0.189]
LST (°C)
26.5
27.0
27.5
28.0
28.5
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=2.40, p=0.302 ns
η²=0.003]
Curah Hujan (mm)
50
100
150
200
250
300
350
400
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KOTA DUMAI

A. Luas Area per Kelas Kedalaman



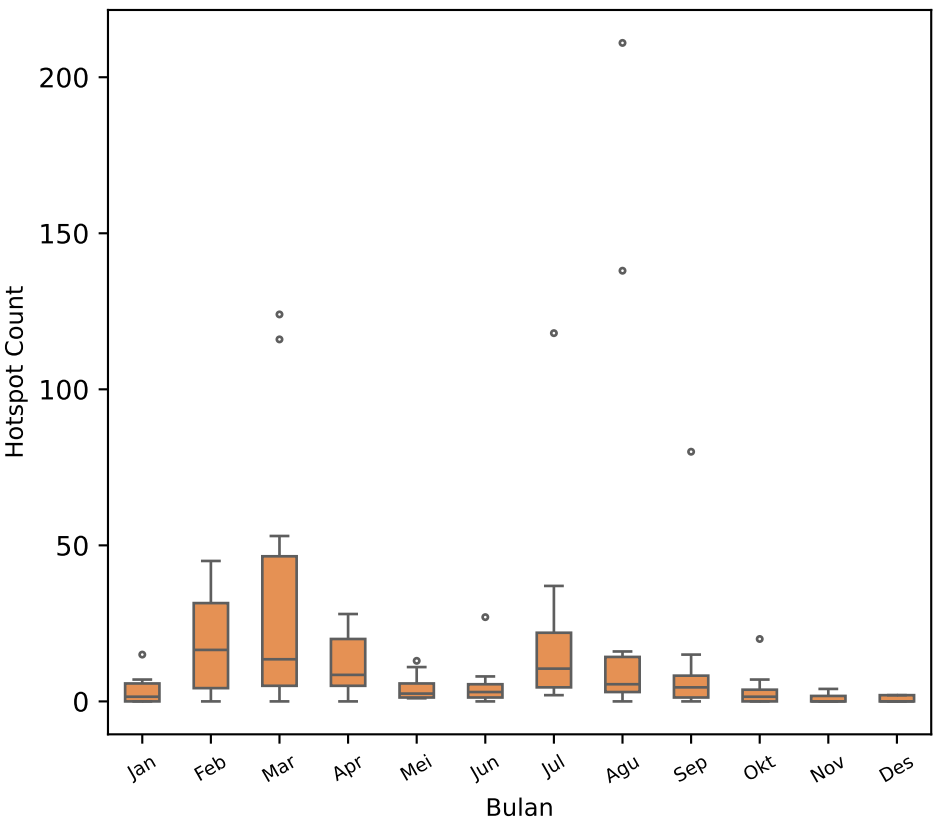
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
Statistik Gambut — Kota Dumai
Total Luas          :      257,356 ha
Kelas 1    50–<100 cm:      15,314 ha  (  6.0%)
Kelas 2   100–<200 cm:      10,356 ha  (  4.0%)
Kelas 3   200–<300 cm:      82,078 ha  ( 31.9%)
Kelas 4   300–<500 cm:      80,498 ha  ( 31.3%)
Kelas 5   500–<700 cm:      25,776 ha  ( 10.0%)
Kelas 6       >700 cm:      43,334 ha  ( 16.8%)
Kelas Dominan       : 200–<300 cm
Median Berbobot (σ) : 300–<500 cm
⚠  Kedalaman gambut = atribut statis
   Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 200–<300 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

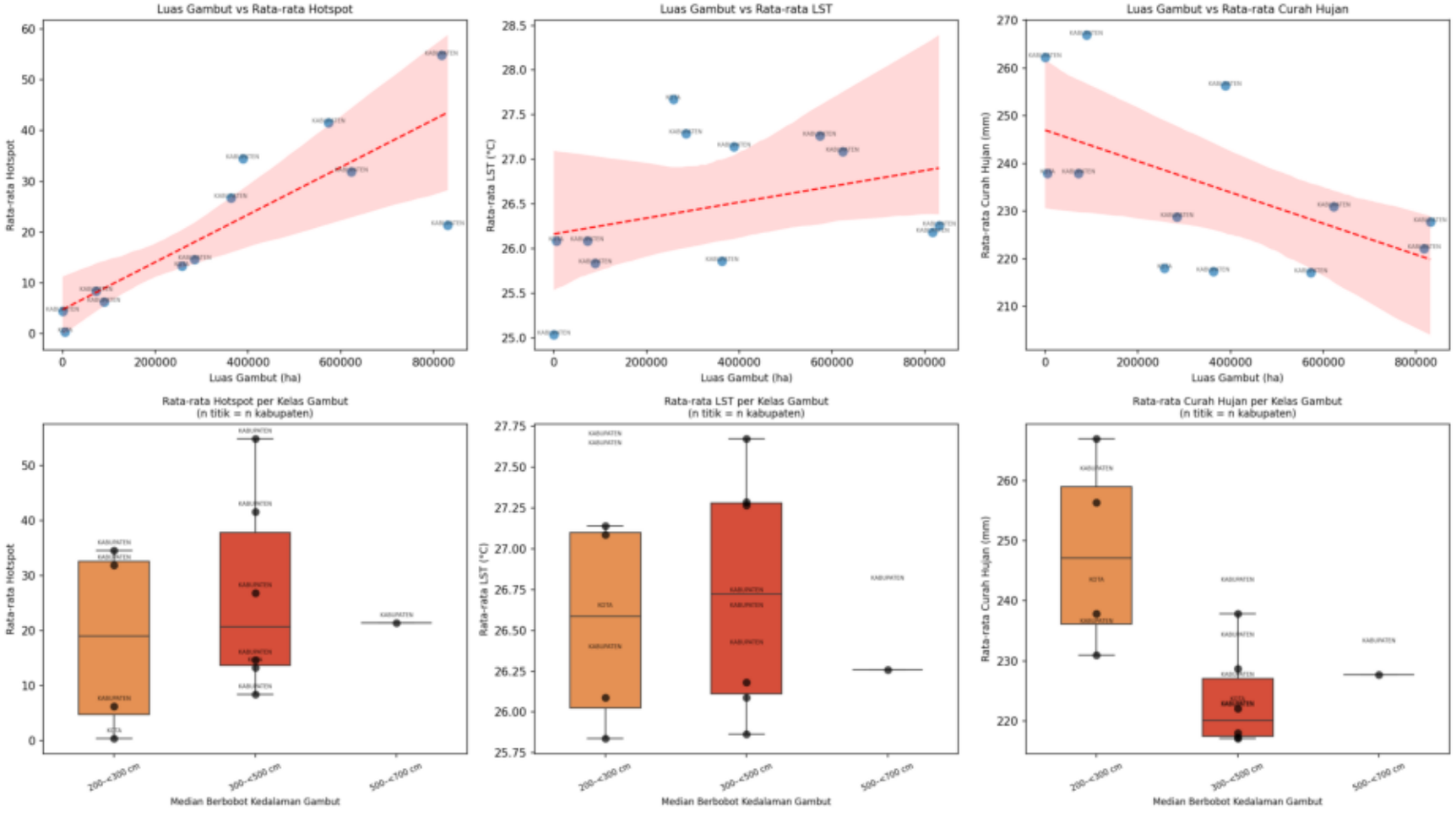
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



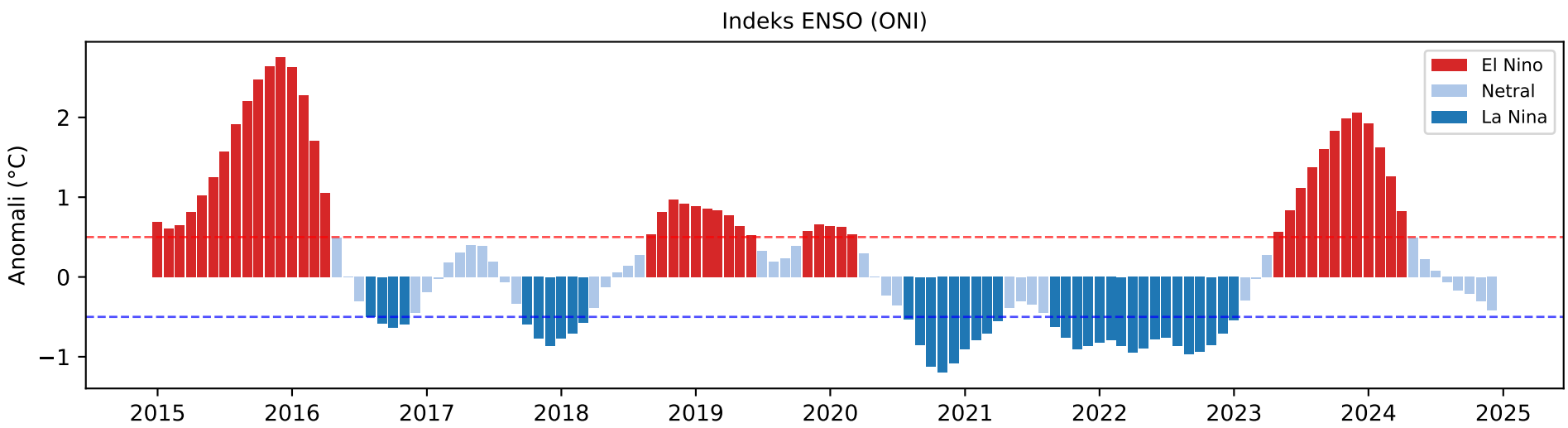
Laporan Korelasi Iklim, Gambut & Hotspot
KOTA PEKANBARU

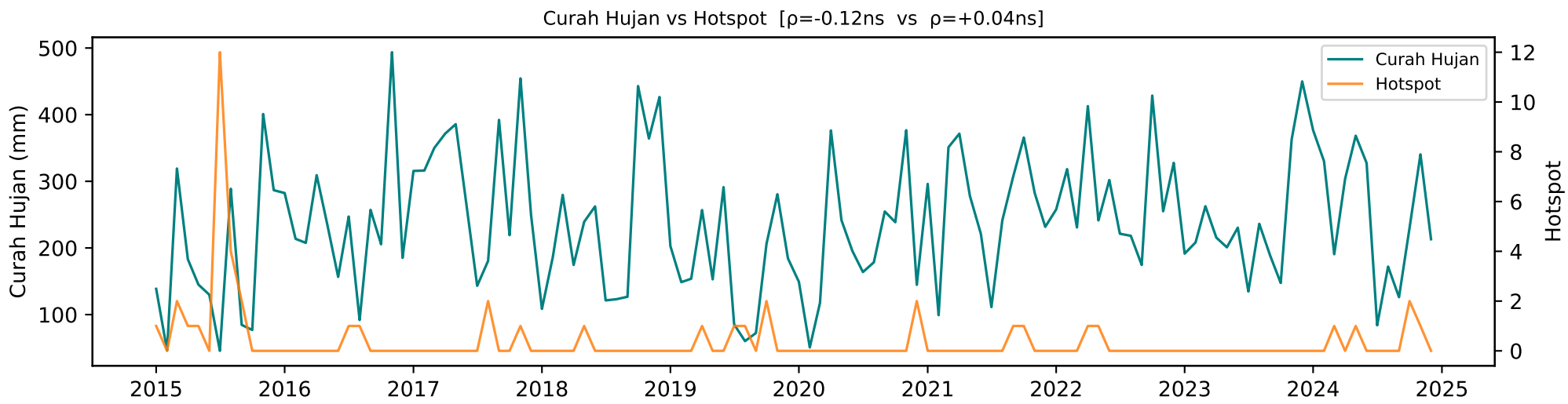
```
═══════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.536 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                            : r=-0.123 ns    (p=0.181, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.123 ns    (p=0.181, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.039 ns    (p=0.672, Spearman ρ, efek=diabaikan, n=120)

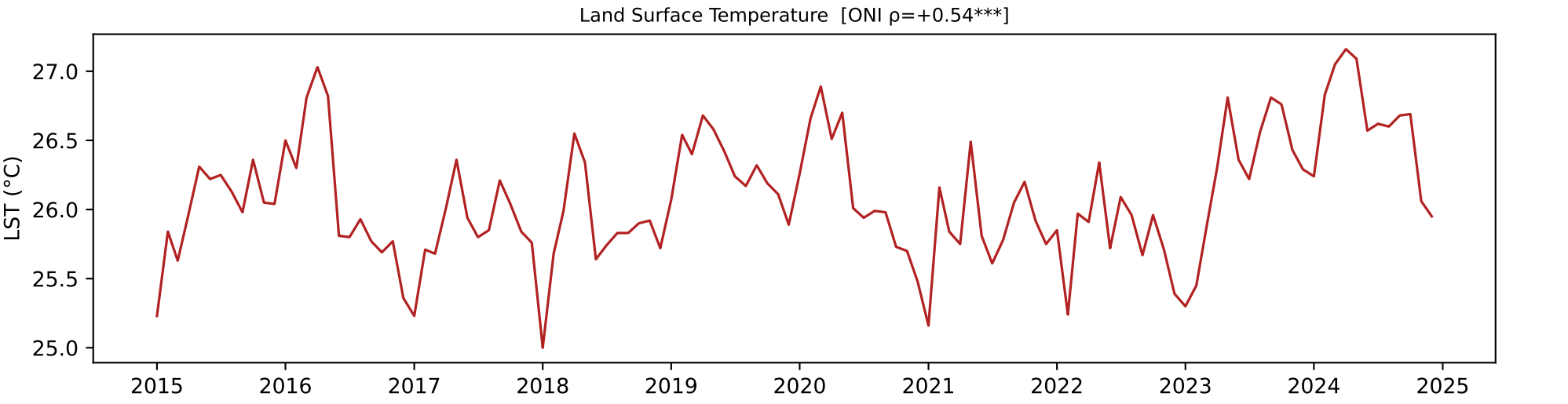
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────
  Hotspot     : H=0.14, p=0.931ns, η²=-0.016
  LST         : H=30.43, p=0.000, η²=0.243
  Curah Hujan : H=4.38, p=0.112, η²=0.020

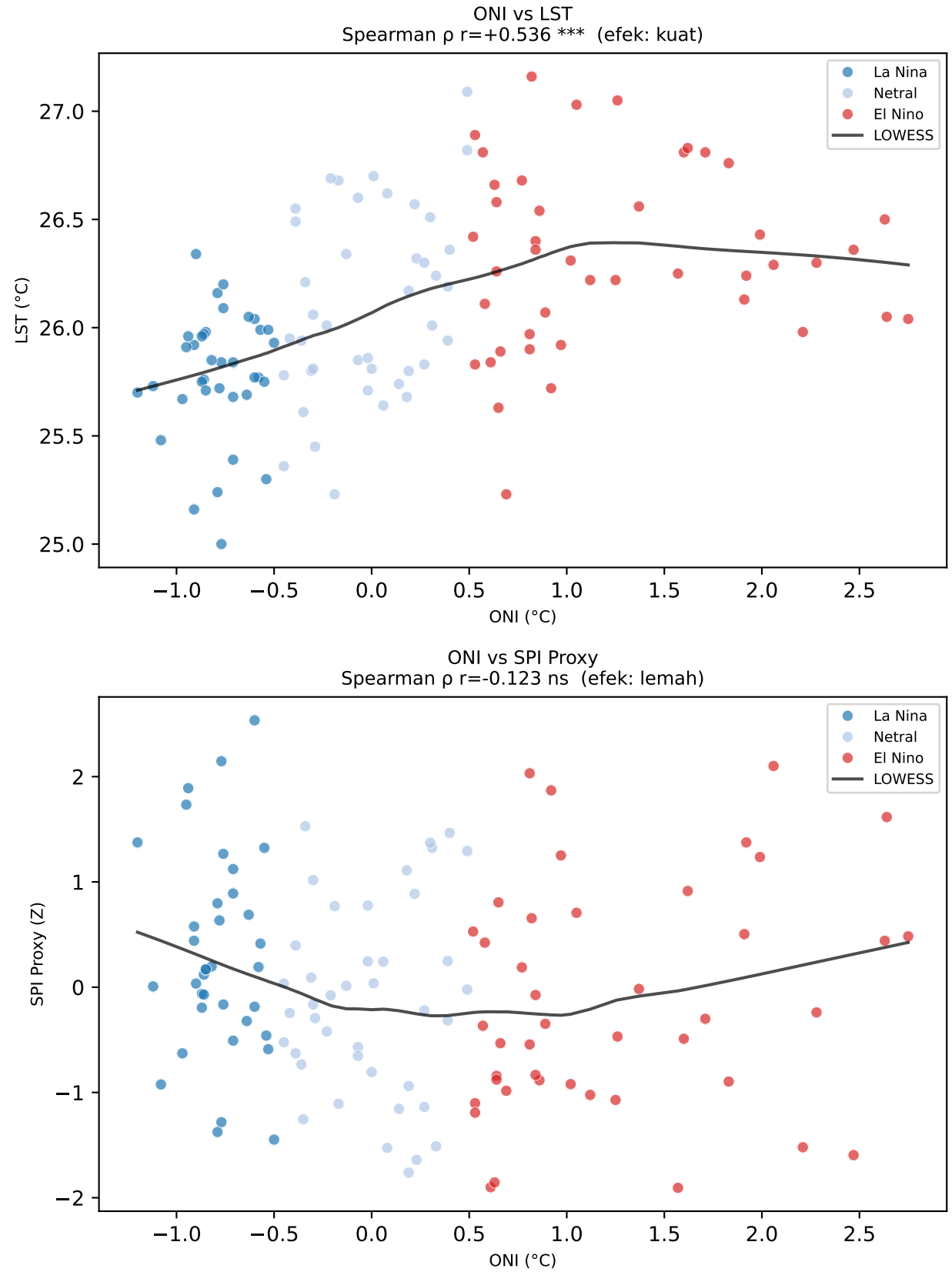
  INFO GAMBUT : total=5,414 ha | kelas dominan=300–<500 cm | median berbobot=200–<300 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

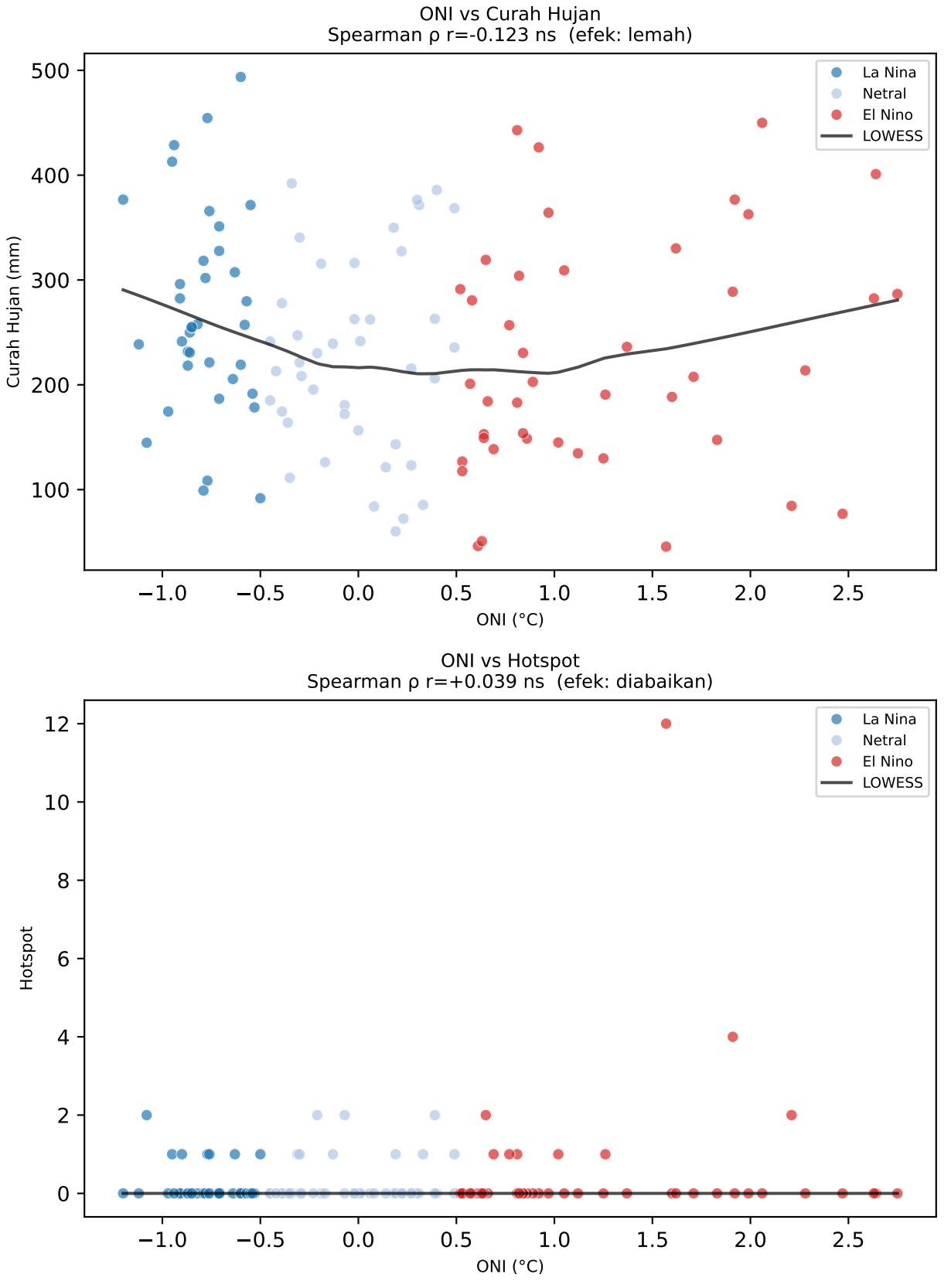


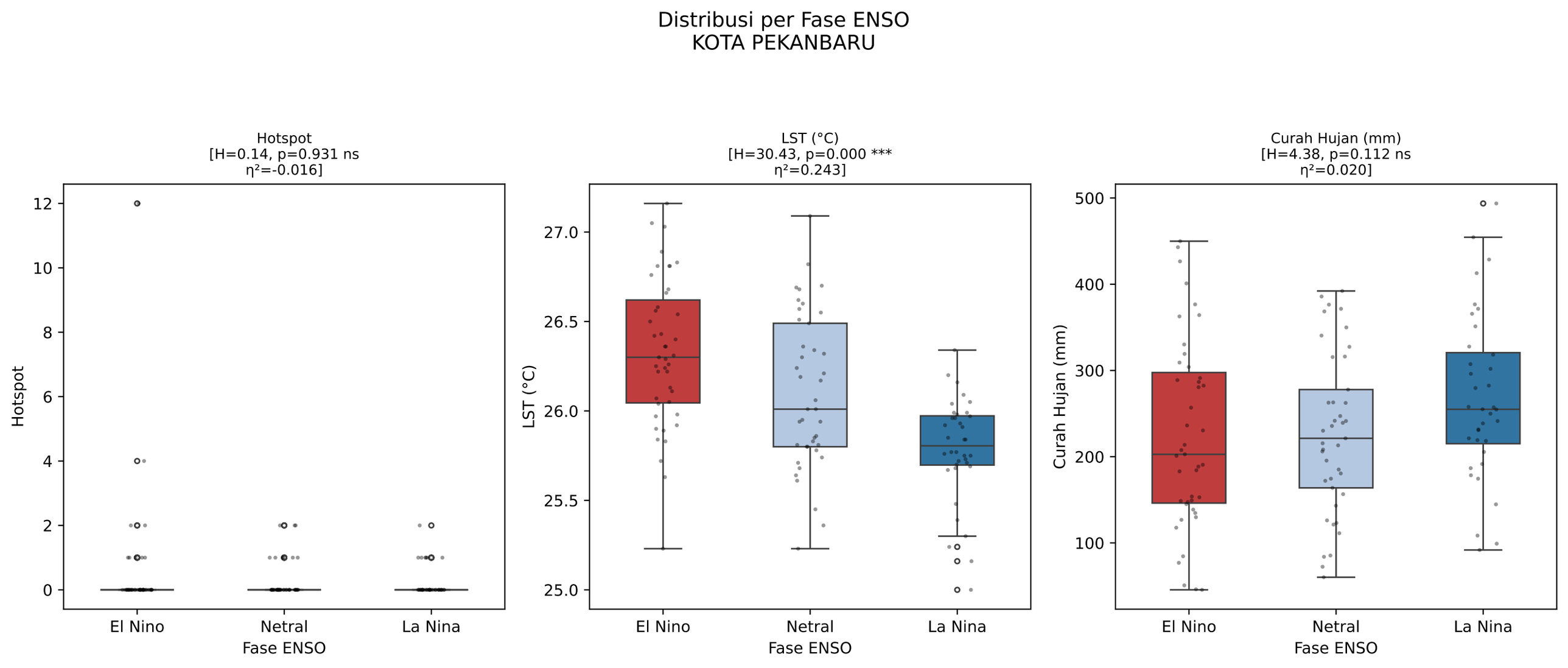




Scatter – ONI vs Variabel Iklim
KOTA PEKANBARU

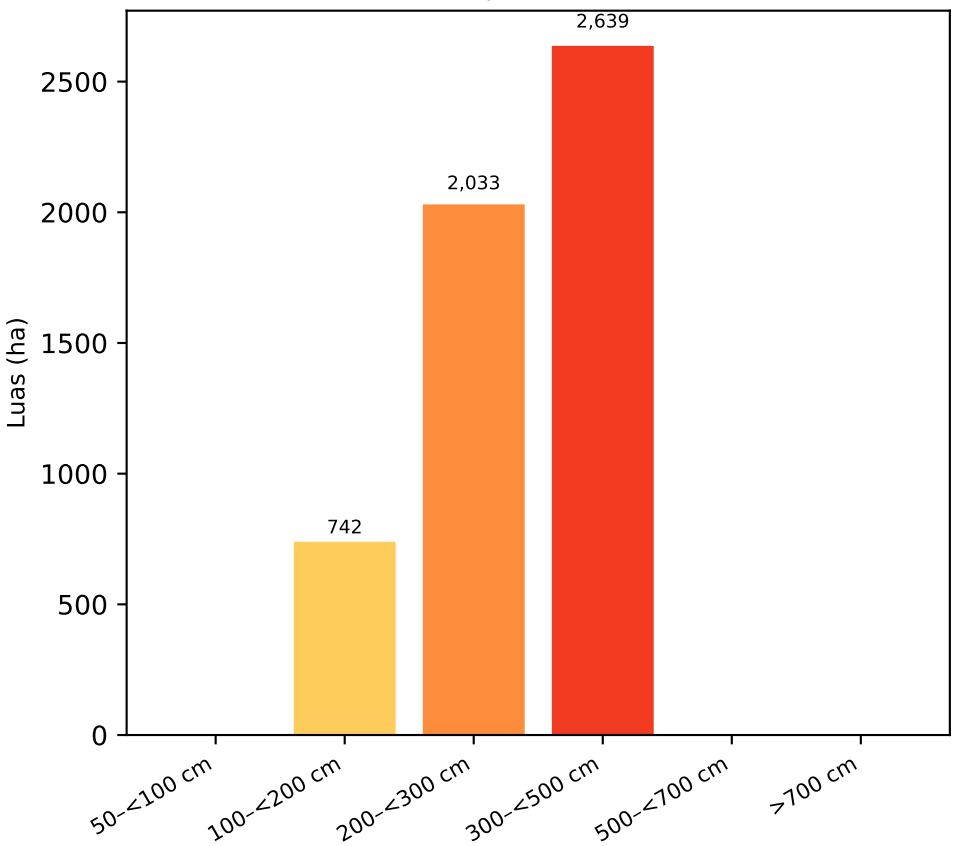




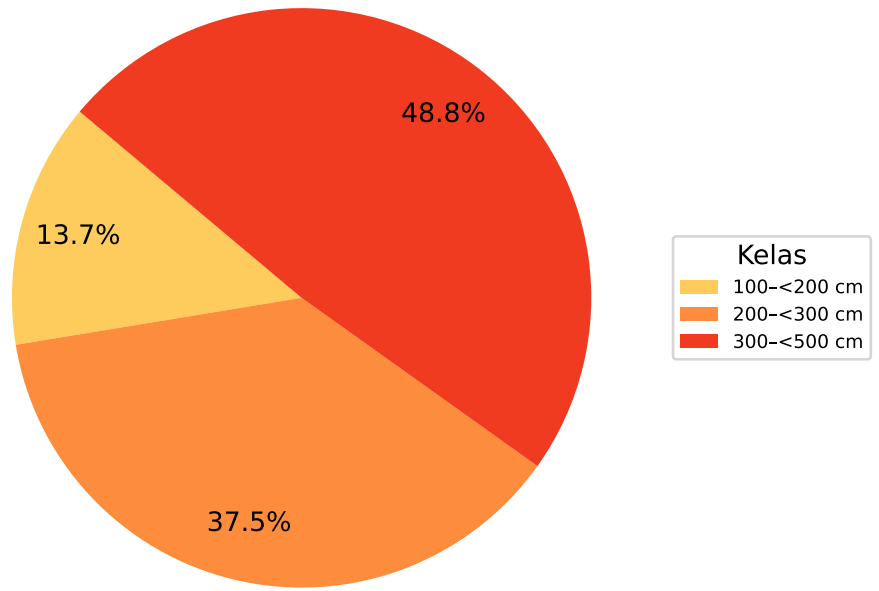

Distribusi per Fase ENSO
KOTA PEKANBARU
Hotspot
[H=0.14, p=0.931 ns
η²=-0.016]
Hotspot
Fase ENSO
El Nino
Netral
La Nina
LST (°C)
[H=30.43, p=0.000 ***
η²=0.243]
LST (°C)
Curah Hujan (mm)
[H=4.38, p=0.112 ns
η²=0.020]
Curah Hujan (mm)

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KOTA PEKANBARU

A. Luas Area per Kelas Kedalaman



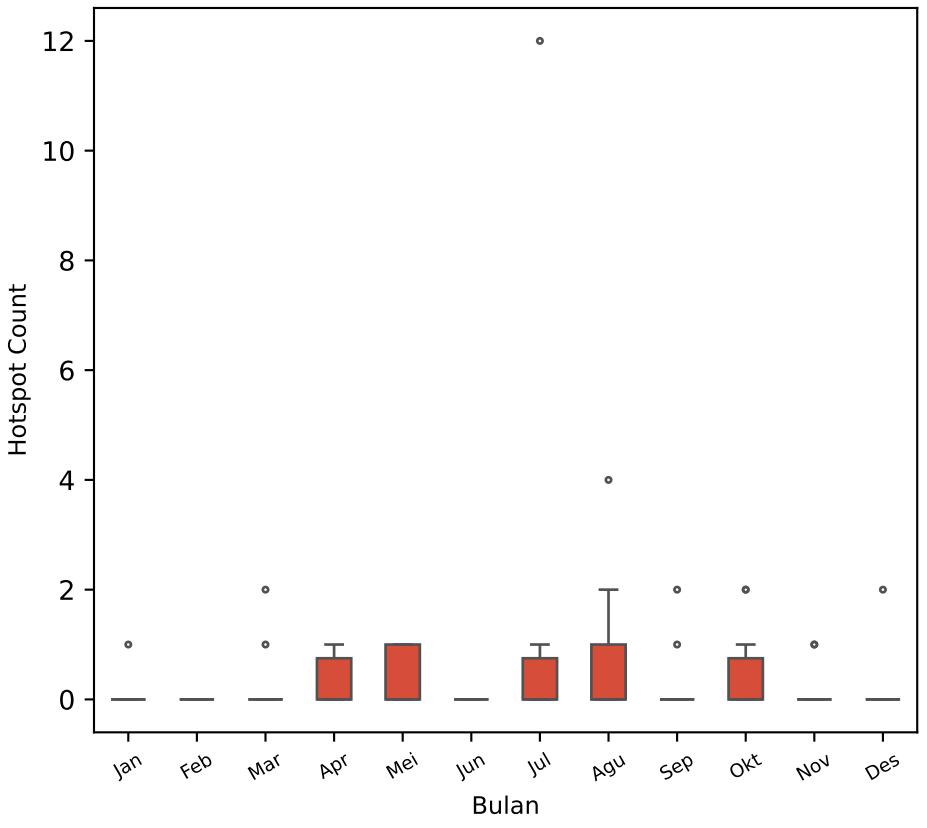
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
Statistik Gambut — Kota Pekanbaru
Total Luas          :        5,414 ha
Kelas 1    50–<100 cm:            0 ha  (  0.0%)
Kelas 2   100–<200 cm:          742 ha  ( 13.7%)
Kelas 3   200–<300 cm:        2,033 ha  ( 37.5%)
Kelas 4   300–<500 cm:        2,639 ha  ( 48.8%)
Kelas 5   500–<700 cm:            0 ha  (  0.0%)
Kelas 6       >700 cm:            0 ha  (  0.0%)
Kelas Dominan       : 300–<500 cm
Median Berbobot (σ) : 200–<300 cm
⚠  Kedalaman gambut = atribut statis
   Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 300–<500 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

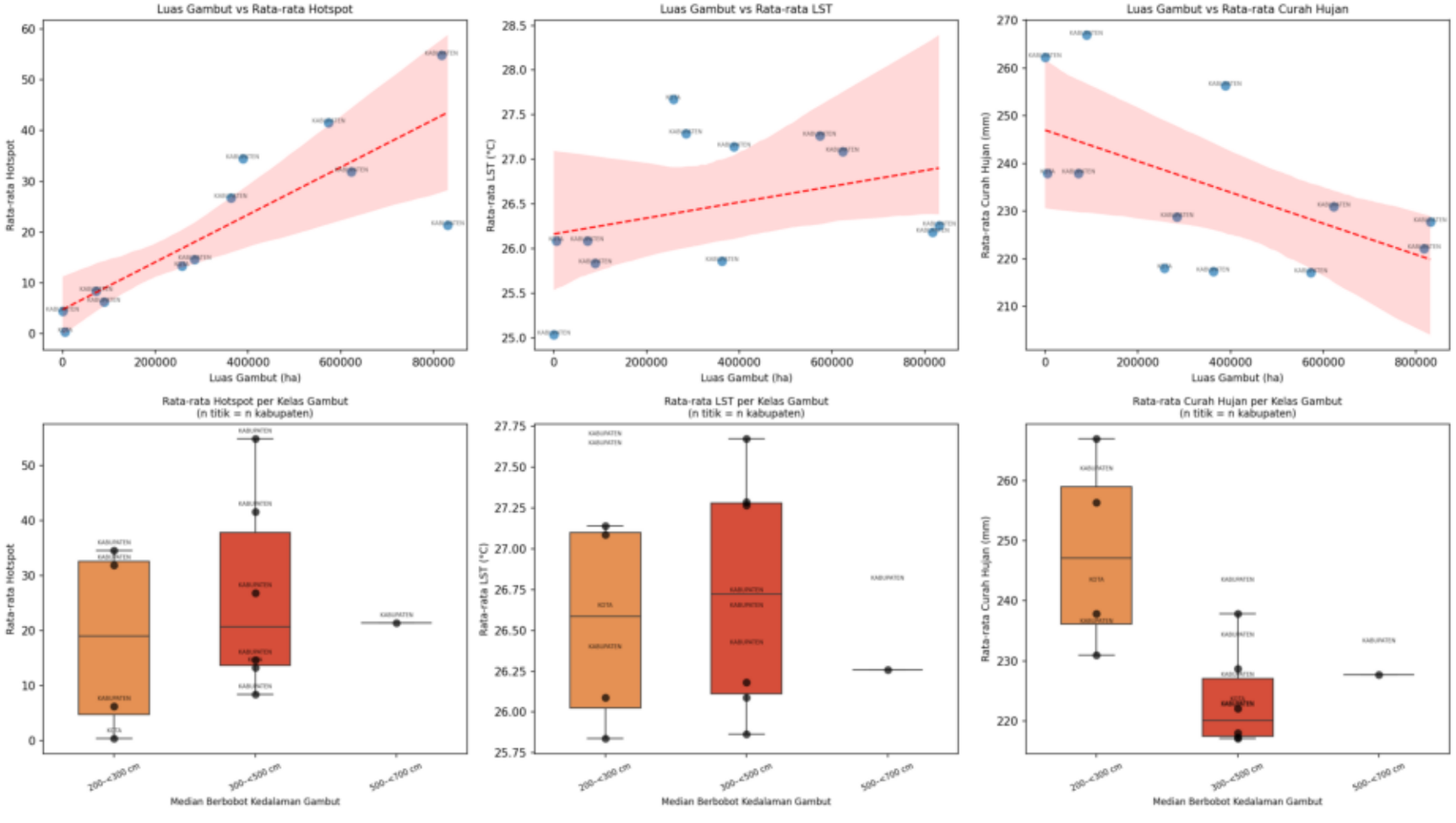
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



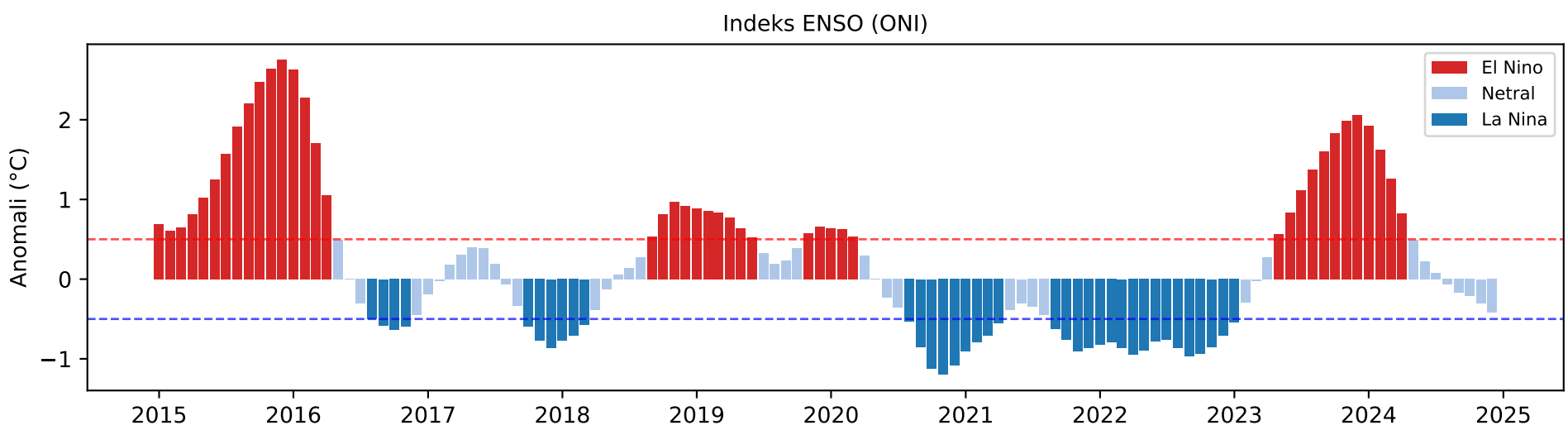
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN ROKAN HILIR

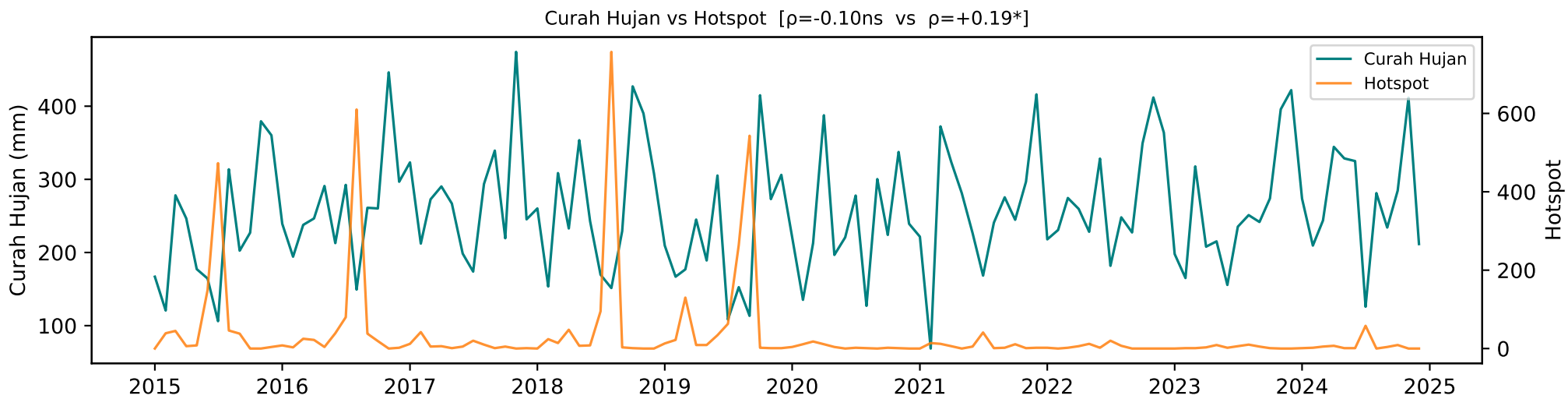
```
═══════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.443 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.101 ns    (p=0.270, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.101 ns    (p=0.270, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.194 *     (p=0.033, Spearman ρ, efek=lemah, n=120)

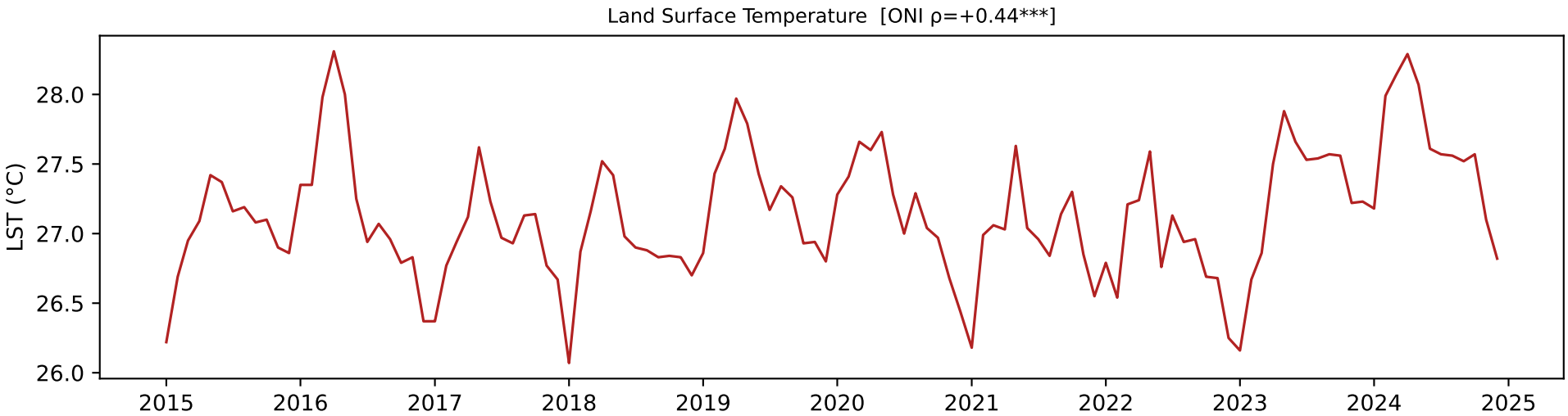
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────
  Hotspot     : H=5.51, p=0.064ns, η²=0.030
  LST         : H=21.57, p=0.000, η²=0.167
  Curah Hujan : H=1.79, p=0.409, η²=-0.002

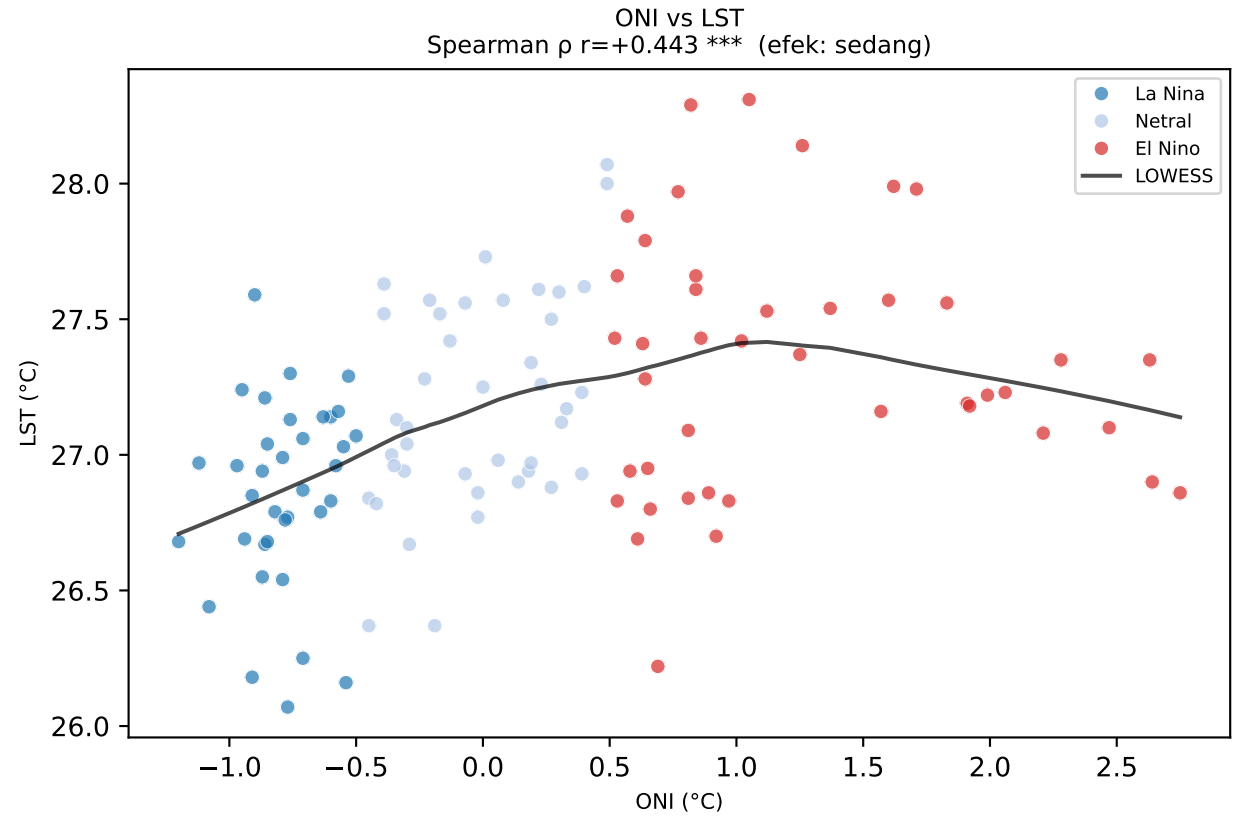
  INFO GAMBUT : total=388,624 ha | kelas dominan=100–<200 cm | median berbobot=200–<300 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

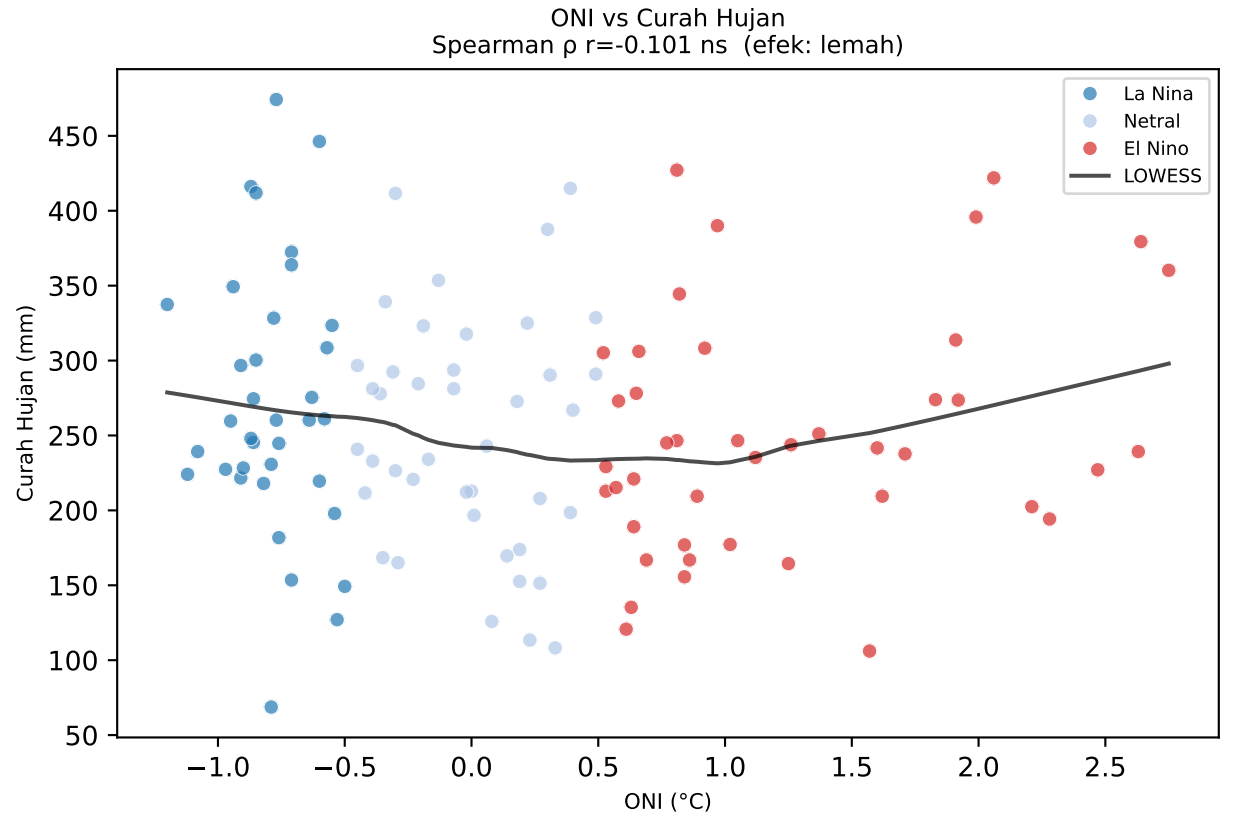


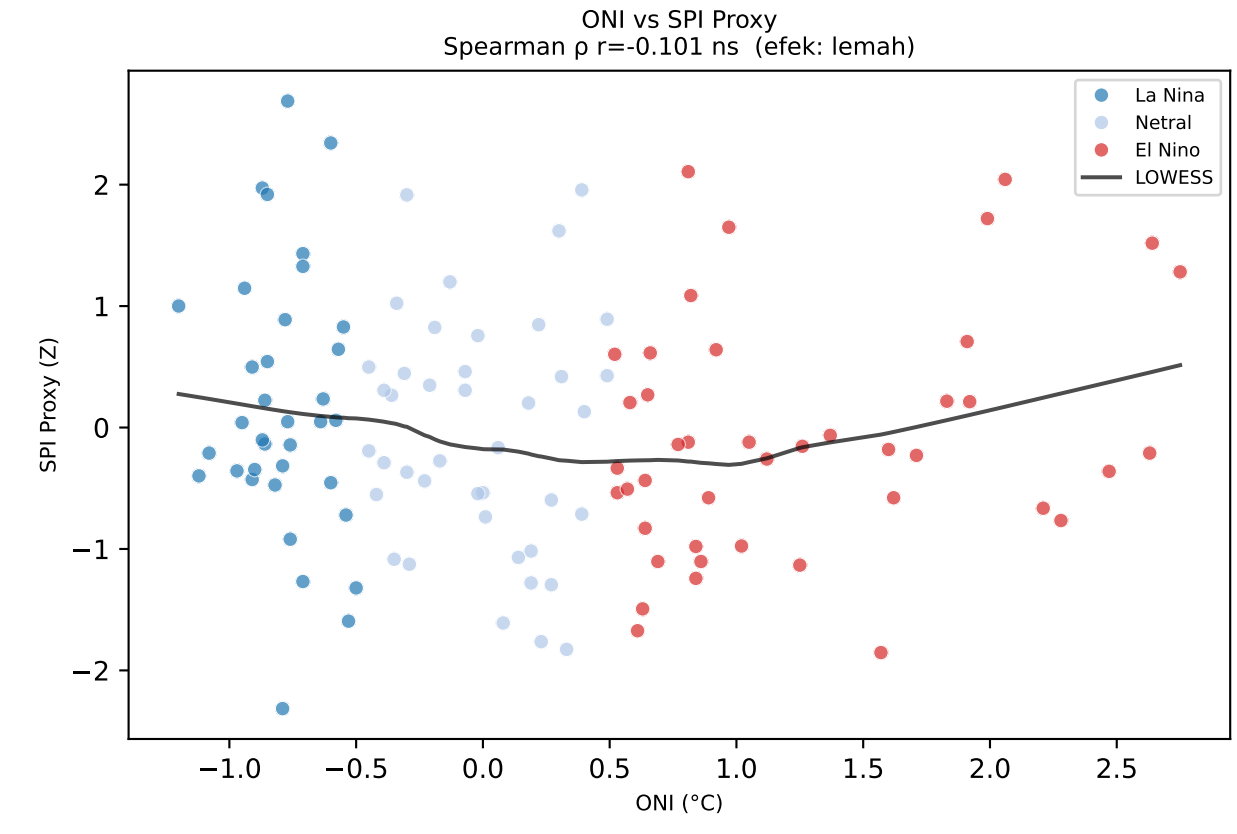


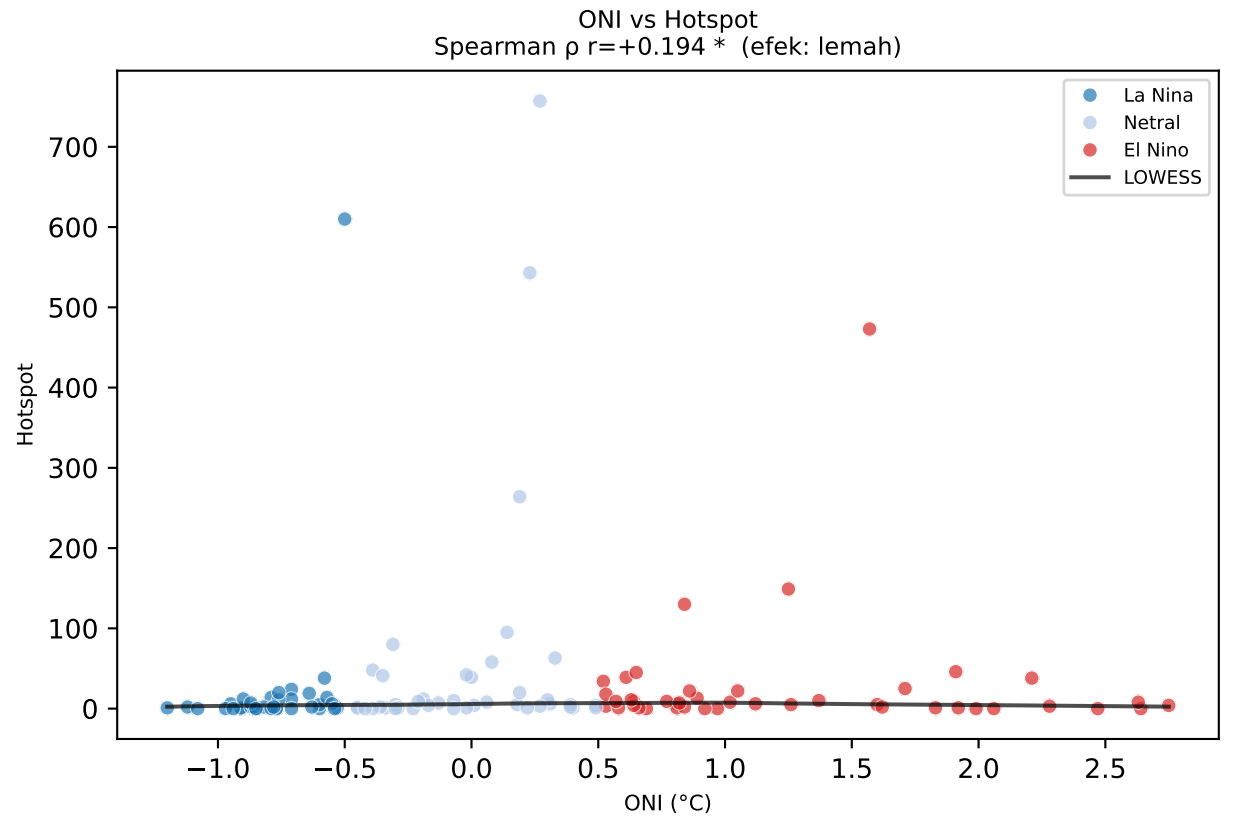


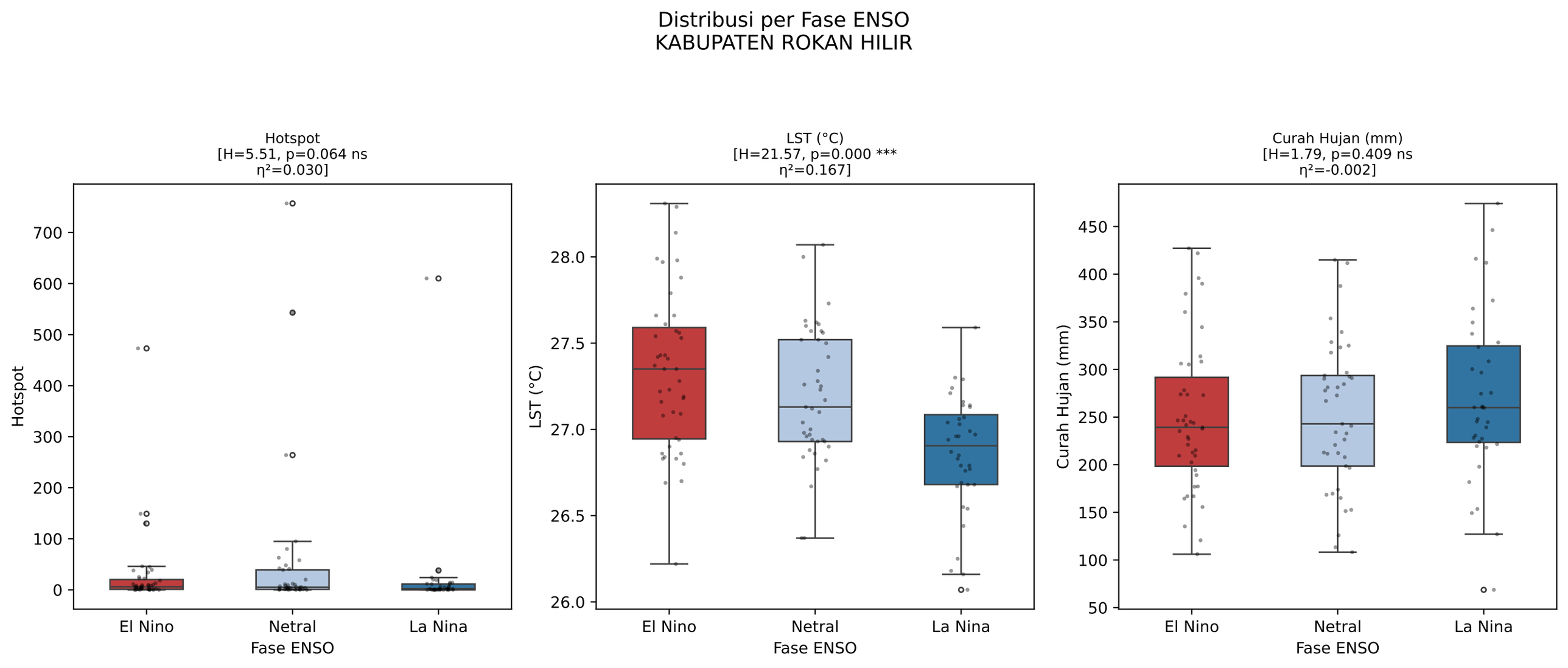
Scatter – ONI vs Variabel Iklim
KABUPATEN ROKAN HILIR





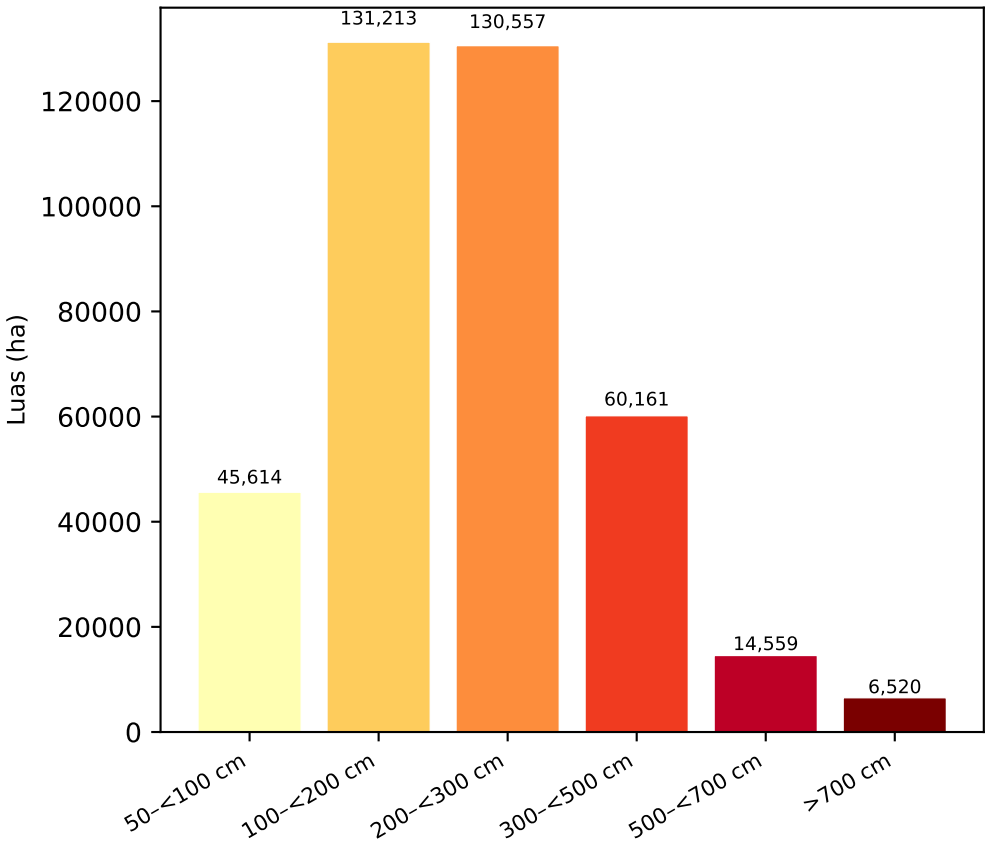




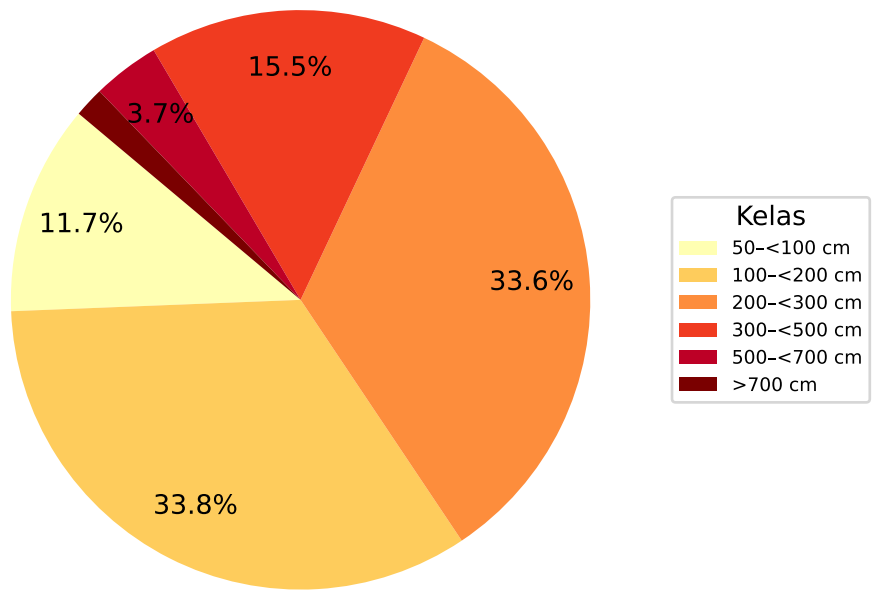

Distribusi per Fase ENSO
KABUPATEN ROKAN HILIR
Hotspot
[H=5.51, p=0.064 ns
η²=0.030]
Hotspot
0
100
200
300
400
500
600
700
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=21.57, p=0.000 ***
η²=0.167]
LST (°C)
26.0
26.5
27.0
27.5
28.0
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=1.79, p=0.409 ns
η²=-0.002]
Curah Hujan (mm)
50
100
150
200
250
300
350
400
450
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN ROKAN HILIR

A. Luas Area per Kelas Kedalaman



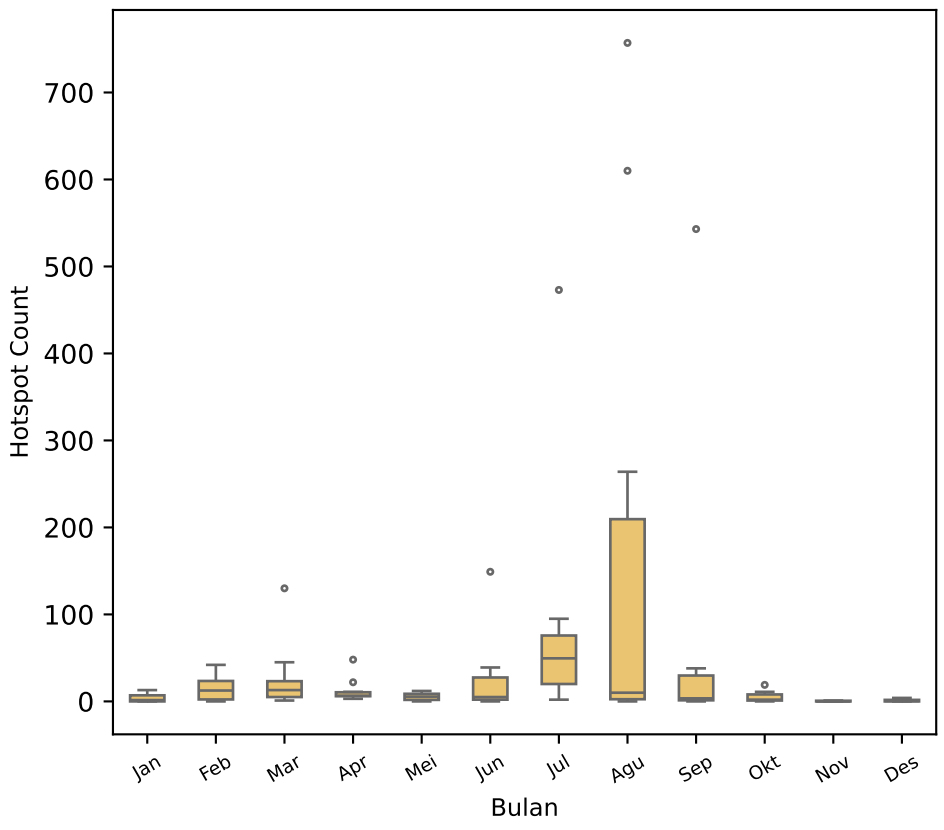
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════
  Statistik Gambut — Kabupaten Rokan Hili
═══════════════════════════════════════════════
  Total Luas          :      388,624 ha
═══════════════════════════════════════════════
  Kelas 1    50–<100 cm:     45,614 ha  ( 11.7%)
  Kelas 2   100–<200 cm:    131,213 ha  ( 33.8%)
  Kelas 3   200–<300 cm:    130,557 ha  ( 33.6%)
  Kelas 4   300–<500 cm:     60,161 ha  ( 15.5%)
  Kelas 5   500–<700 cm:     14,559 ha  (  3.7%)
  Kelas 6       >700 cm:      6,520 ha  (  1.7%)
═══════════════════════════════════════════════
  Kelas Dominan       : 100–<200 cm
  Median Berbobot (σ) : 200–<300 cm
═══════════════════════════════════════════════
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 100–<200 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

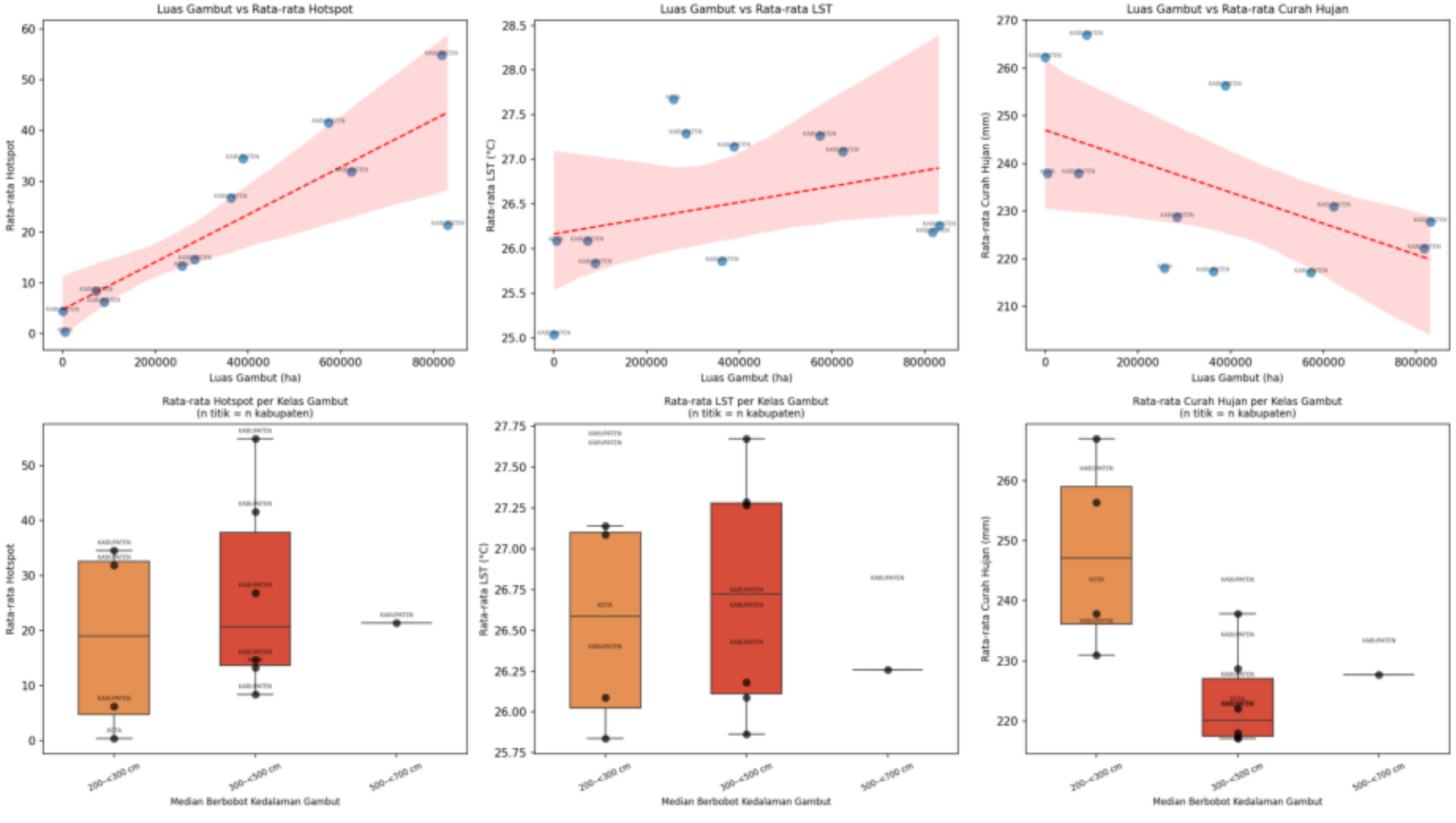
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



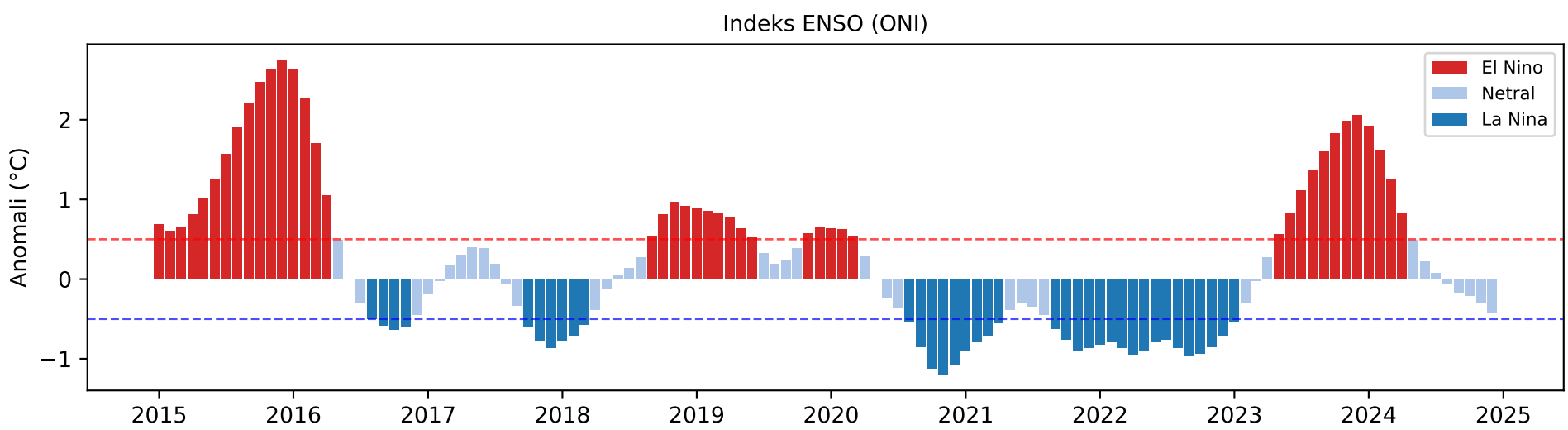
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN PELALAWAN

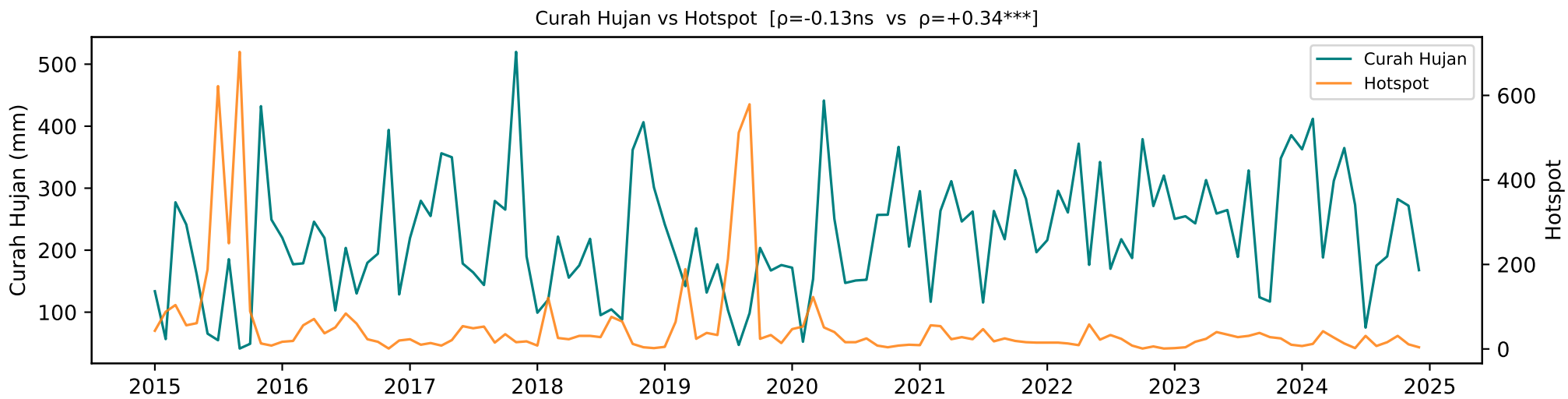
```
═══════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.509 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                            : r=-0.133 ns    (p=0.147, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.133 ns    (p=0.147, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.343 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

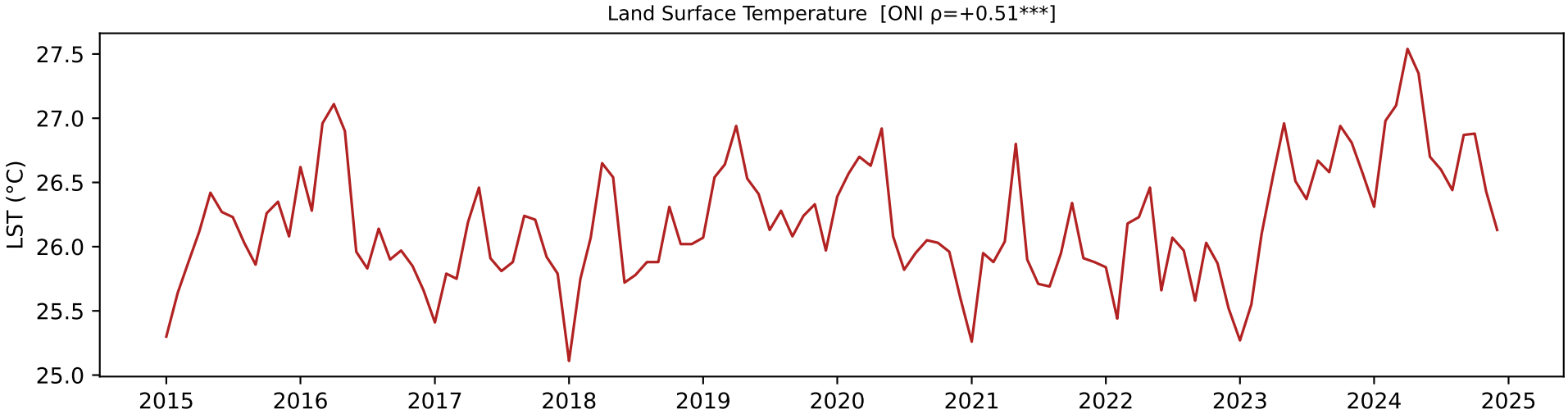
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────
  Hotspot     : H=13.64, p=0.001**, η²=0.099
  LST         : H=28.08, p=0.000, η²=0.223
  Curah Hujan : H=4.94, p=0.085, η²=0.025

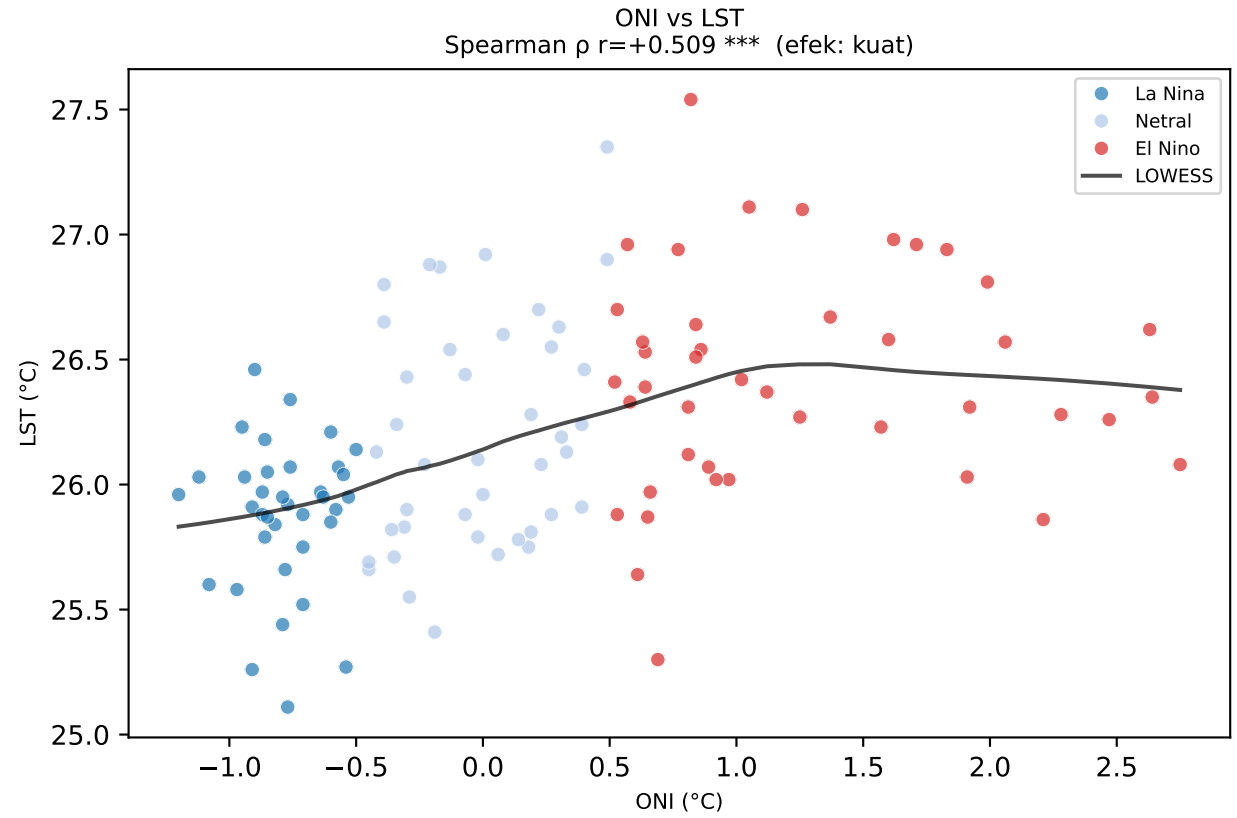
  INFO GAMBUT : total=817,315 ha | kelas dominan=500–<700 cm | median berbobot=300–<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

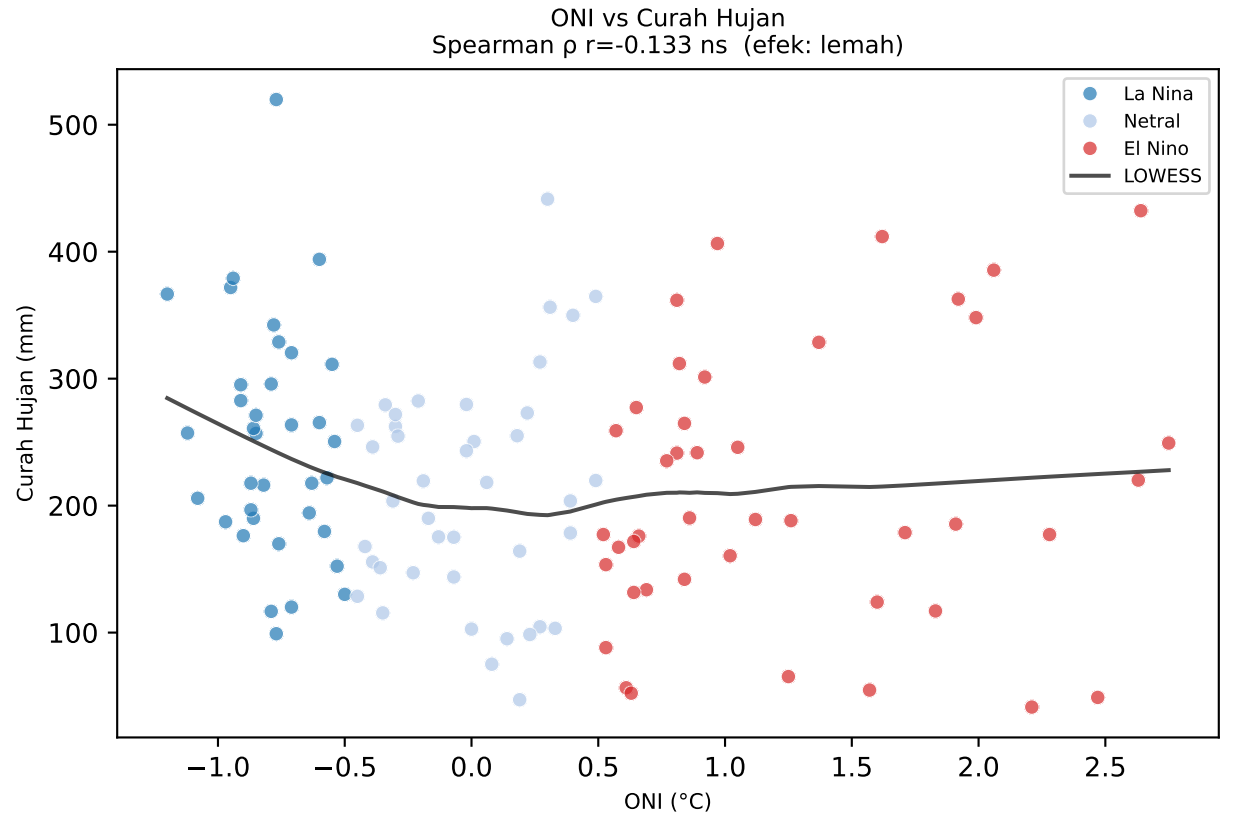


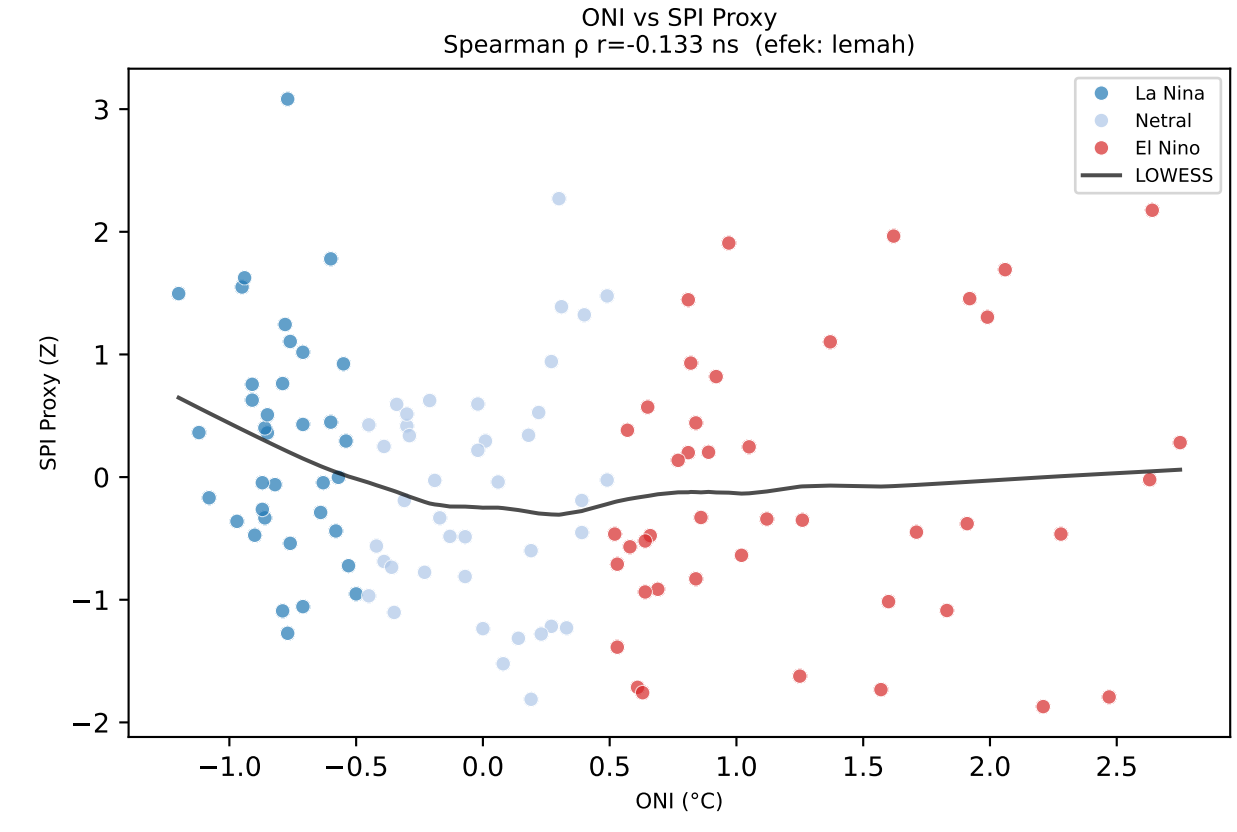


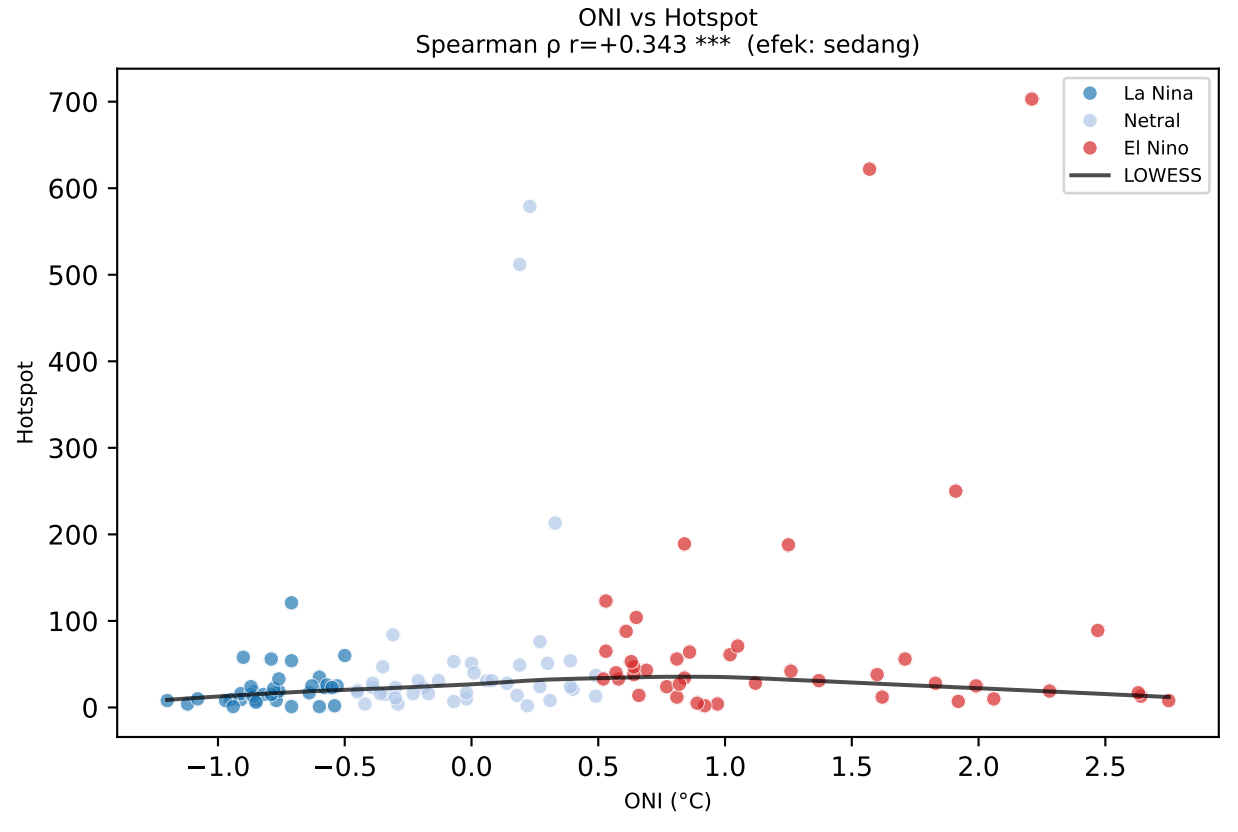


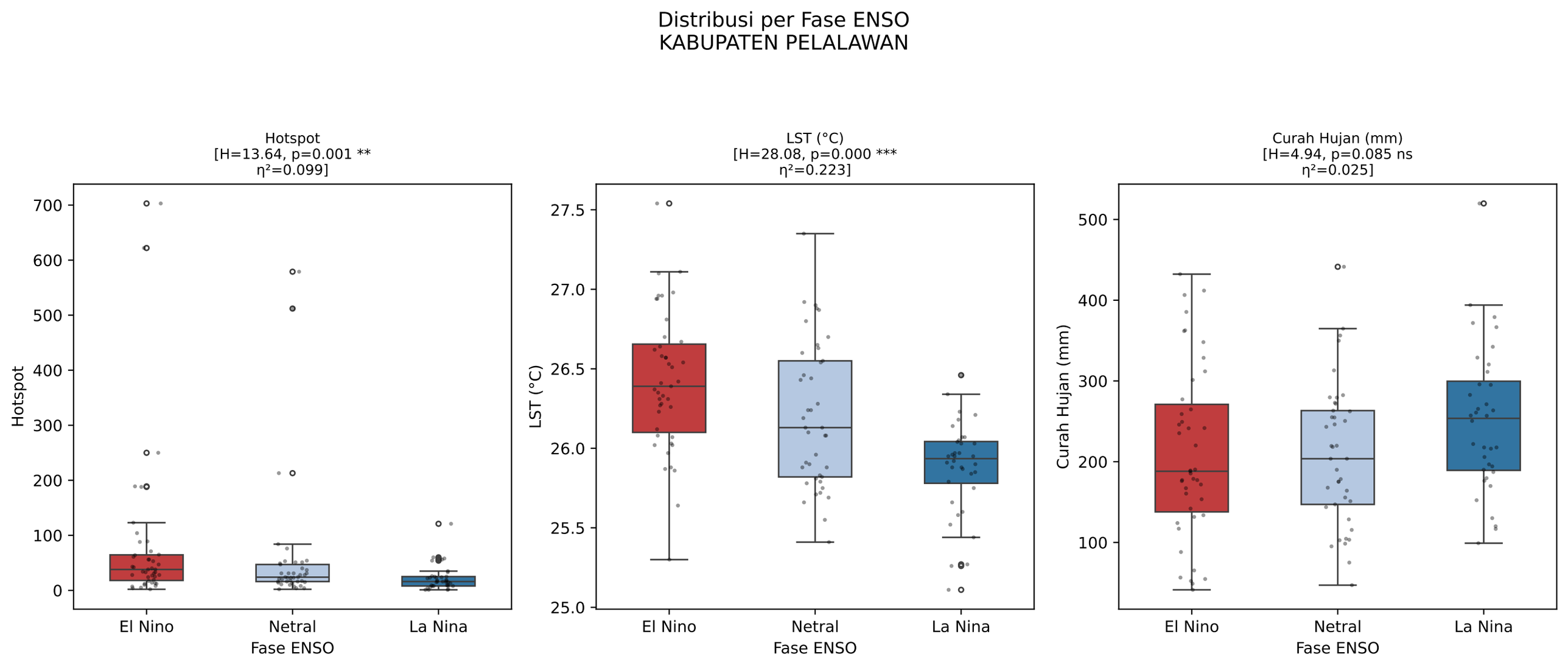
Scatter – ONI vs Variabel Iklim
KABUPATEN PELALAWAN





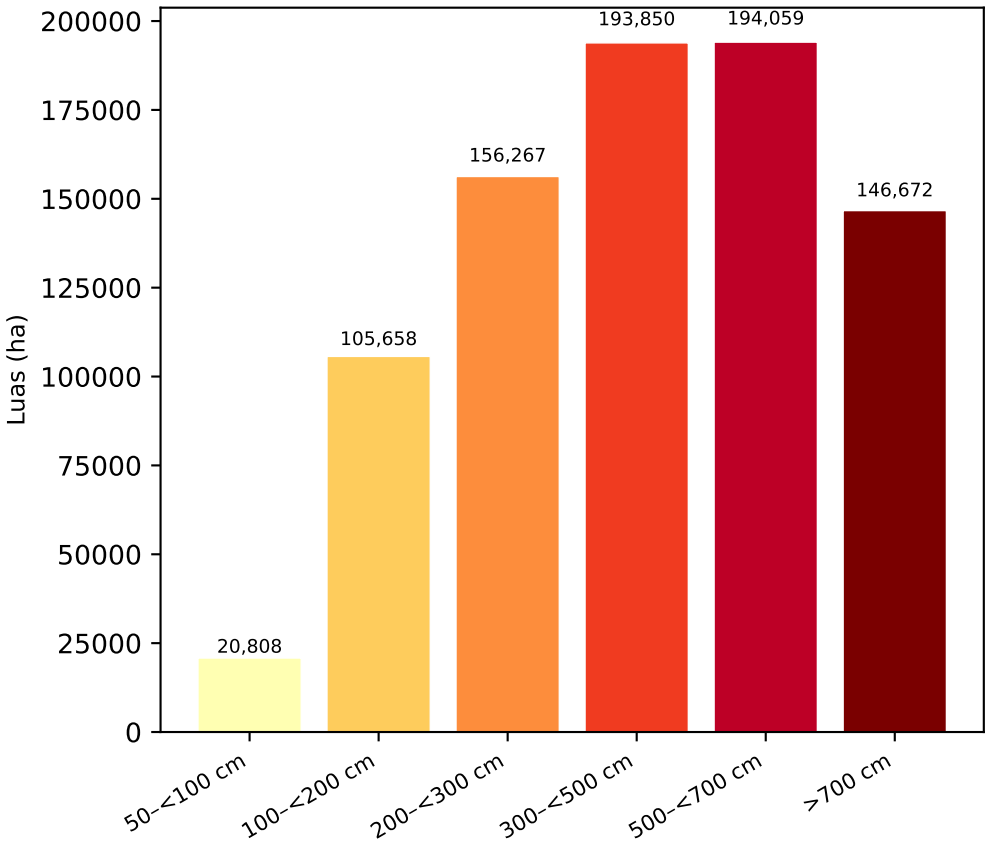




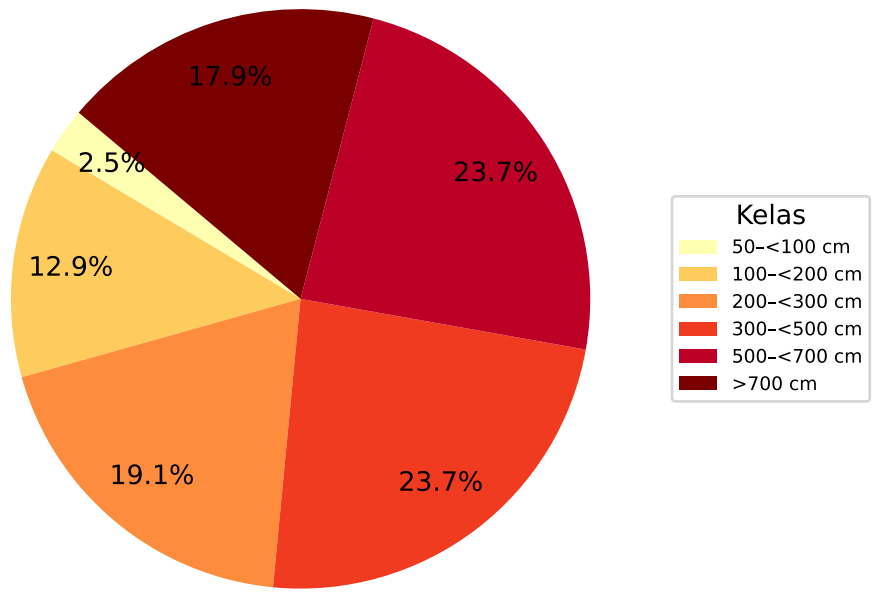

Distribusi per Fase ENSO
KABUPATEN PELALAWAN
Hotspot
[H=13.64, p=0.001 **
η²=0.099]
Hotspot
0
100
200
300
400
500
600
700
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=28.08, p=0.000 ***
η²=0.223]
LST (°C)
25.0
25.5
26.0
26.5
27.0
27.5
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=4.94, p=0.085 ns
η²=0.025]
Curah Hujan (mm)
100
200
300
400
500
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN PELALAWAN

A. Luas Area per Kelas Kedalaman



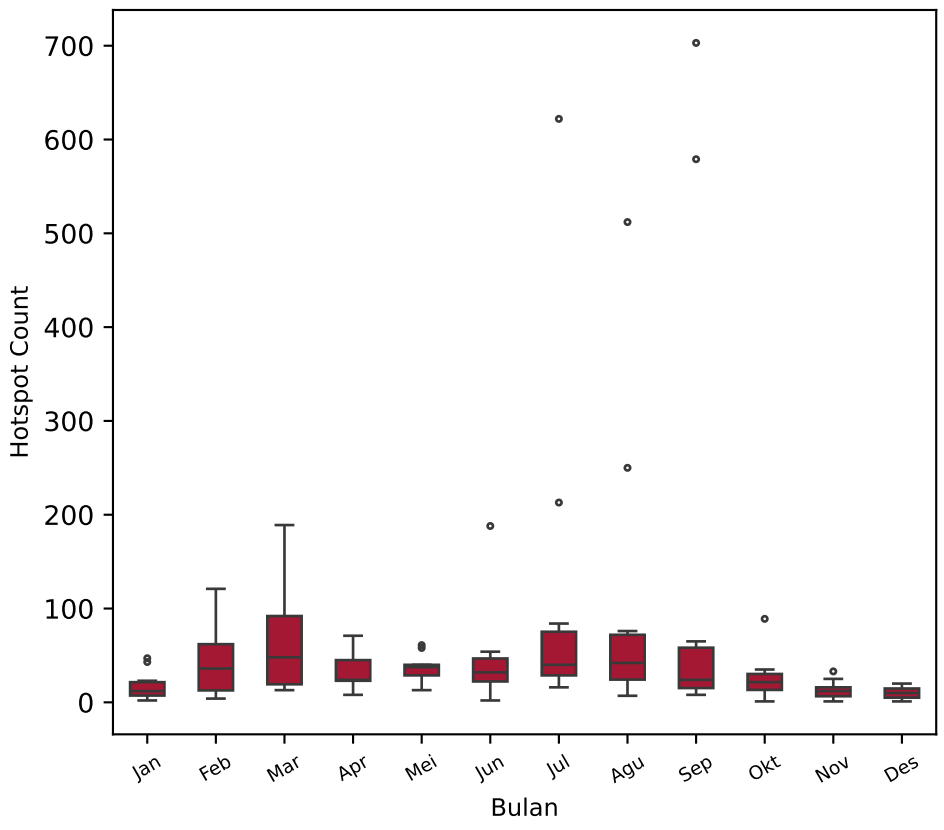
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════
  Statistik Gambut — Kabupaten Pelalawan
───────────────────────────────────────────────
  Total Luas          :      817,315 ha
───────────────────────────────────────────────
  Kelas 1    50–<100 cm:     20,808 ha  (  2.5%)
  Kelas 2   100–<200 cm:    105,658 ha  ( 12.9%)
  Kelas 3   200–<300 cm:    156,267 ha  ( 19.1%)
  Kelas 4   300–<500 cm:    193,850 ha  ( 23.7%)
  Kelas 5   500–<700 cm:    194,059 ha  ( 23.7%)
  Kelas 6       >700 cm:    146,672 ha  ( 17.9%)
───────────────────────────────────────────────
  Kelas Dominan       : 500–<700 cm
  Median Berbobot (σ) : 300–<500 cm
───────────────────────────────────────────────
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 500–<700 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

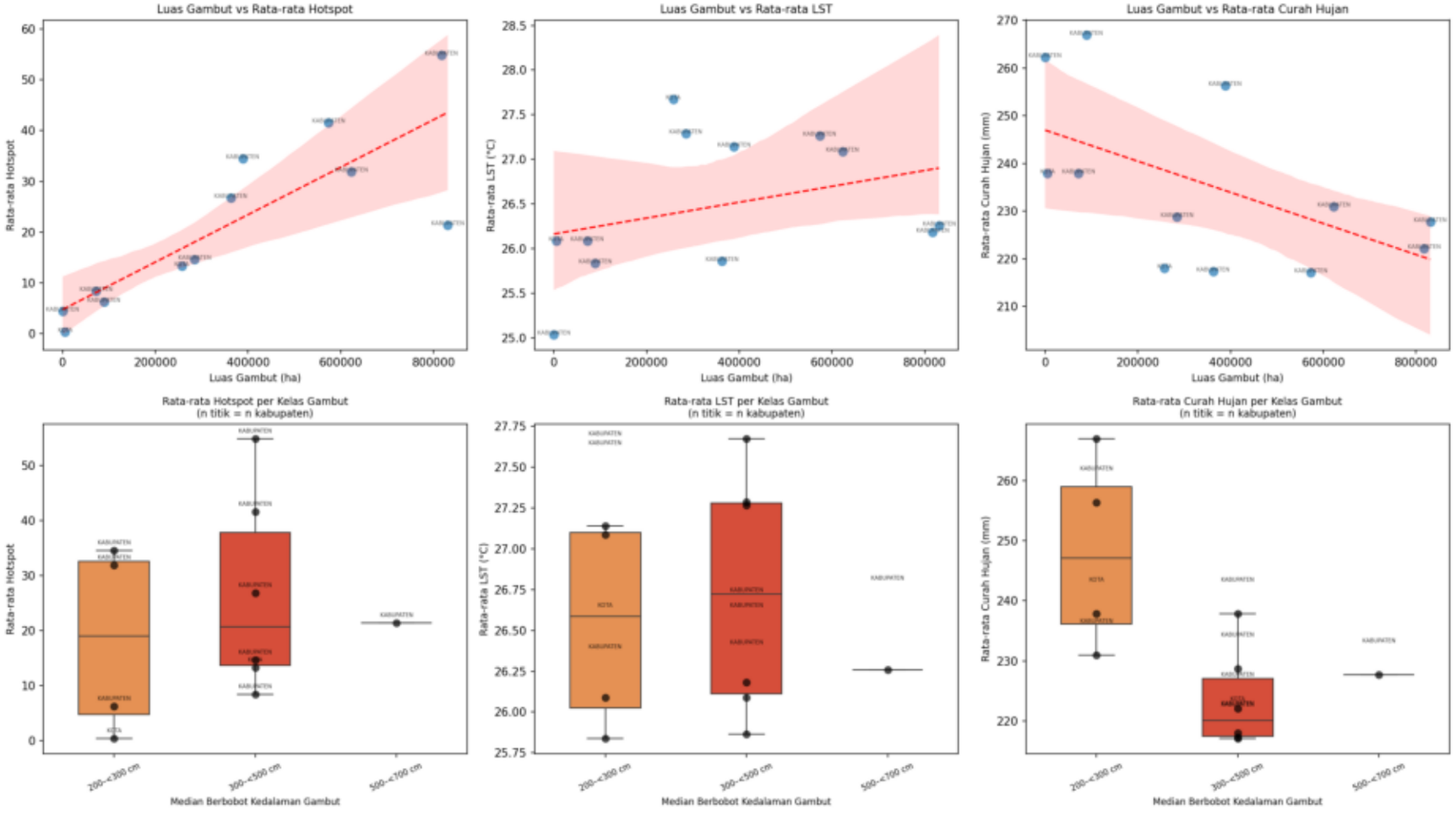
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



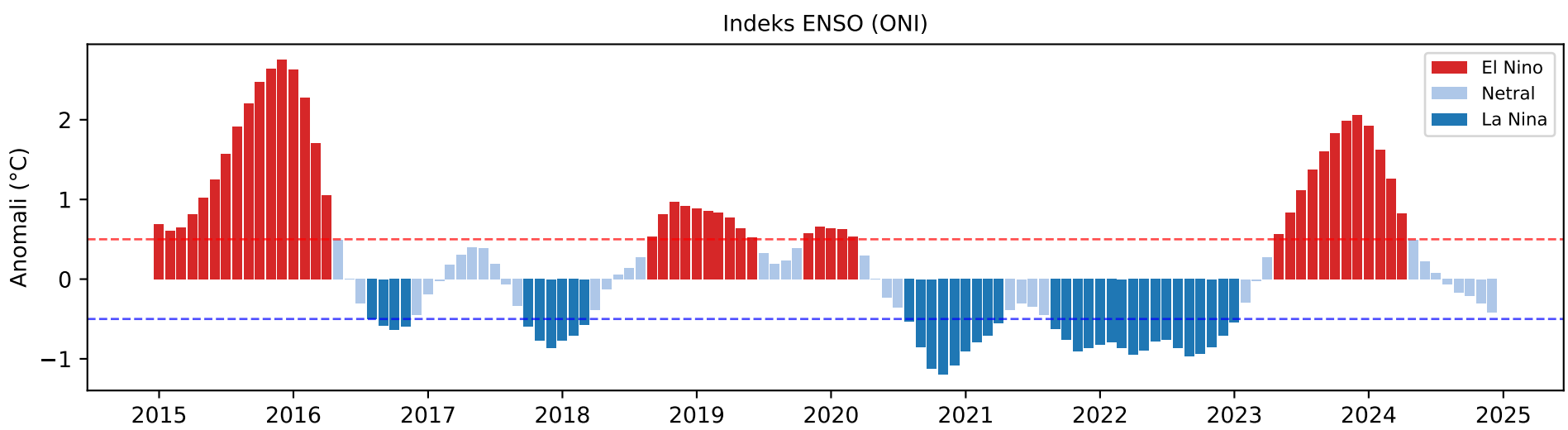
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN KUANTAN SINGINGI

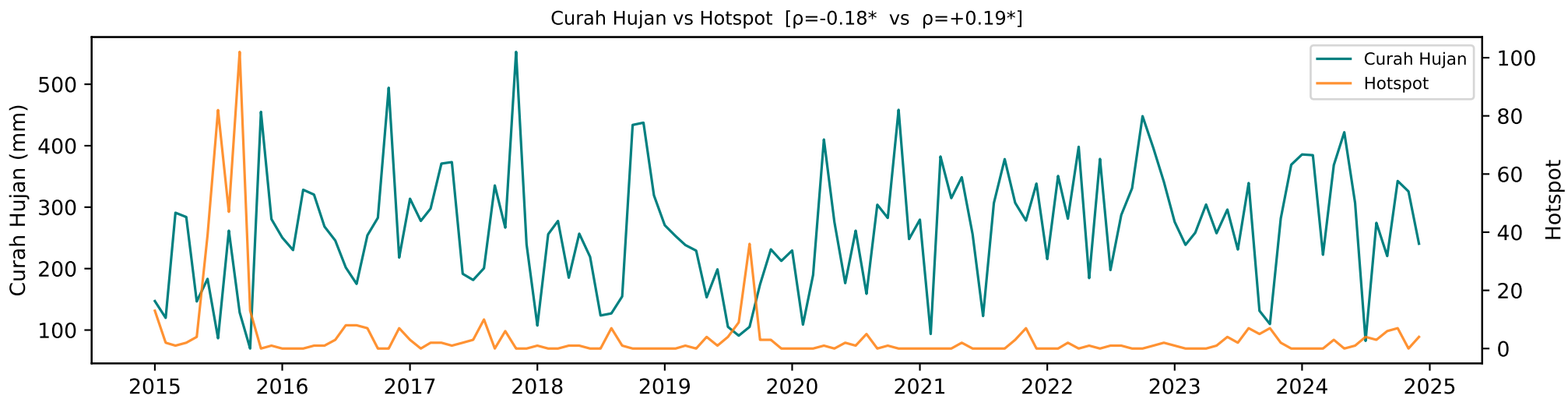
```
═══════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.527 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                            : r=-0.182 *     (p=0.046, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.182 *     (p=0.046, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.193 *     (p=0.035, Spearman ρ, efek=lemah, n=120)

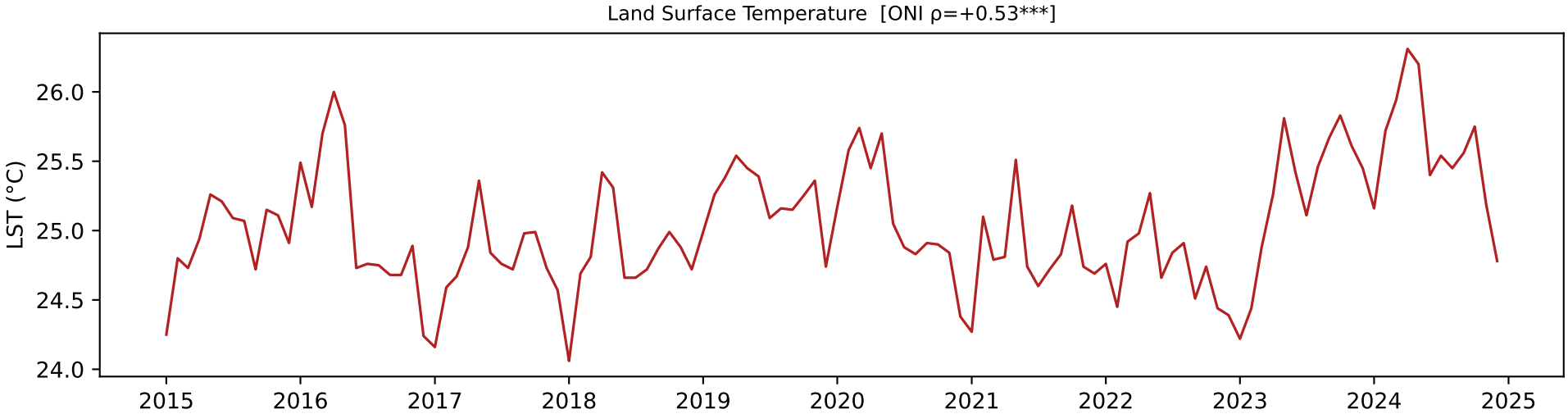
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────
  Hotspot     : H=6.49, p=0.039*, η²=0.038
  LST         : H=31.48, p=0.000, η²=0.252
  Curah Hujan : H=7.39, p=0.025, η²=0.046

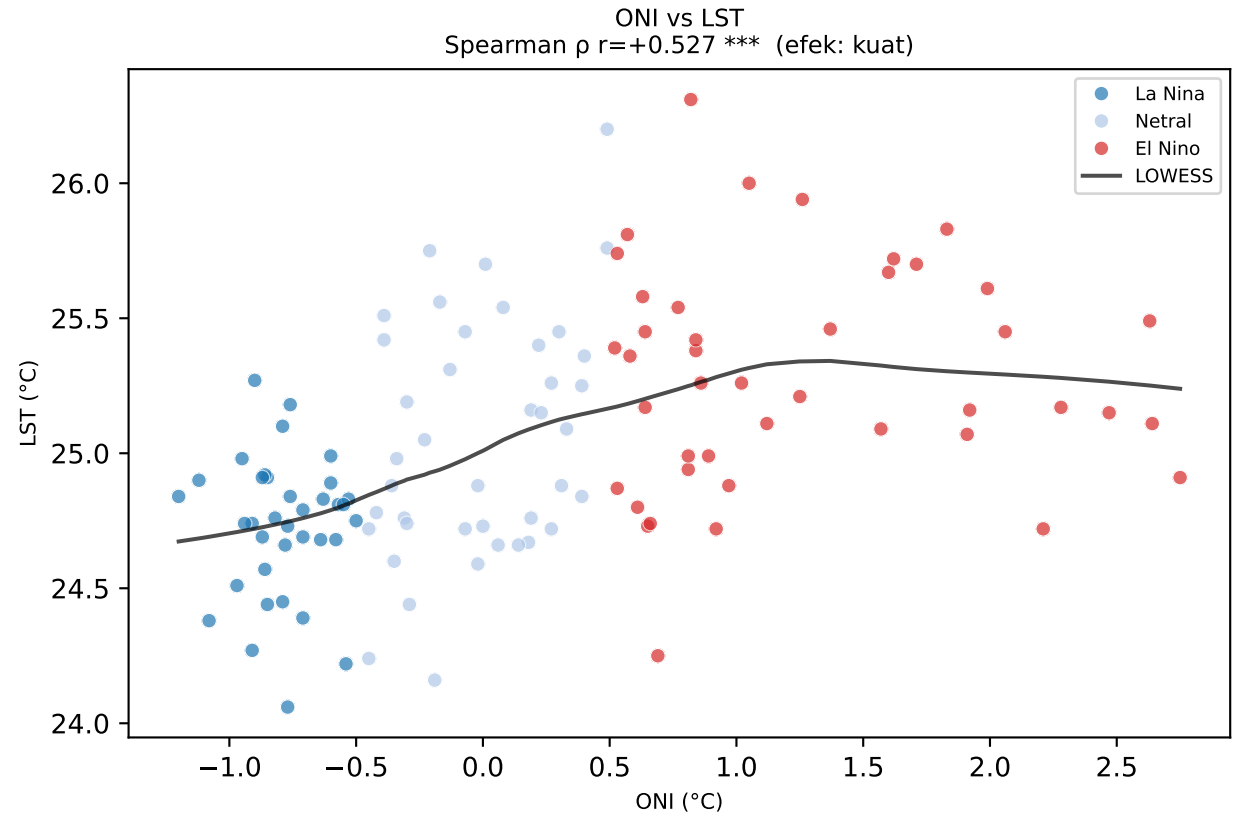
  INFO GAMBUT : total=0 ha | kelas dominan=Tidak Ada | median berbobot=Tidak Ada
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

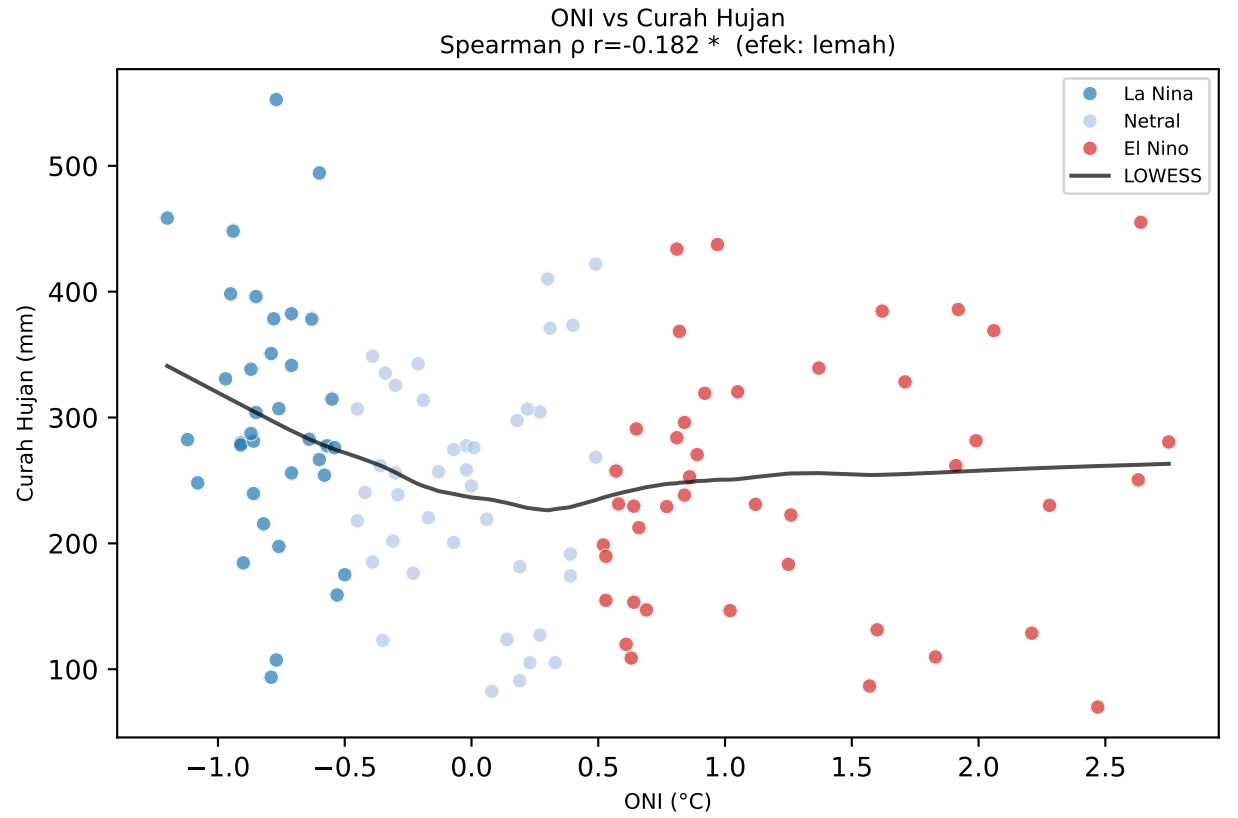


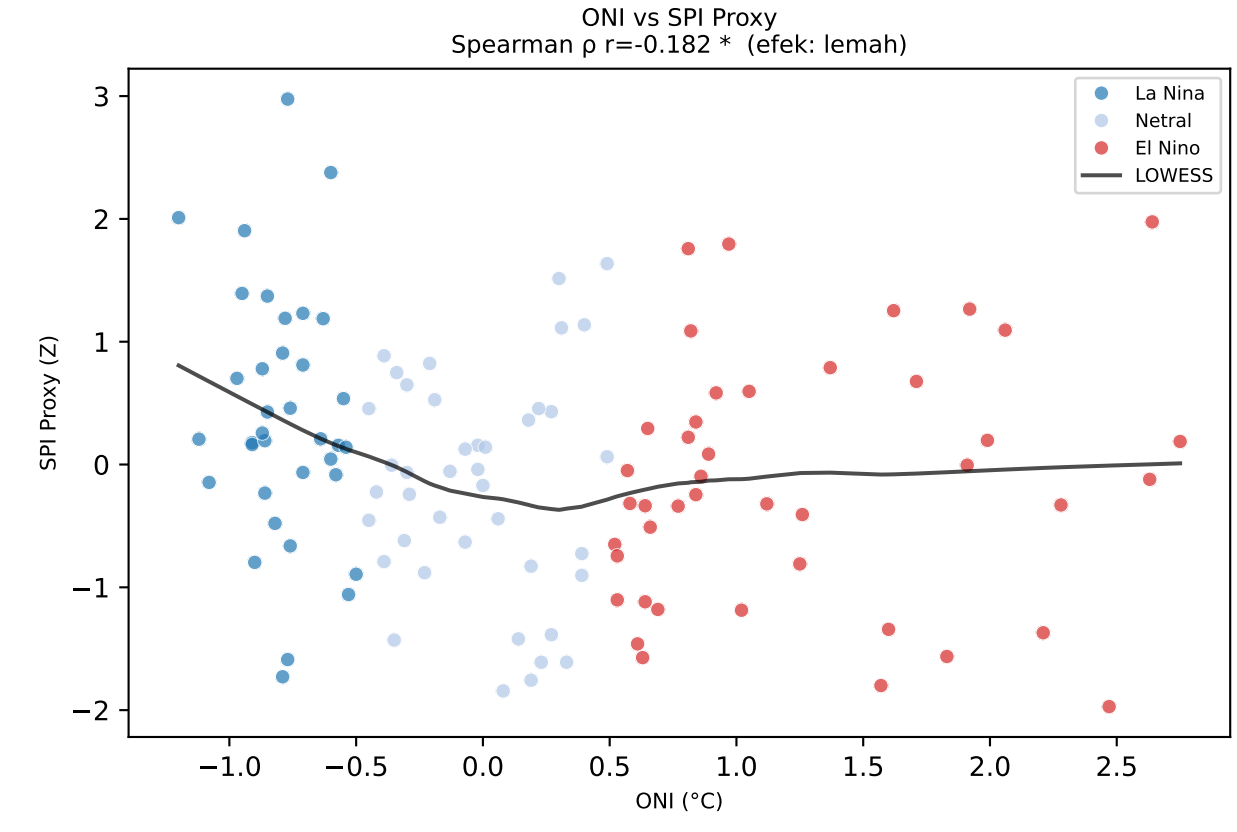


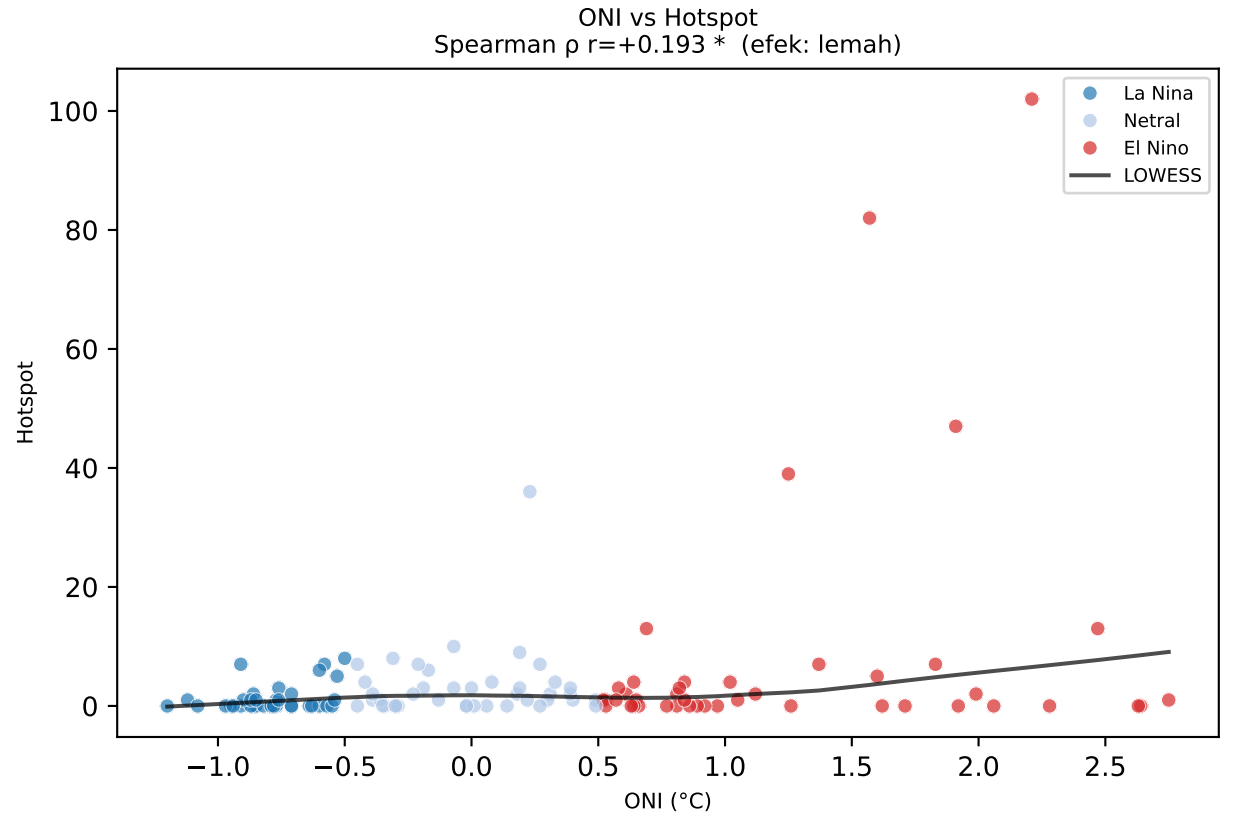


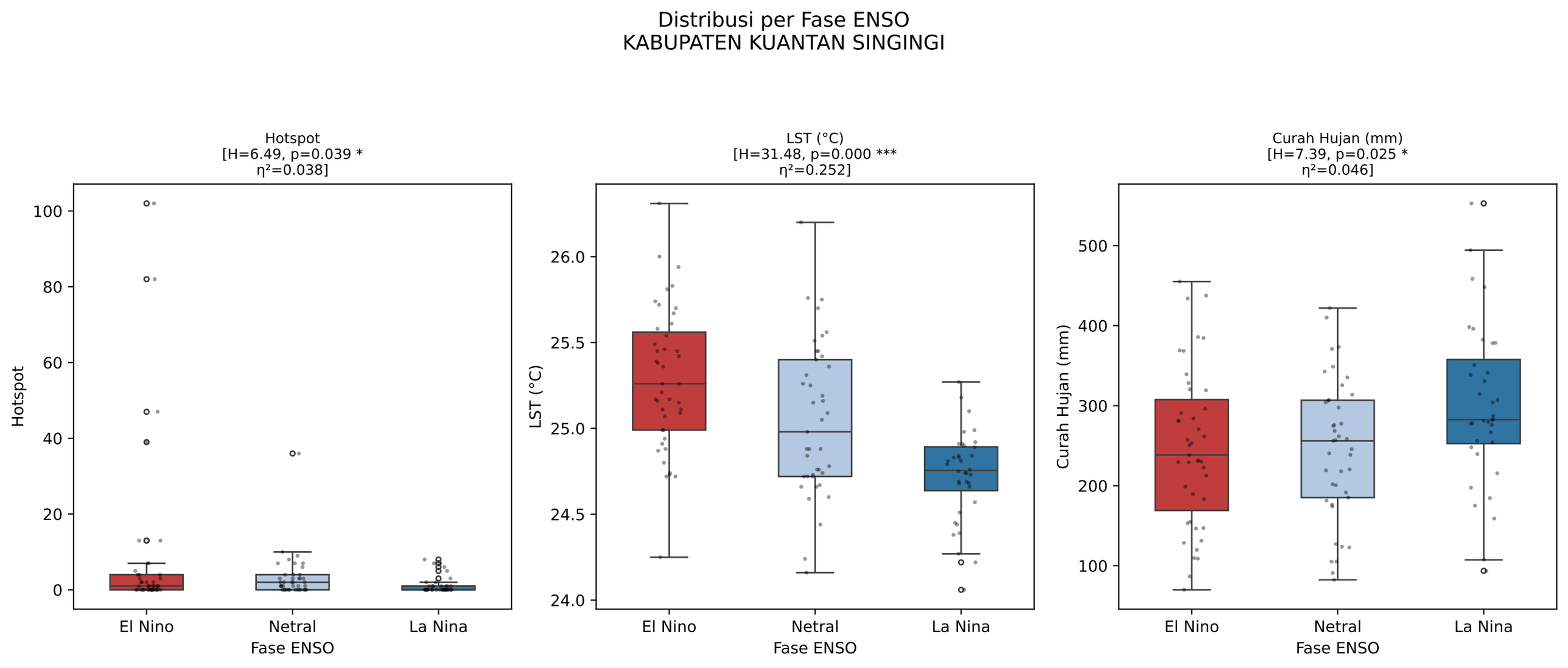
Scatter – ONI vs Variabel Iklim
KABUPATEN KUANTAN SINGINGI





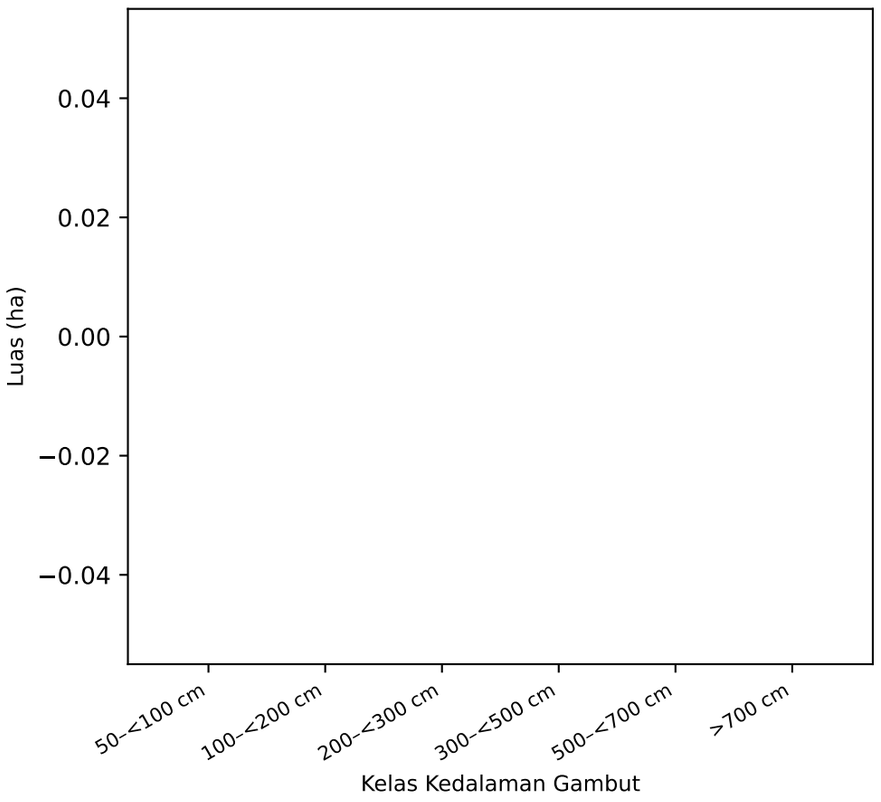




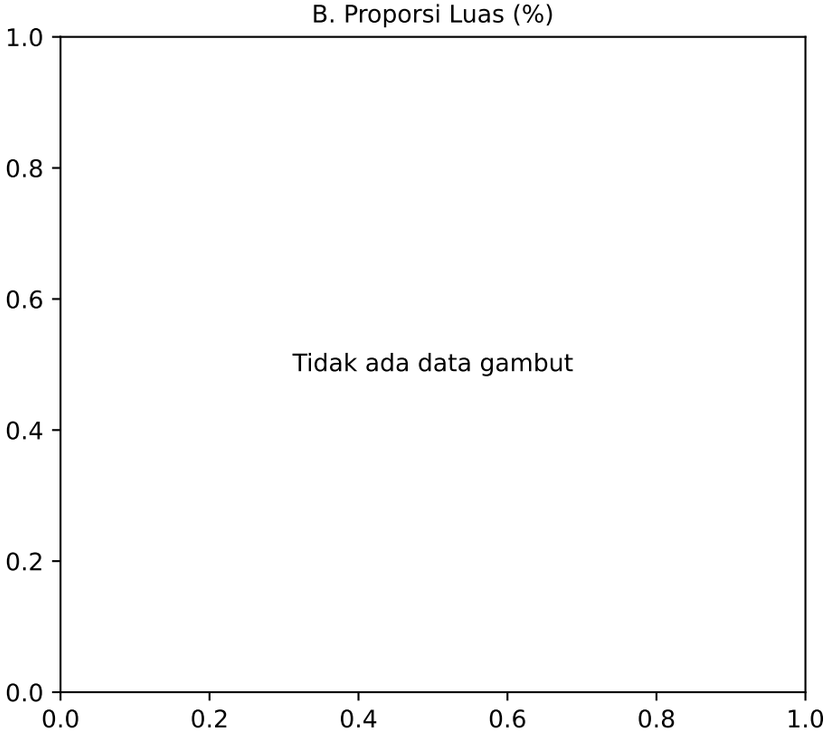

Distribusi per Fase ENSO
KABUPATEN KUANTAN SINGINGI
Hotspot
[H=6.49, p=0.039 *
η²=0.038]
Hotspot
0
20
40
60
80
100
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=31.48, p=0.000 ***
η²=0.252]
LST (°C)
24.0
24.5
25.0
25.5
26.0
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=7.39, p=0.025 *
η²=0.046]
Curah Hujan (mm)
100
200
300
400
500
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN KUANTAN SINGINGI

A. Luas Area per Kelas Kedalaman



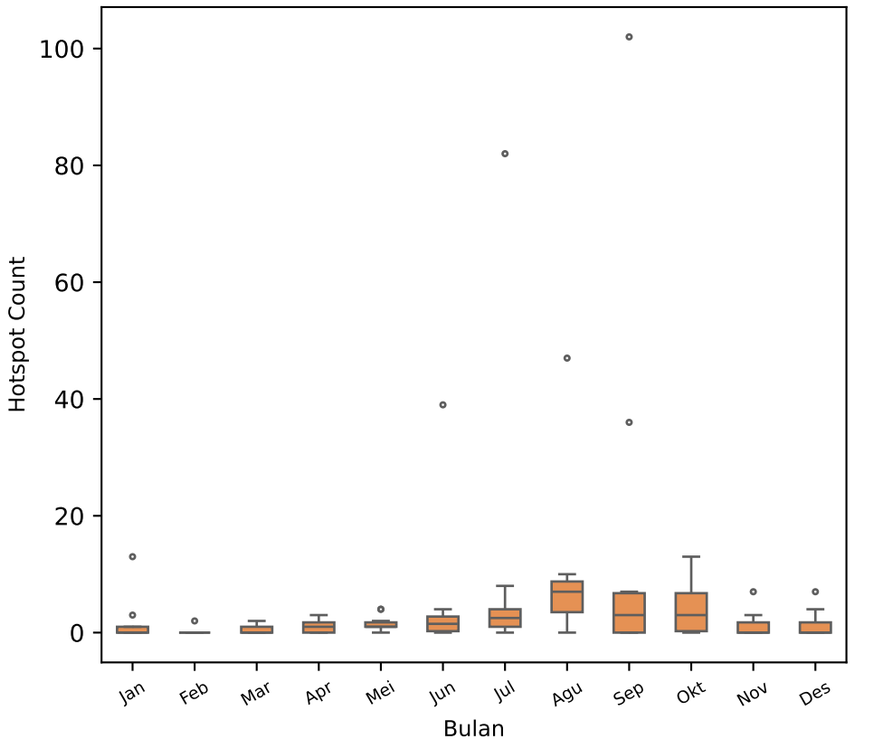
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════
  Statistik Gambut — Kabupaten Kuantan Si
═══════════════════════════════════════════
  Total Luas          :            0 ha
═══════════════════════════════════════════
  Kelas 1    50–<100 cm:          0 ha  (  0.0%)
  Kelas 2   100–<200 cm:          0 ha  (  0.0%)
  Kelas 3   200–<300 cm:          0 ha  (  0.0%)
  Kelas 4   300–<500 cm:          0 ha  (  0.0%)
  Kelas 5   500–<700 cm:          0 ha  (  0.0%)
  Kelas 6       >700 cm:          0 ha  (  0.0%)
═══════════════════════════════════════════
  Kelas Dominan       : Tidak Ada
  Median Berbobot (σ) : Tidak Ada
═══════════════════════════════════════════
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: Tidak Ada)



Korelasi Peatland – Lintas Kabupaten (Global)

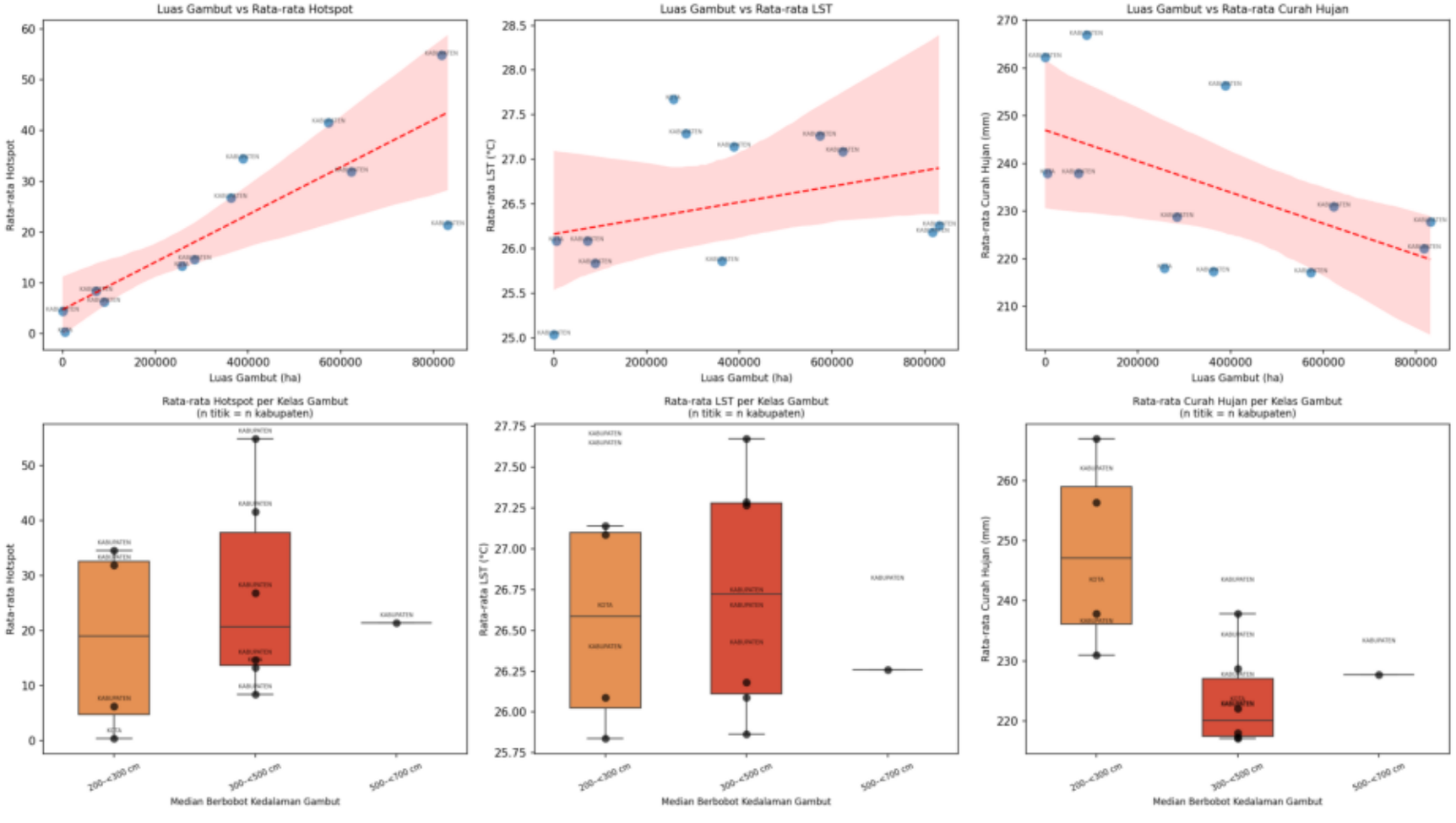
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



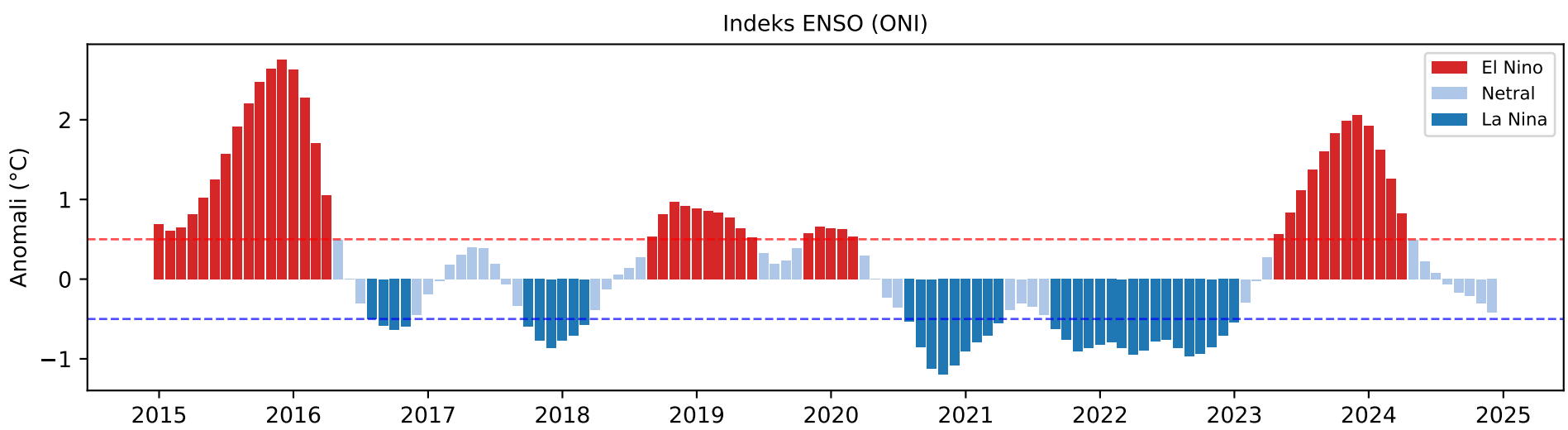
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN KEPULAUAN MERANTI

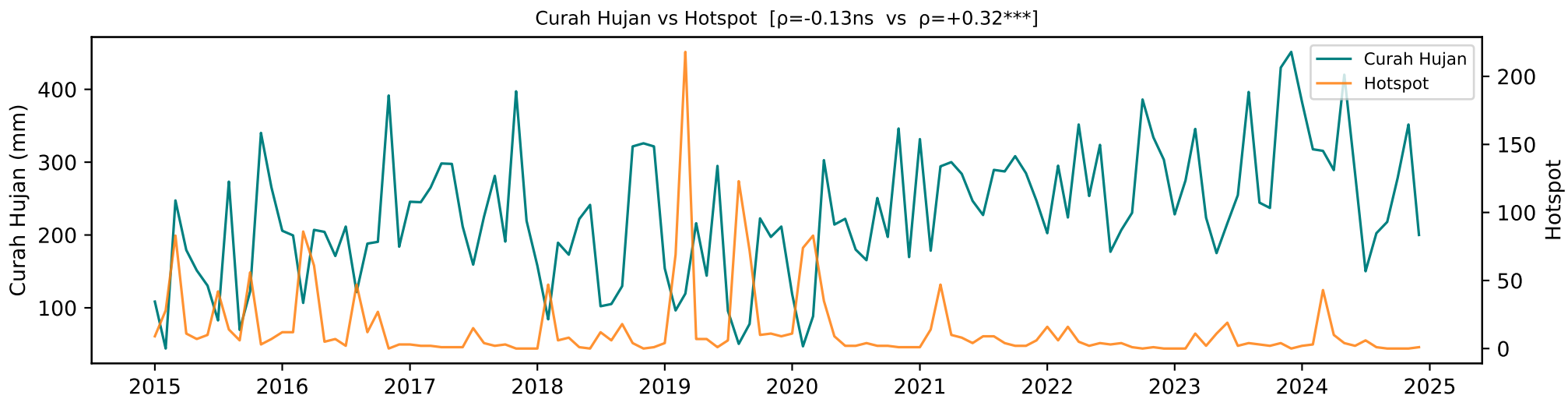
```
═══════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.465 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.134 ns    (p=0.145, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.134 ns    (p=0.145, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.323 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

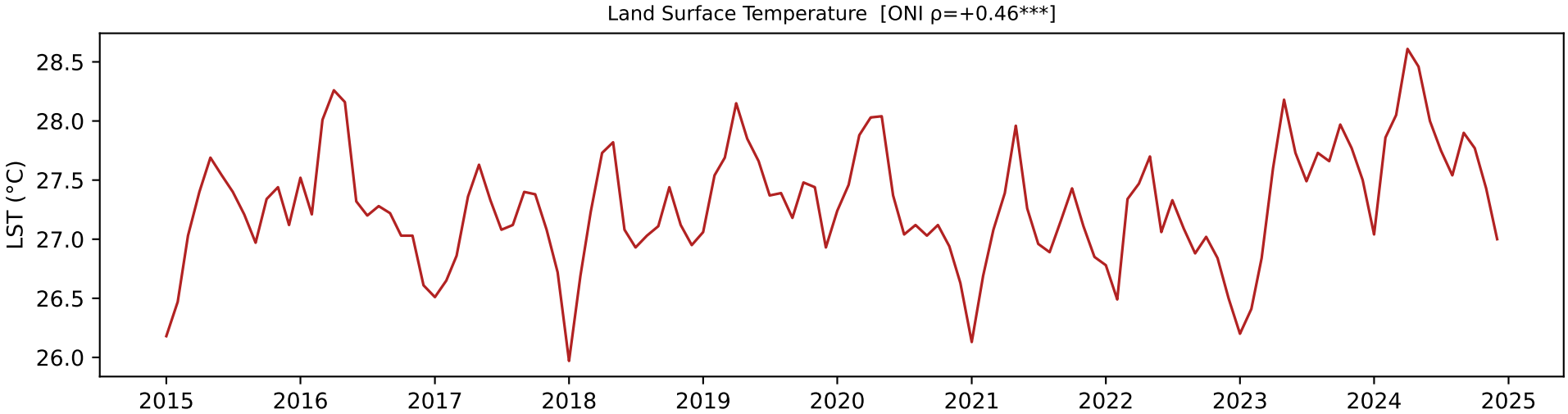
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────
  Hotspot     : H=14.67, p=0.001***, η²=0.108
  LST         : H=24.04, p=0.000, η²=0.188
  Curah Hujan : H=3.35, p=0.187, η²=0.012

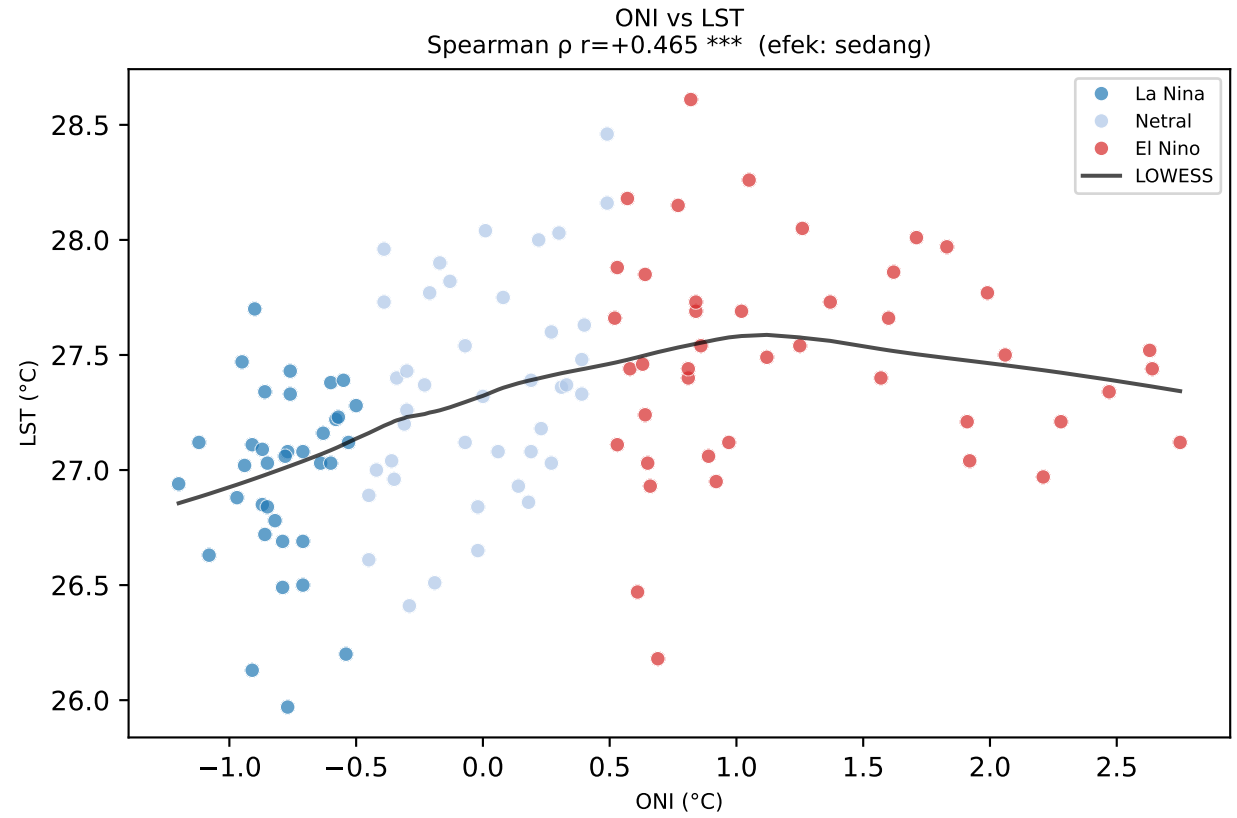
  INFO GAMBUT : total=284,441 ha | kelas dominan=300–<500 cm | median berbobot=300–<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

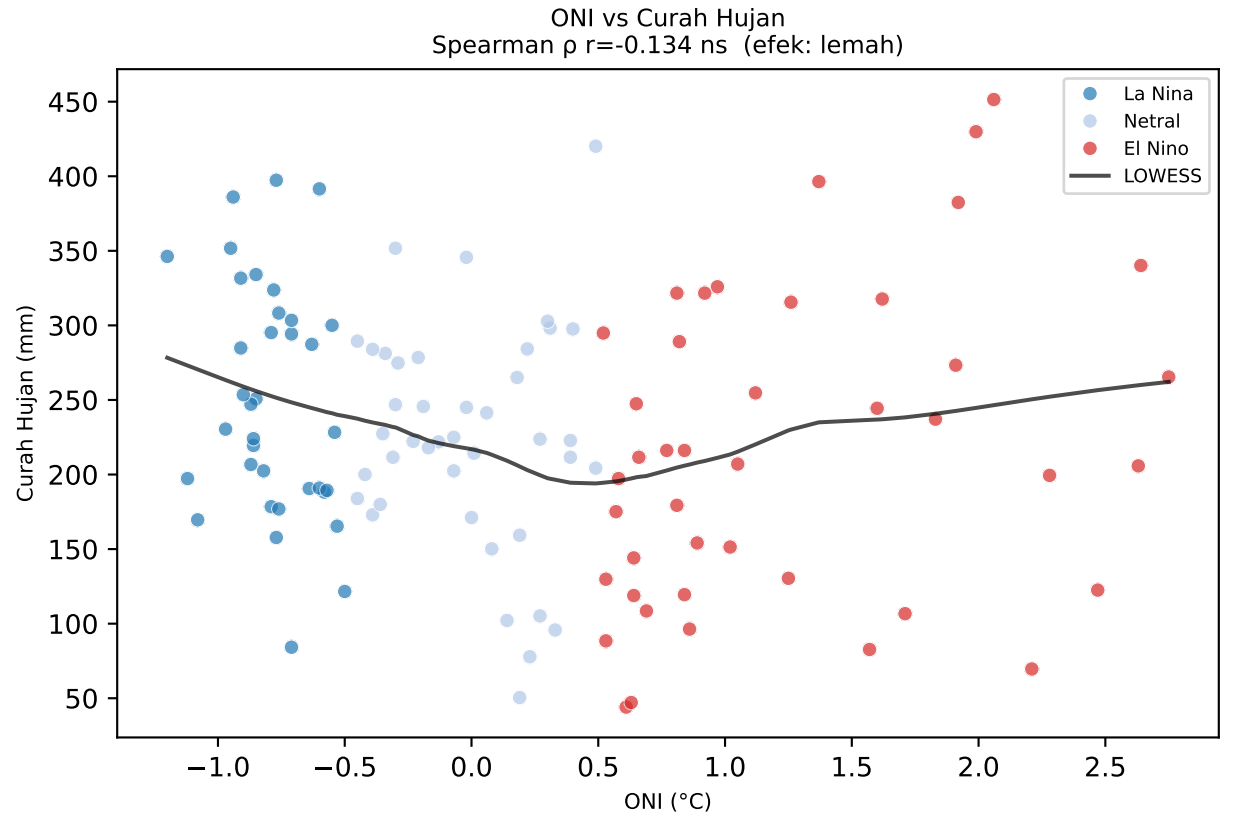


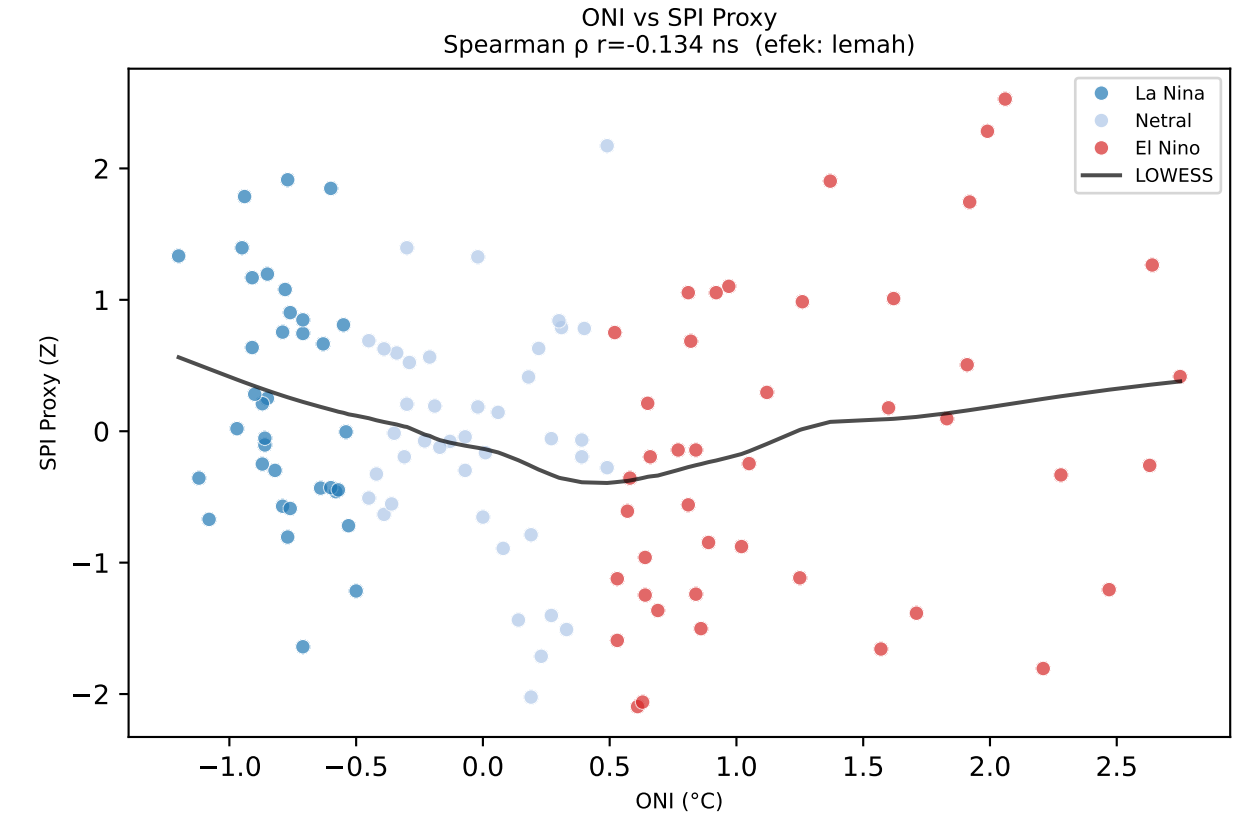


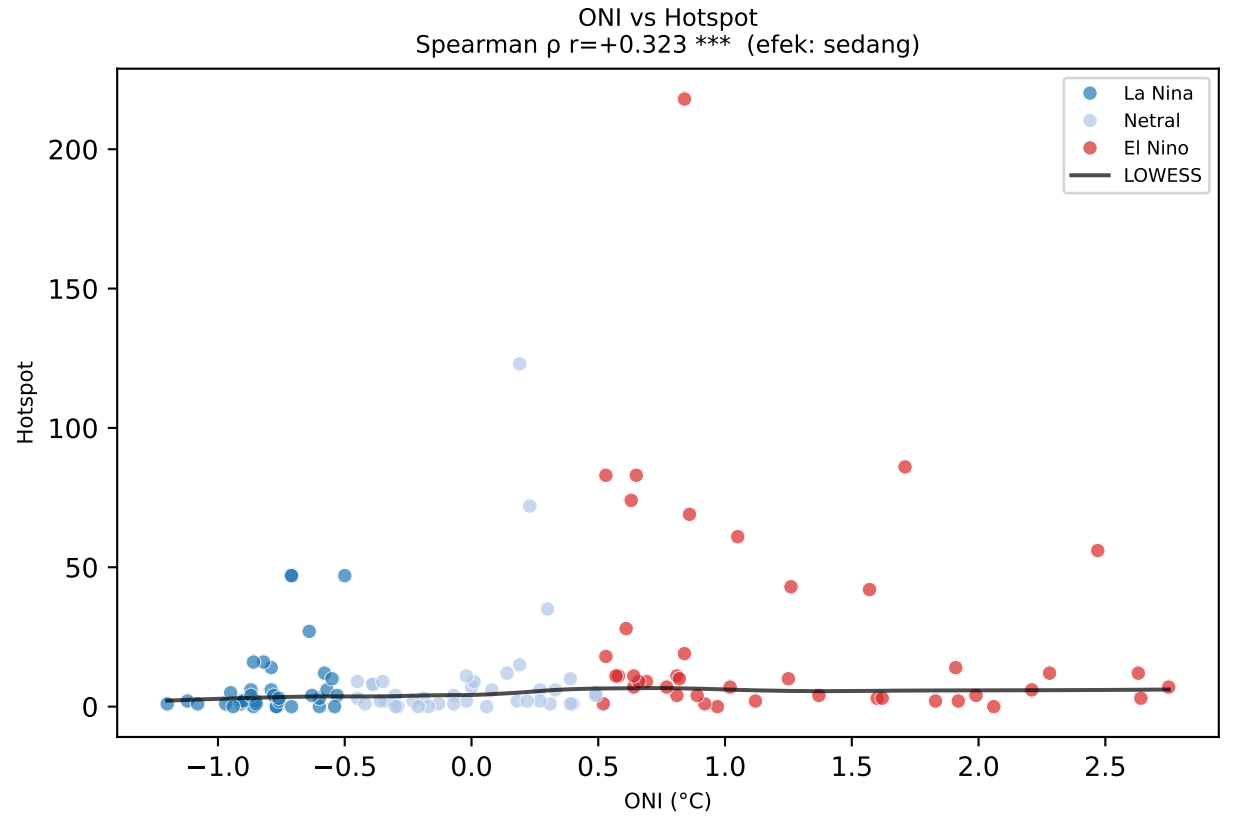


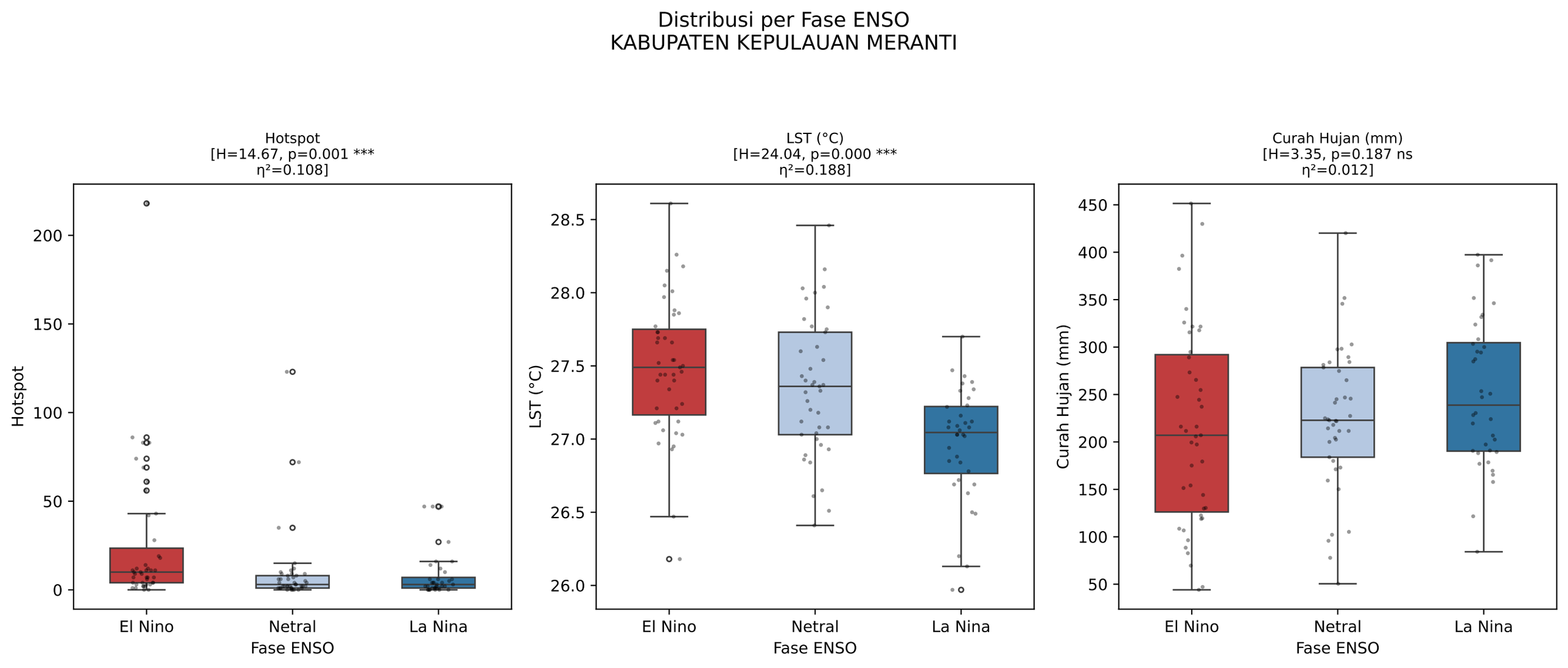
Scatter – ONI vs Variabel Iklim
KABUPATEN KEPULAUAN MERANTI





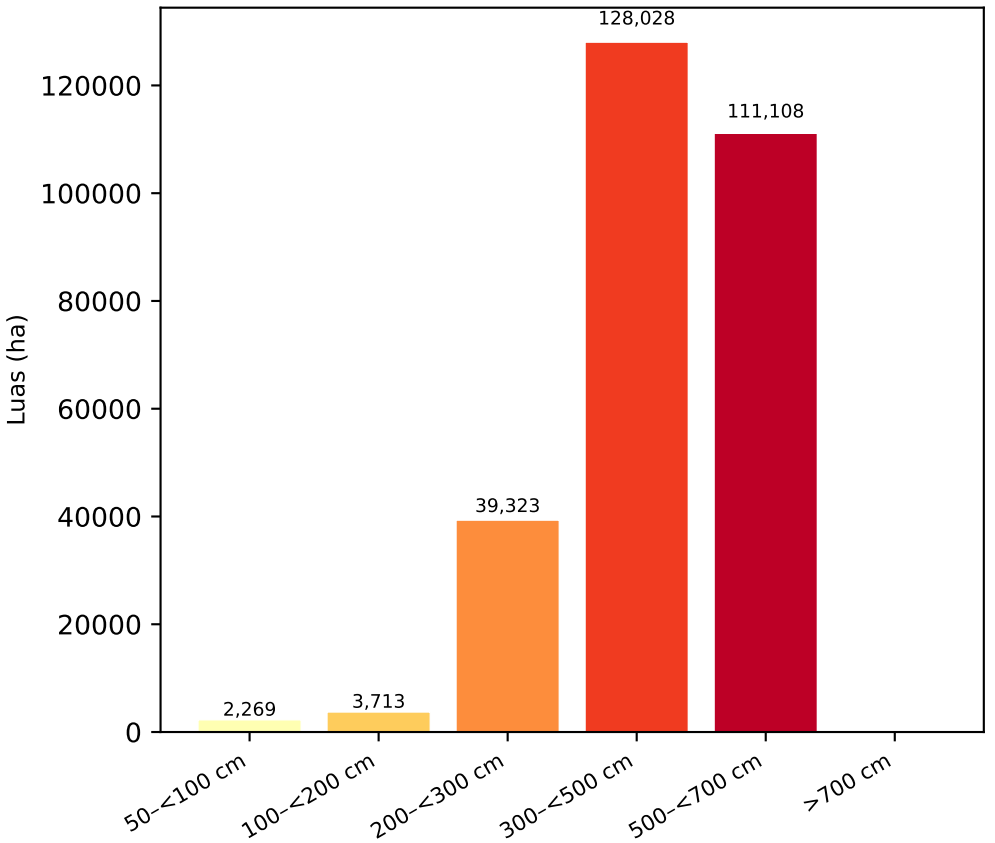




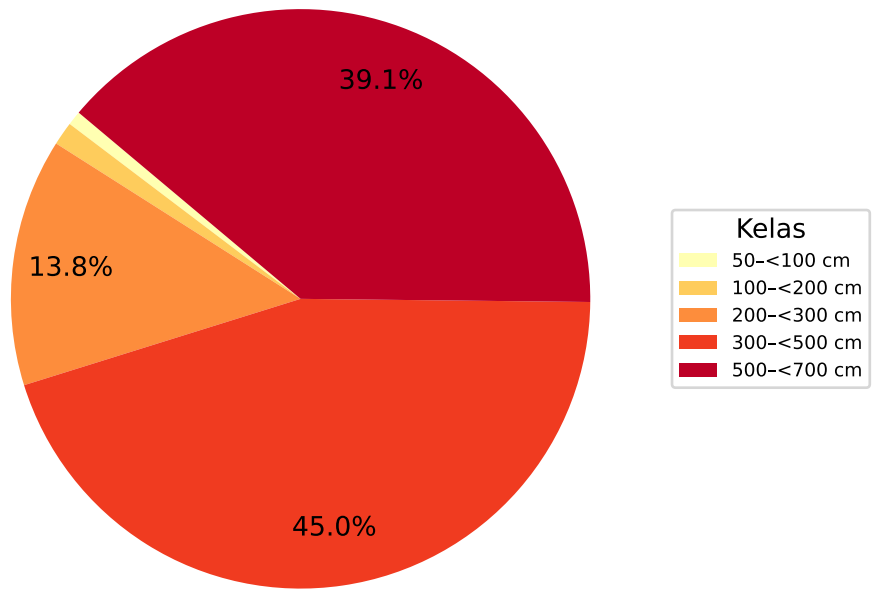

Distribusi per Fase ENSO
KABUPATEN KEPULAUAN MERANTI
Hotspot
[H=14.67, p=0.001 ***
η²=0.108]
Hotspot
0
50
100
150
200
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=24.04, p=0.000 ***
η²=0.188]
LST (°C)
26.0
26.5
27.0
27.5
28.0
28.5
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=3.35, p=0.187 ns
η²=0.012]
Curah Hujan (mm)
50
100
150
200
250
300
350
400
450
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN KEPULAUAN MERANTI

A. Luas Area per Kelas Kedalaman



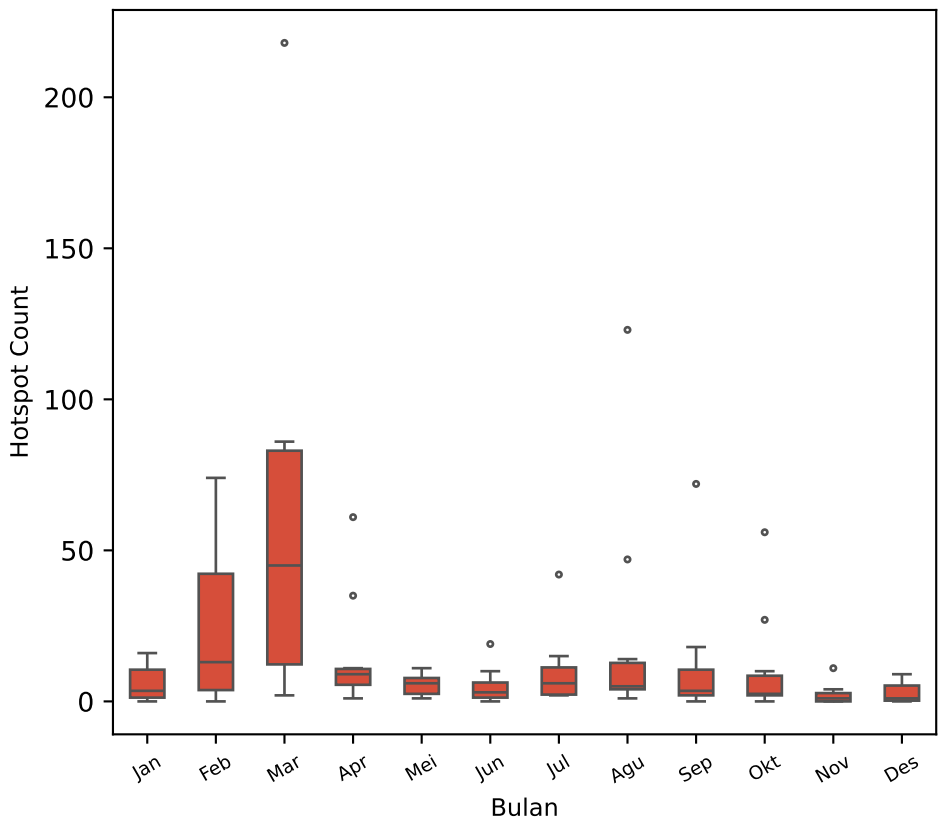
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════════
  Statistik Gambut — Kabupaten Kepulauan
═══════════════════════════════════════════════════
  Total Luas          :      284,441 ha
═══════════════════════════════════════════════════
  Kelas 1    50–<100 cm:       2,269 ha  (  0.8%)
  Kelas 2   100–<200 cm:       3,713 ha  (  1.3%)
  Kelas 3   200–<300 cm:      39,323 ha  ( 13.8%)
  Kelas 4   300–<500 cm:     128,028 ha  ( 45.0%)
  Kelas 5   500–<700 cm:     111,108 ha  ( 39.1%)
  Kelas 6       >700 cm:           0 ha  (  0.0%)
═══════════════════════════════════════════════════
  Kelas Dominan       : 300–<500 cm
  Median Berbobot (σ) : 300–<500 cm
═══════════════════════════════════════════════════
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 300–<500 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

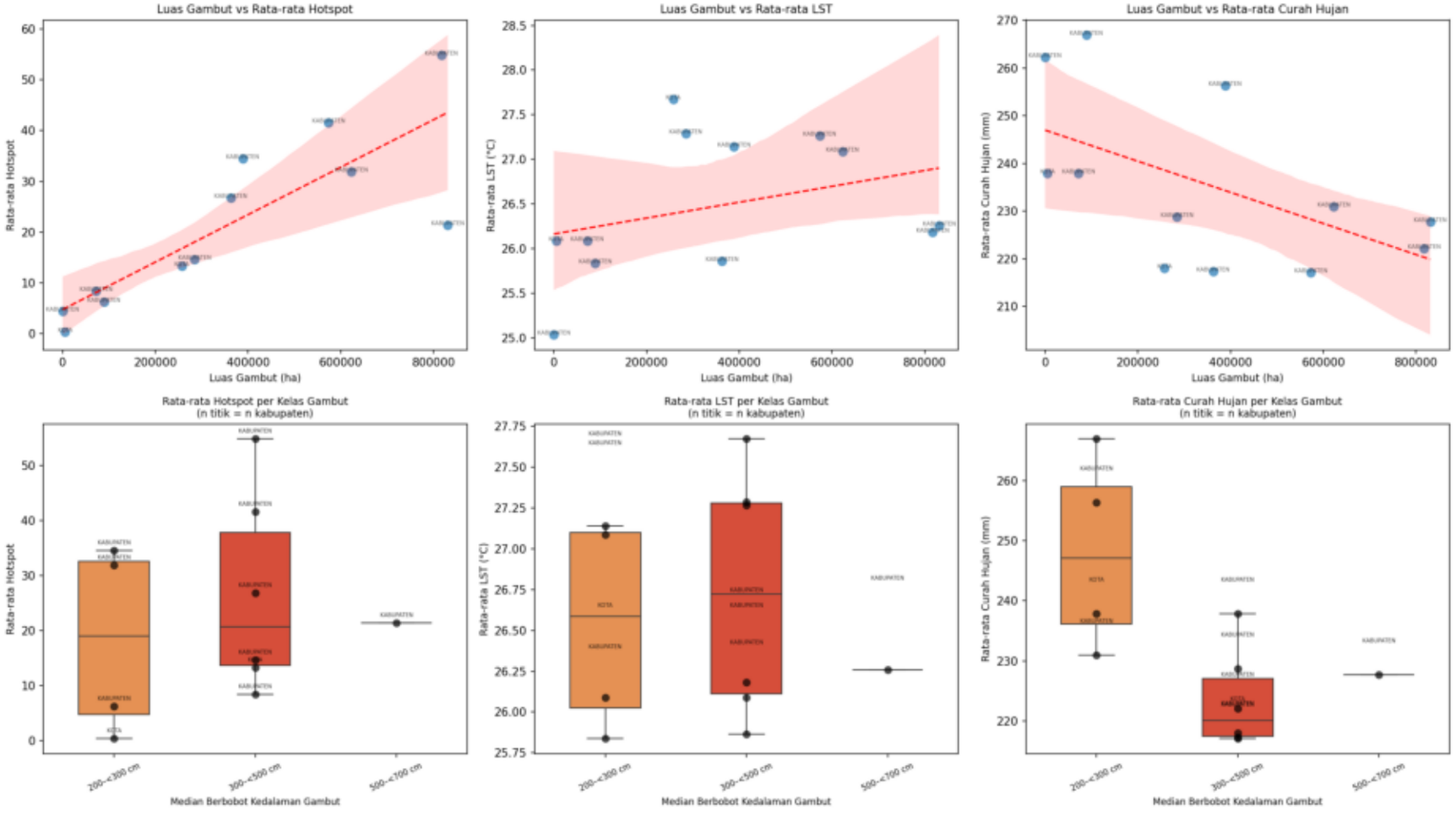
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



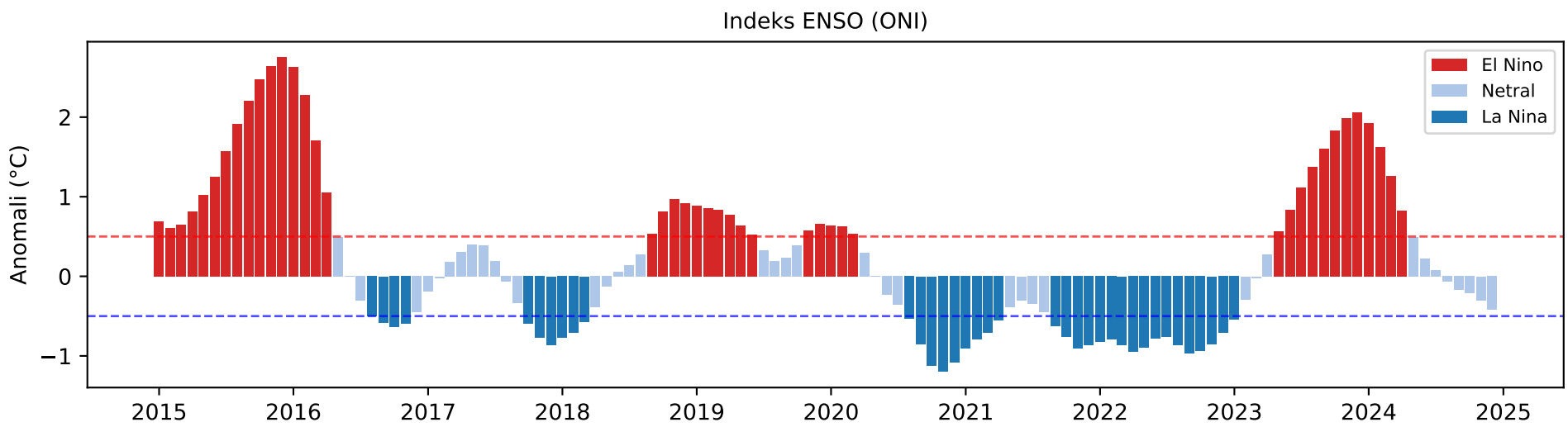
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN KAMPAR

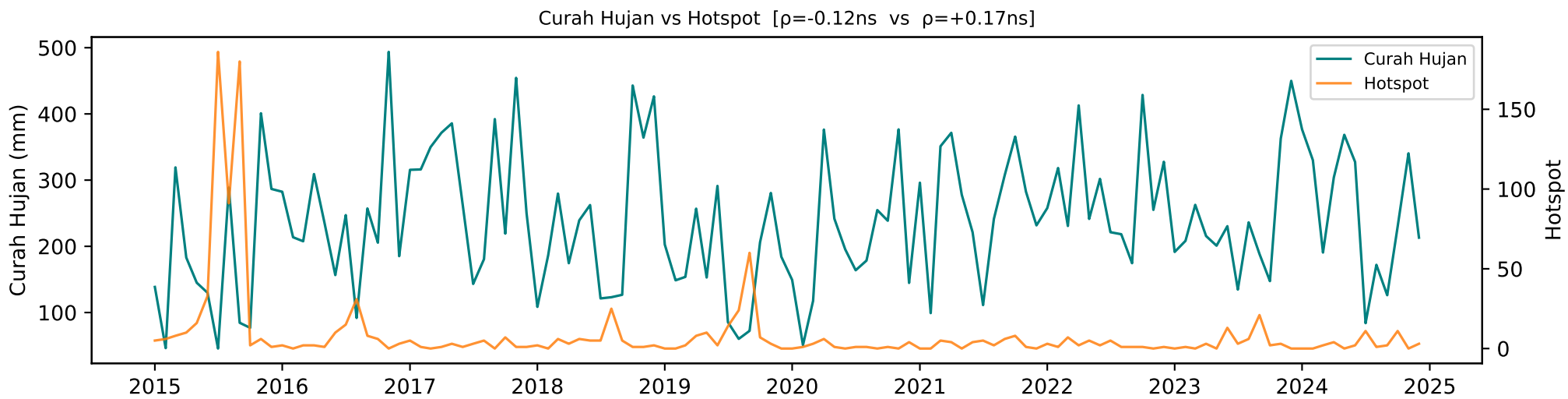
```
═══════════════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────────────
  ONI vs LST                                        : r=+0.536 ***   (p=0.000, Spearman ρ, efek=kuat, n=120)
  ONI vs Curah Hujan                                : r=-0.123 ns    (p=0.181, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                     : r=-0.123 ns    (p=0.181, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                    : r=+0.168 ns    (p=0.066, Spearman ρ, efek=lemah, n=120)

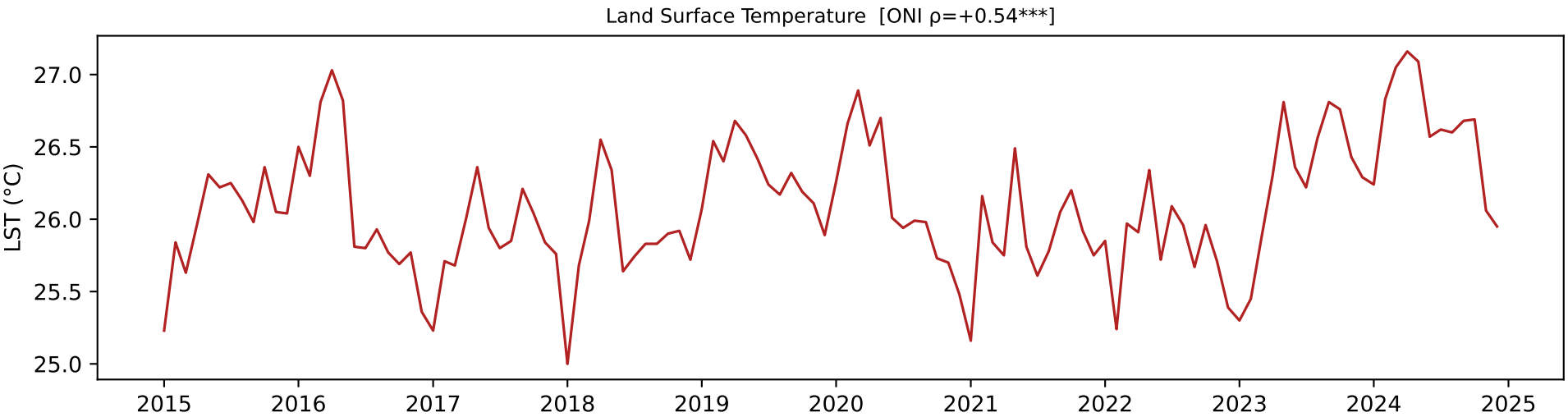
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                           : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────────────
  Hotspot     : H=2.63, p=0.269ns, η²=0.005
  LST         : H=30.43, p=0.000, η²=0.243
  Curah Hujan : H=4.38, p=0.112, η²=0.020

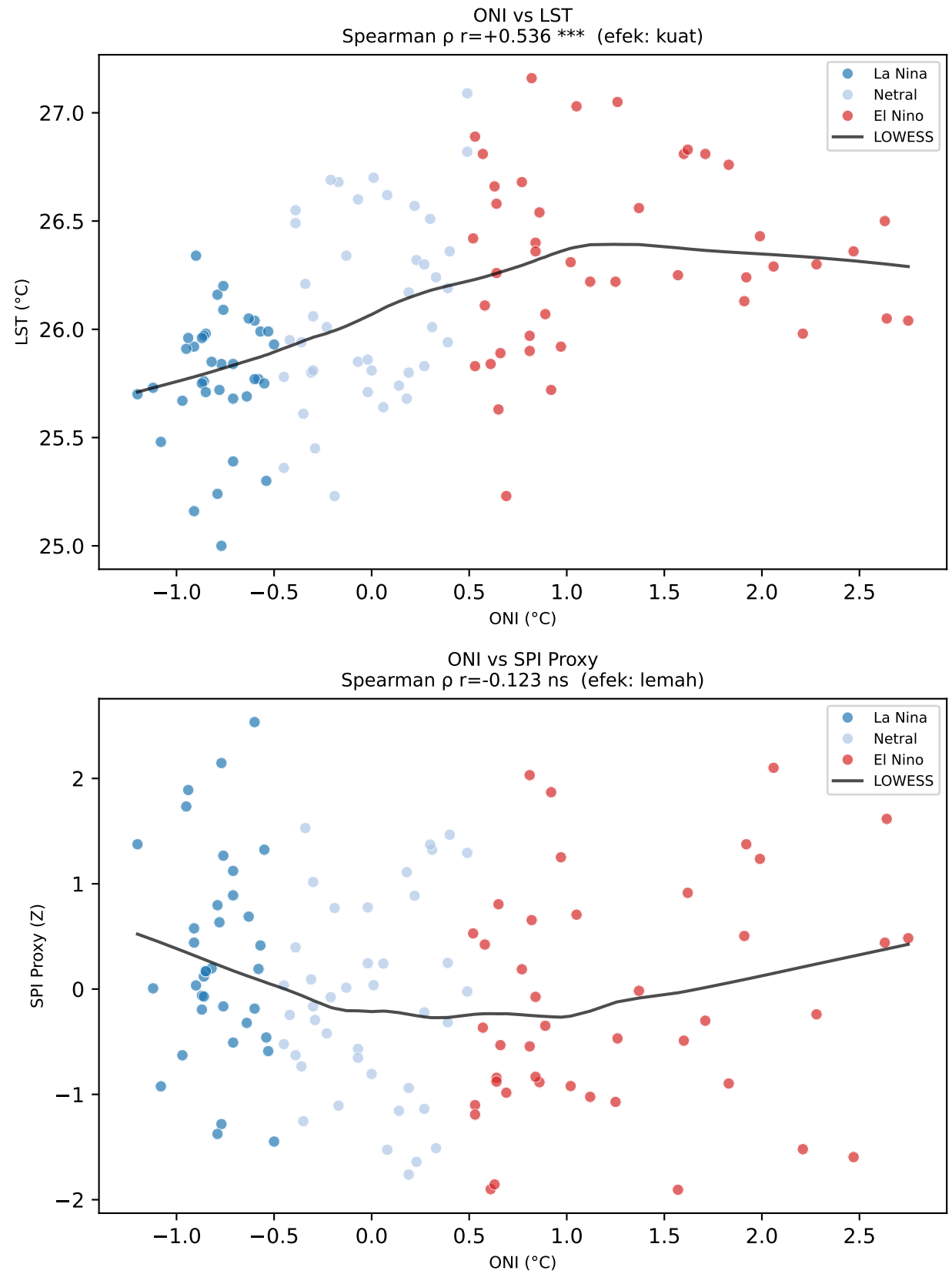
  INFO GAMBUT : total=71,972 ha | kelas dominan=300—<500 cm | median berbobot=300—<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

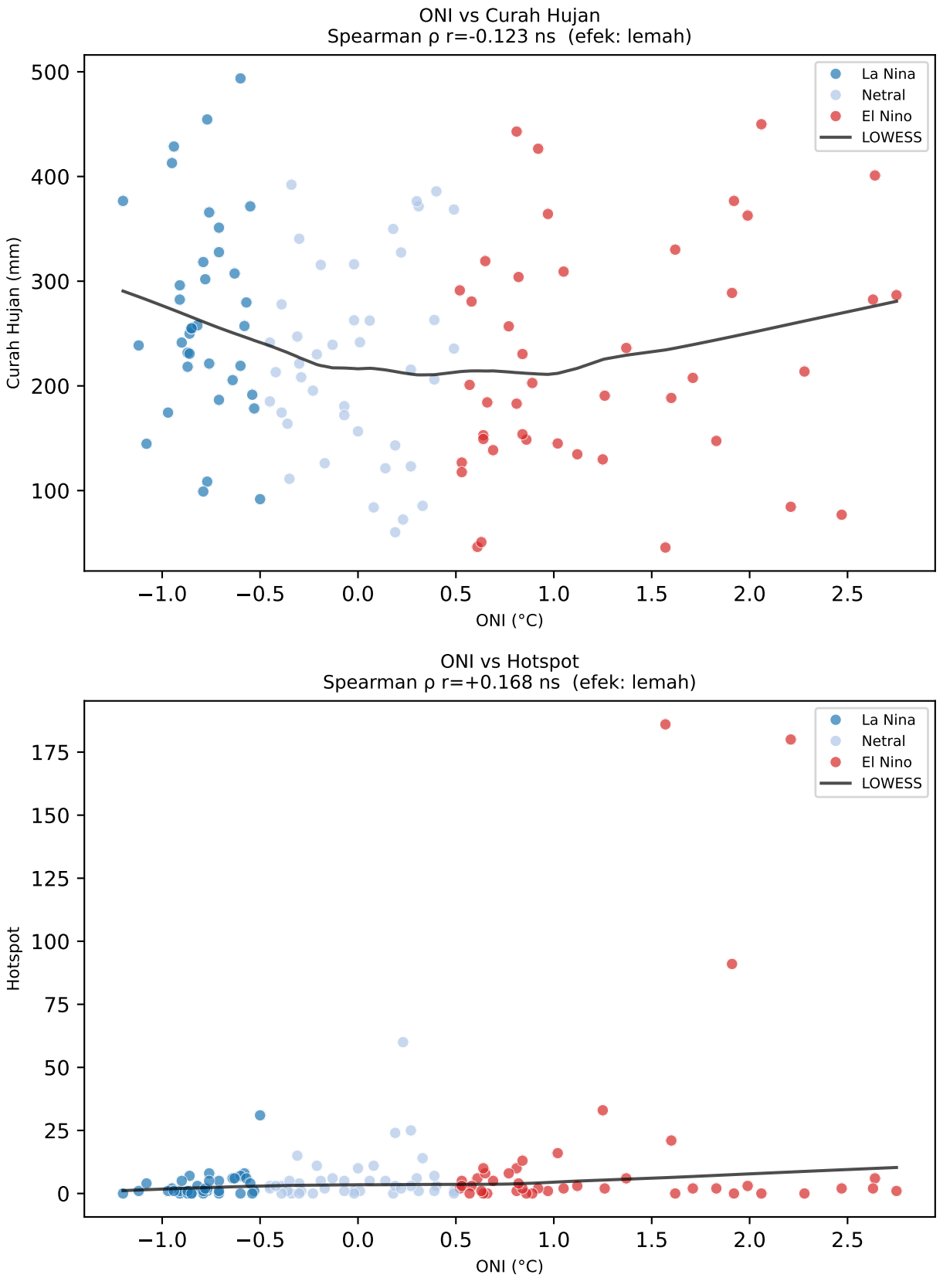


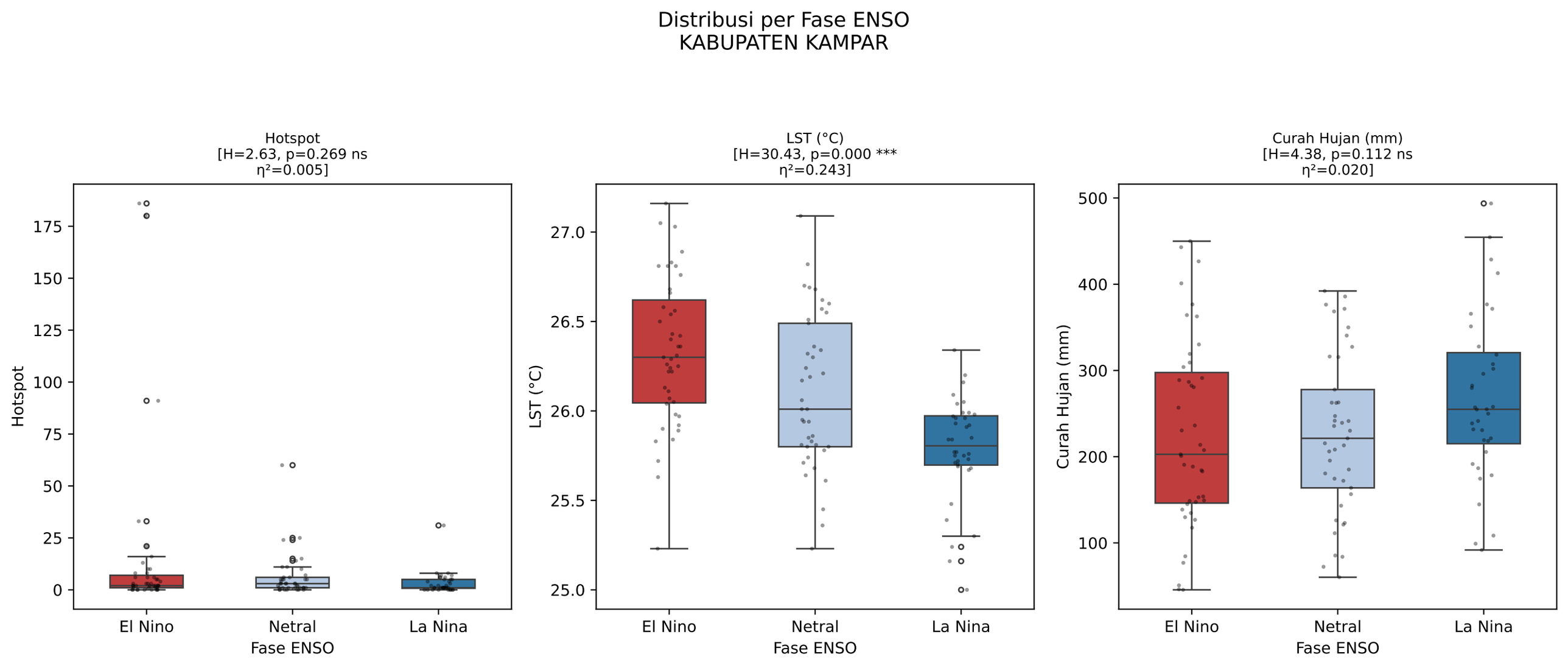




Scatter – ONI vs Variabel Iklim
KABUPATEN KAMPAR

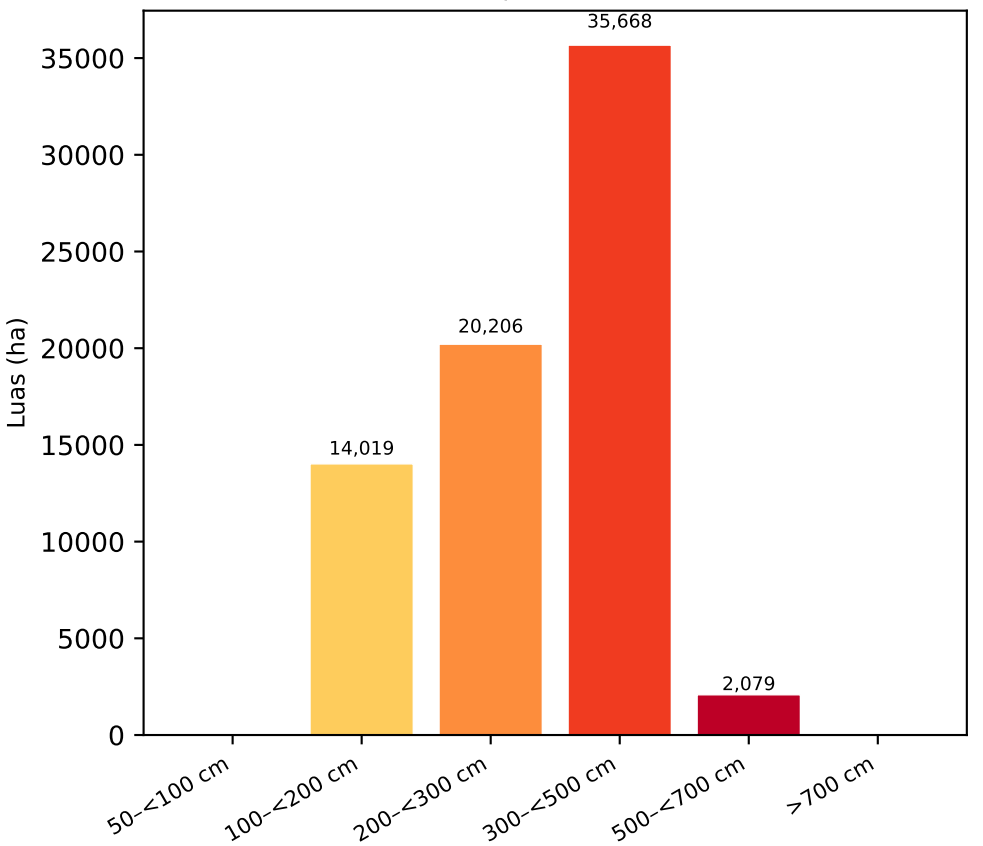




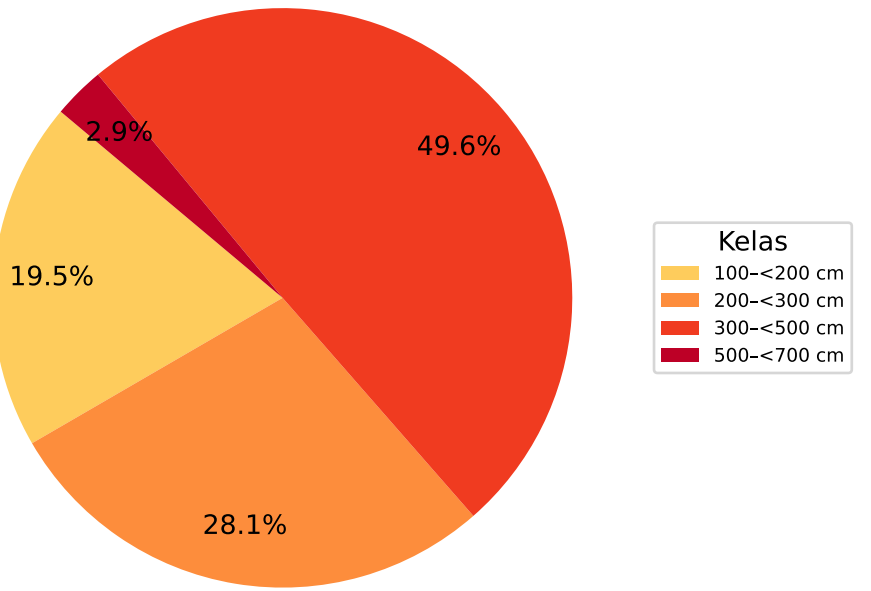

Distribusi per Fase ENSO
KABUPATEN KAMPAR
Hotspot
[H=2.63, p=0.269 ns
η²=0.005]
Hotspot
0
25
50
75
100
125
150
175
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=30.43, p=0.000 ***
η²=0.243]
LST (°C)
25.0
25.5
26.0
26.5
27.0
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=4.38, p=0.112 ns
η²=0.020]
Curah Hujan (mm)
100
200
300
400
500
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN KAMPAR

A. Luas Area per Kelas Kedalaman



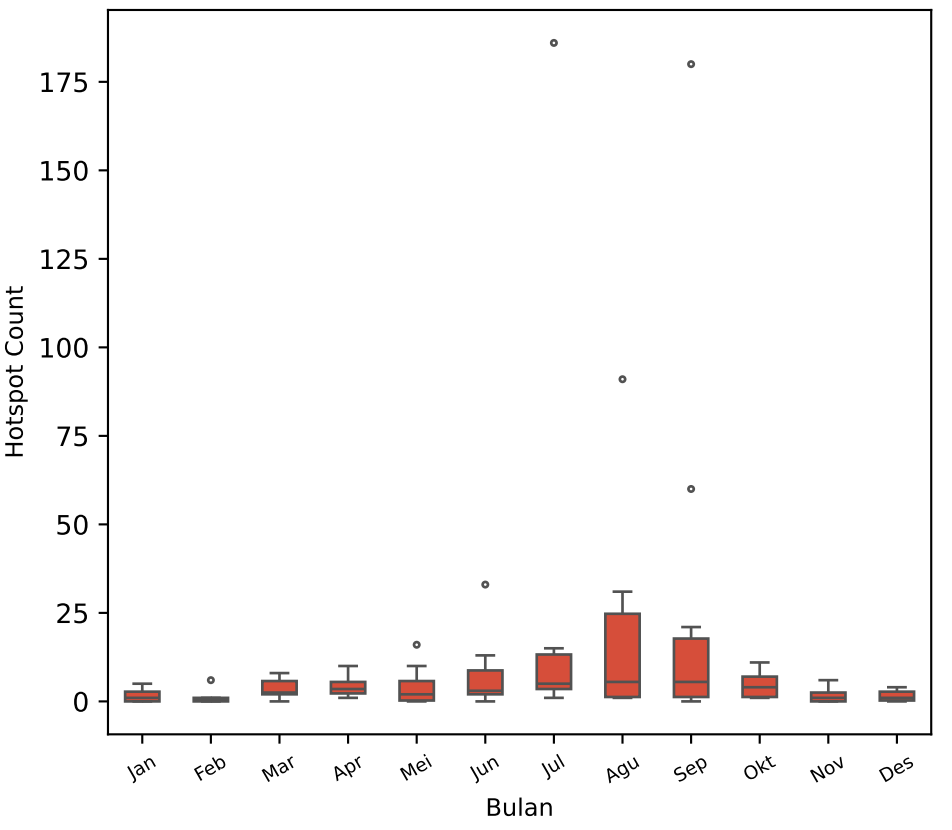
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
Statistik Gambut — Kabupaten Kampar
Total Luas          :       71,972 ha
Kelas 1    50–<100 cm:           0 ha  (  0.0%)
Kelas 2   100–<200 cm:      14,019 ha  ( 19.5%)
Kelas 3   200–<300 cm:      20,206 ha  ( 28.1%)
Kelas 4   300–<500 cm:      35,668 ha  ( 49.6%)
Kelas 5   500–<700 cm:       2,079 ha  (  2.9%)
Kelas 6       >700 cm:           0 ha  (  0.0%)
Kelas Dominan       : 300–<500 cm
Median Berbobot (σ) : 300–<500 cm
⚠  Kedalaman gambut = atribut statis
   Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 300–<500 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

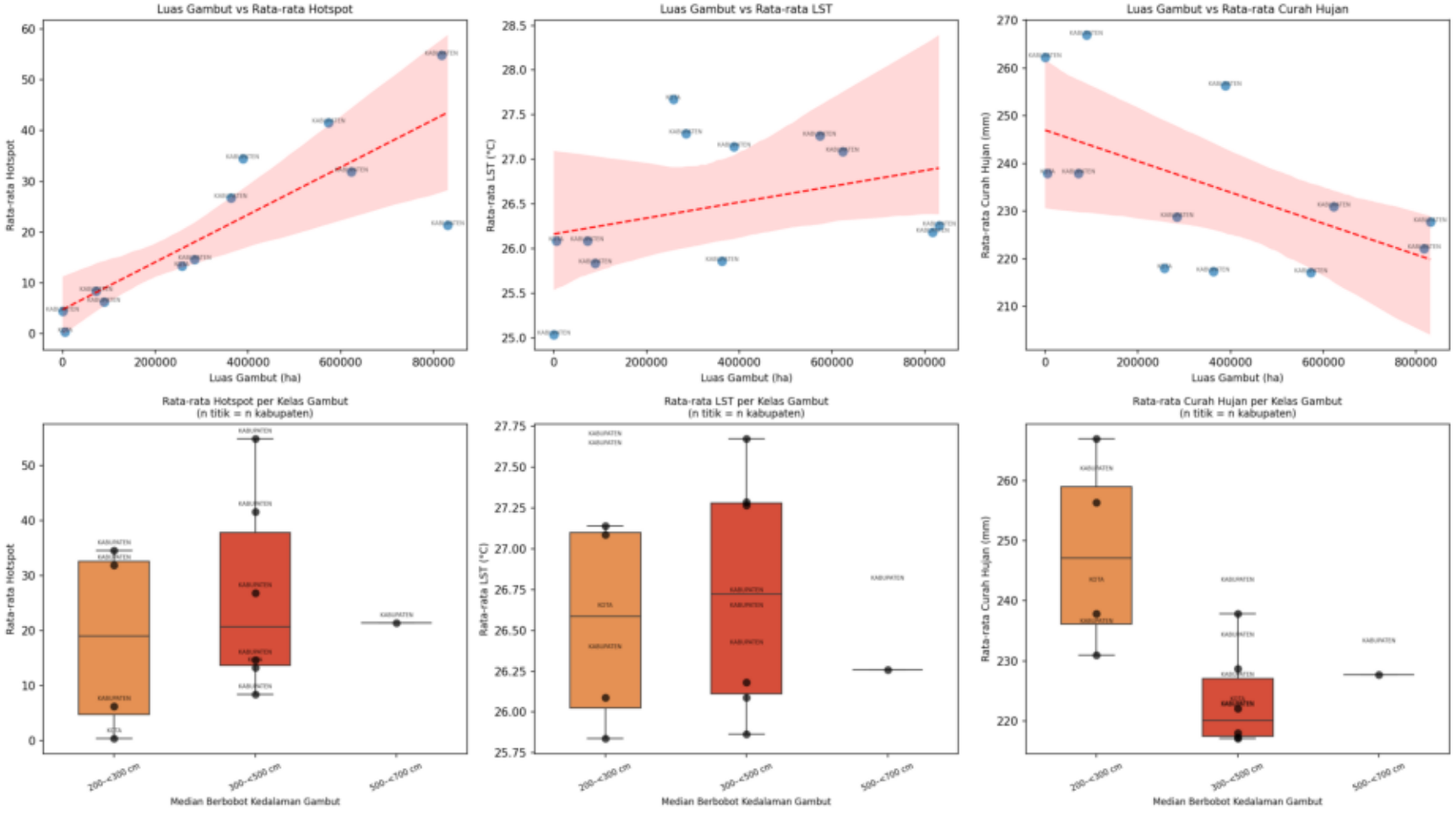
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



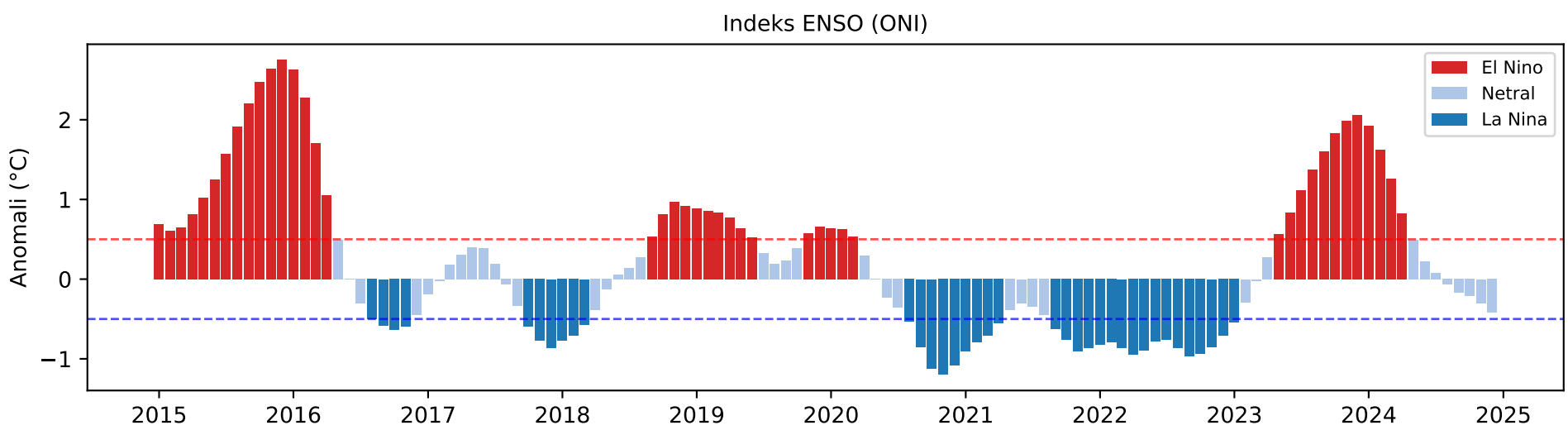
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN INDRAGIRI HULU

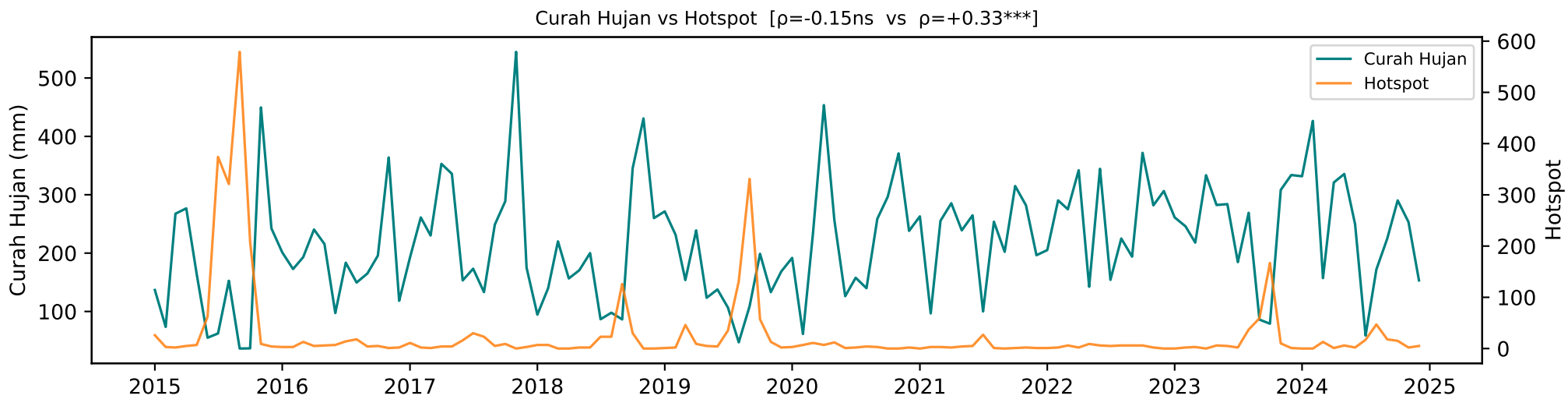
```
═══════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.491 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.147 ns    (p=0.108, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.147 ns    (p=0.108, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.327 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

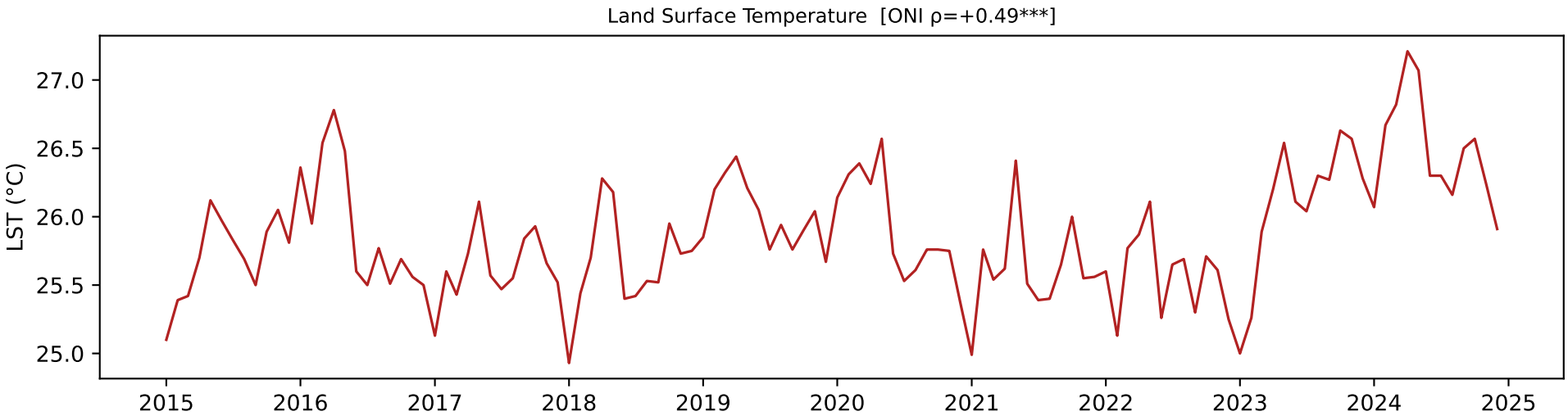
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────
  Hotspot     : H=13.40, p=0.001**, η²=0.097
  LST         : H=26.85, p=0.000, η²=0.212
  Curah Hujan : H=5.87, p=0.053, η²=0.033

  INFO GAMBUT : total=362,630 ha | kelas dominan=300–<500 cm | median berbobot=300–<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

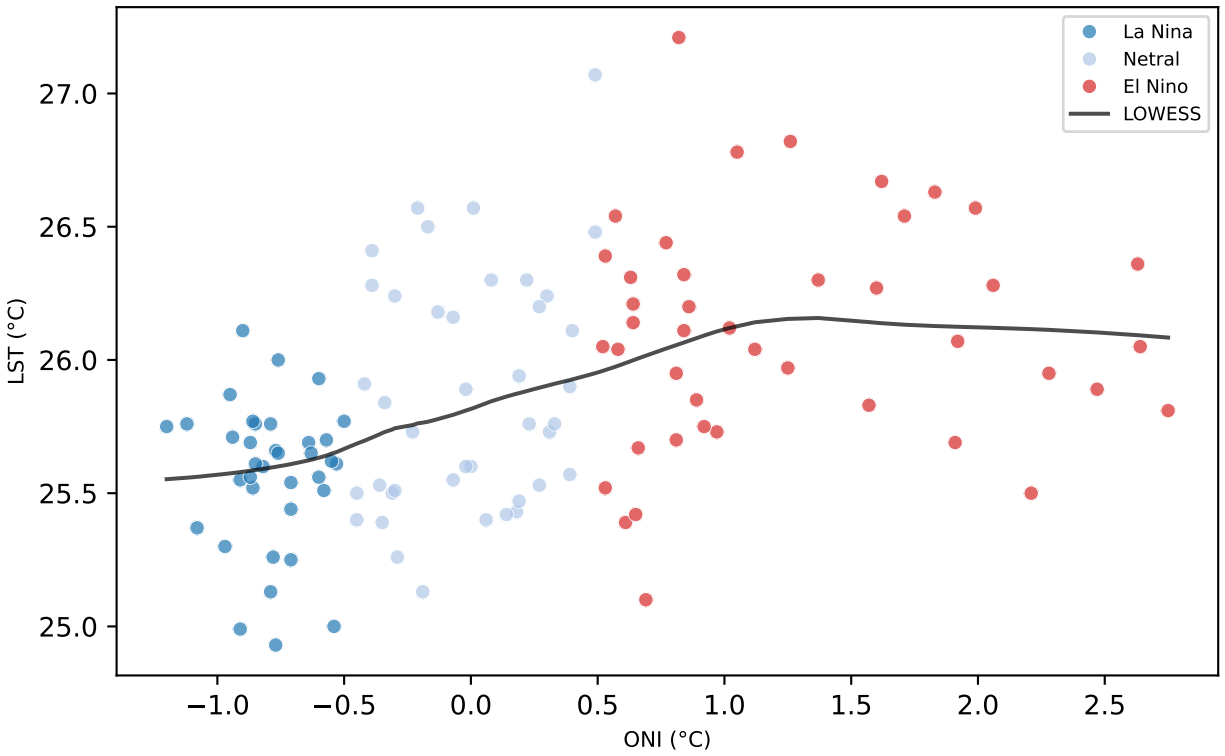




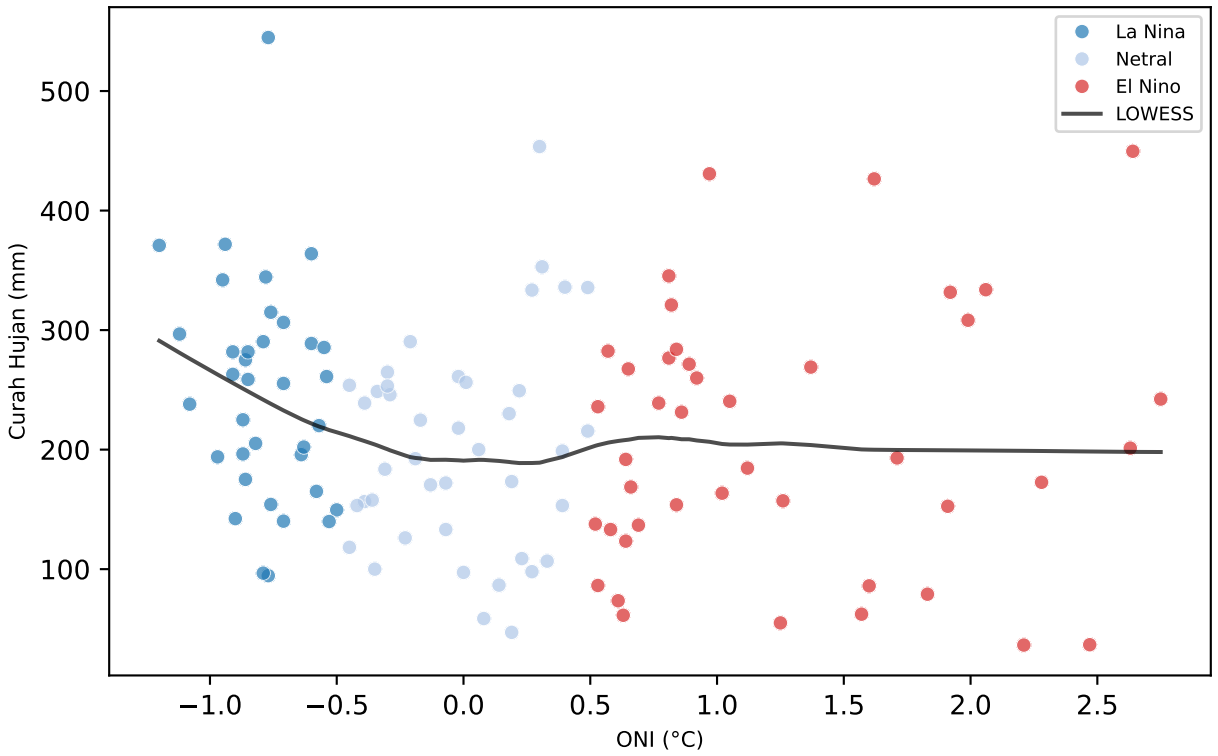


Scatter – ONI vs Variabel Iklim
KABUPATEN INDRAGIRI HULU

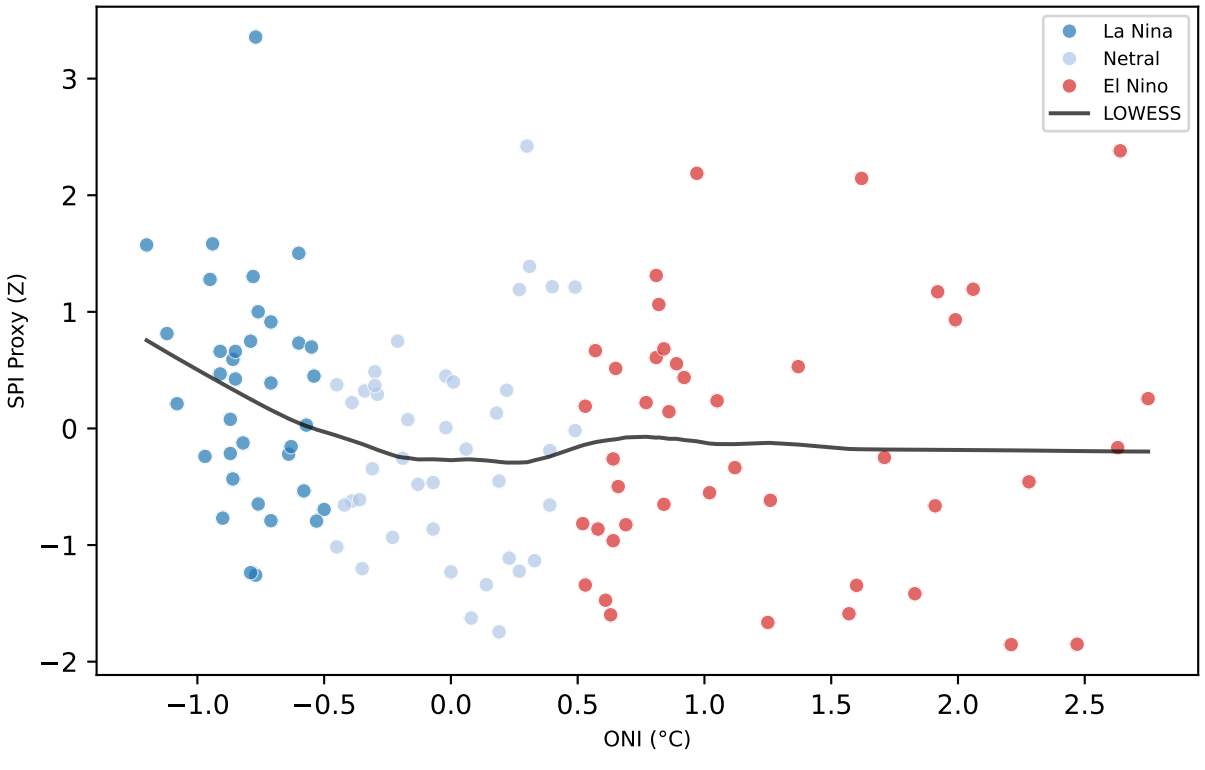
ONI vs LST
Spearman ρ r=+0.491 *** (efek: sedang)



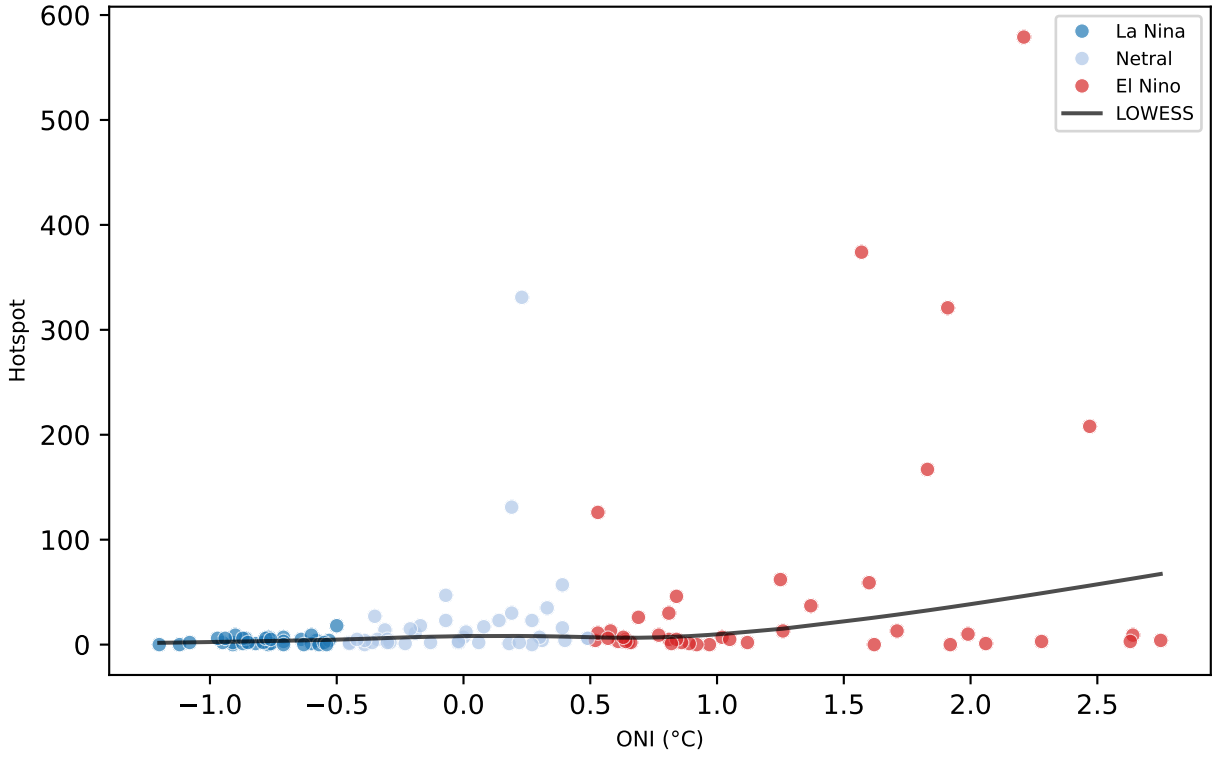
ONI vs Curah Hujan
Spearman ρ r=-0.147 ns (efek: lemah)

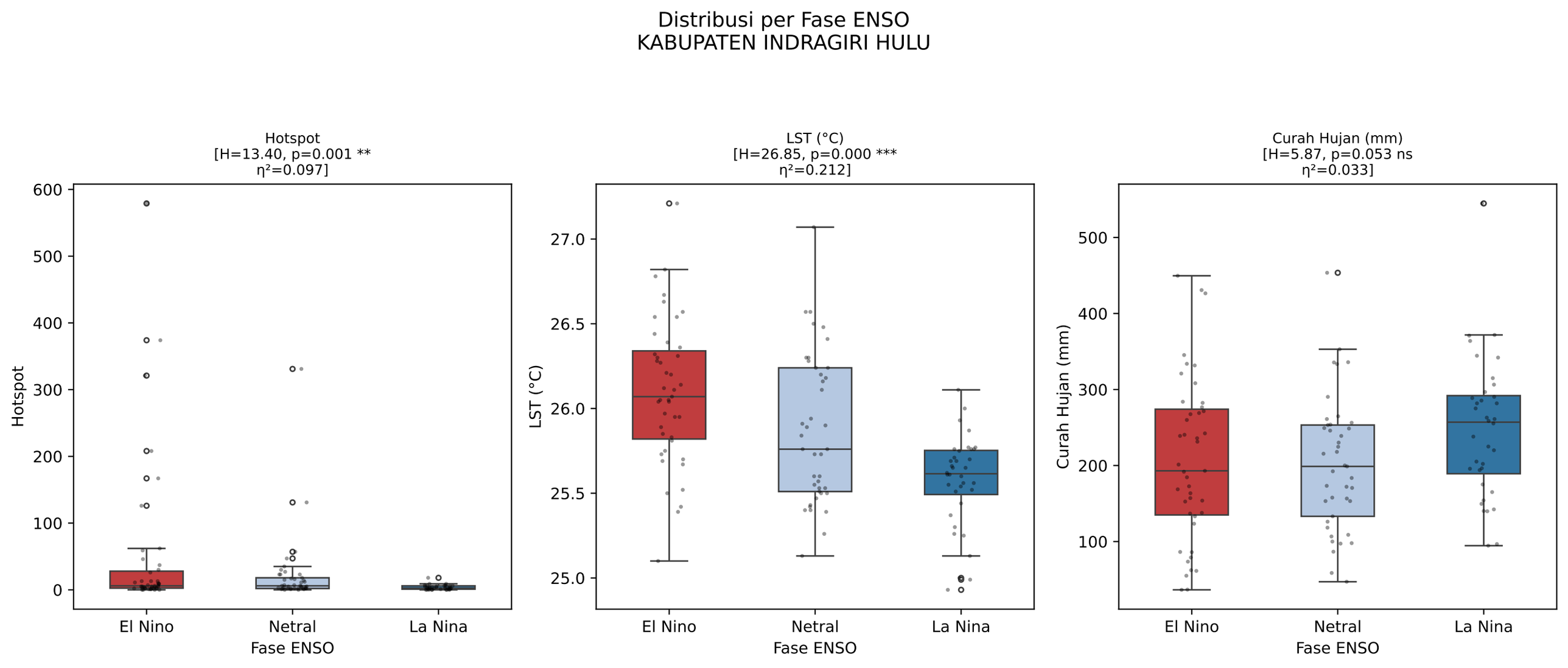


ONI vs SPI Proxy
Spearman ρ r=-0.147 ns (efek: lemah)



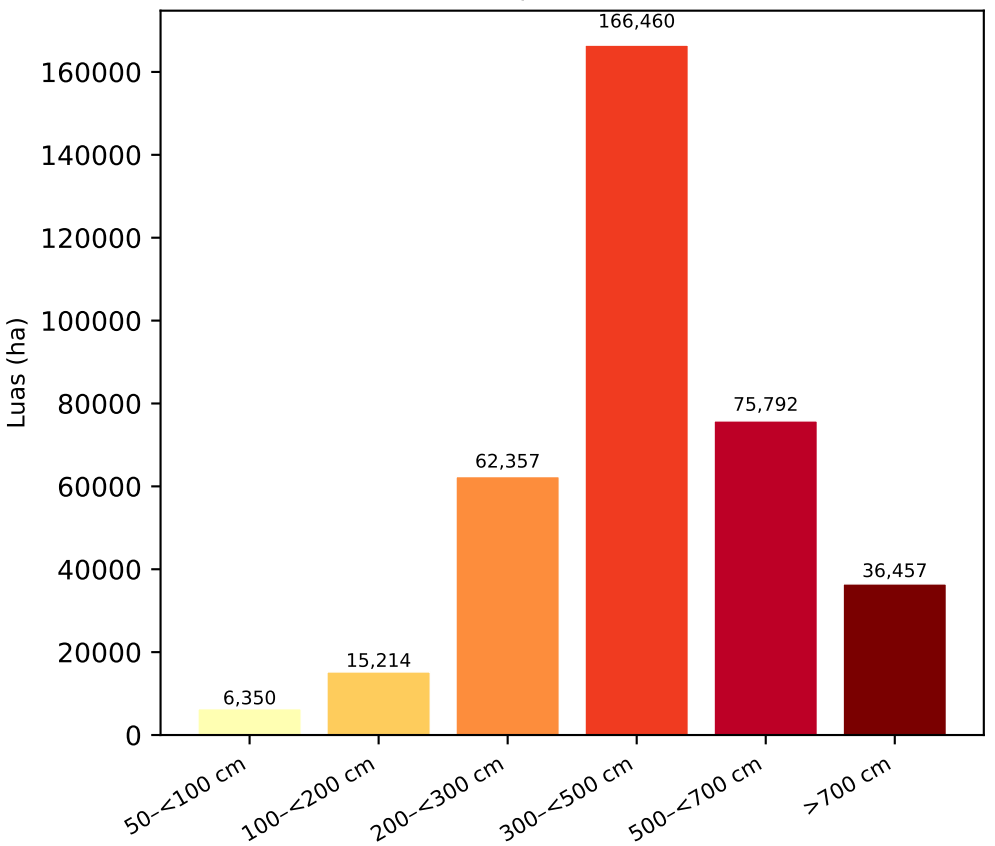
ONI vs Hotspot
Spearman ρ r=+0.327 *** (efek: sedang)



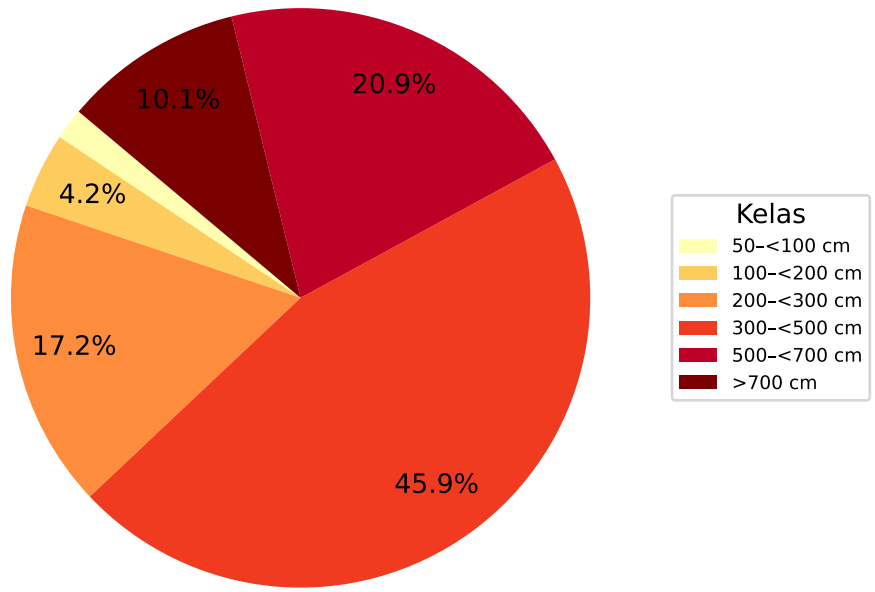

Distribusi per Fase ENSO
KABUPATEN INDRAGIRI HULU
Hotspot
[H=13.40, p=0.001 **
η²=0.097]
Hotspot
600
500
400
300
200
100
0
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=26.85, p=0.000 ***
η²=0.212]
LST (°C)
27.0
26.5
26.0
25.5
25.0
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=5.87, p=0.053 ns
η²=0.033]
Curah Hujan (mm)
500
400
300
200
100
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN INDRAGIRI HULU

A. Luas Area per Kelas Kedalaman



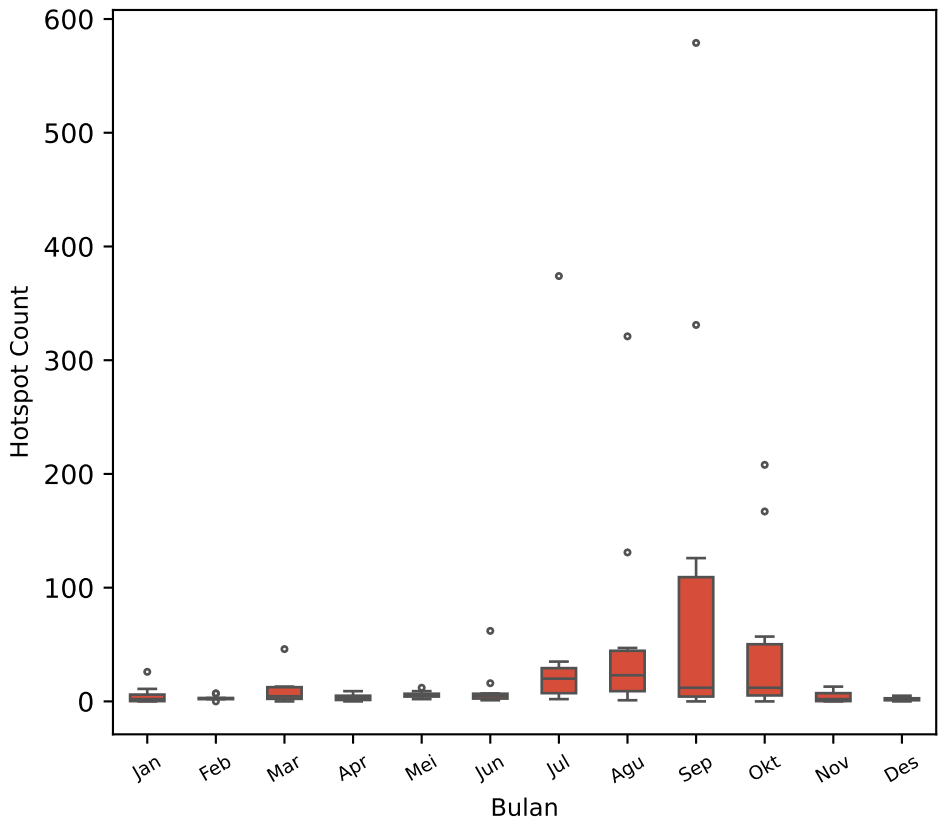
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════════
  Statistik Gambut — Kabupaten Indragiri
═══════════════════════════════════════════════════
  Total Luas          :      362,630 ha
═══════════════════════════════════════════════════
  Kelas 1    50–<100 cm:      6,350 ha  (  1.8%)
  Kelas 2   100–<200 cm:     15,214 ha  (  4.2%)
  Kelas 3   200–<300 cm:     62,357 ha  ( 17.2%)
  Kelas 4   300–<500 cm:    166,460 ha  ( 45.9%)
  Kelas 5   500–<700 cm:     75,792 ha  ( 20.9%)
  Kelas 6       >700 cm:     36,457 ha  ( 10.1%)
═══════════════════════════════════════════════════
  Kelas Dominan       : 300–<500 cm
  Median Berbobot (σ) : 300–<500 cm
═══════════════════════════════════════════════════
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 300–<500 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

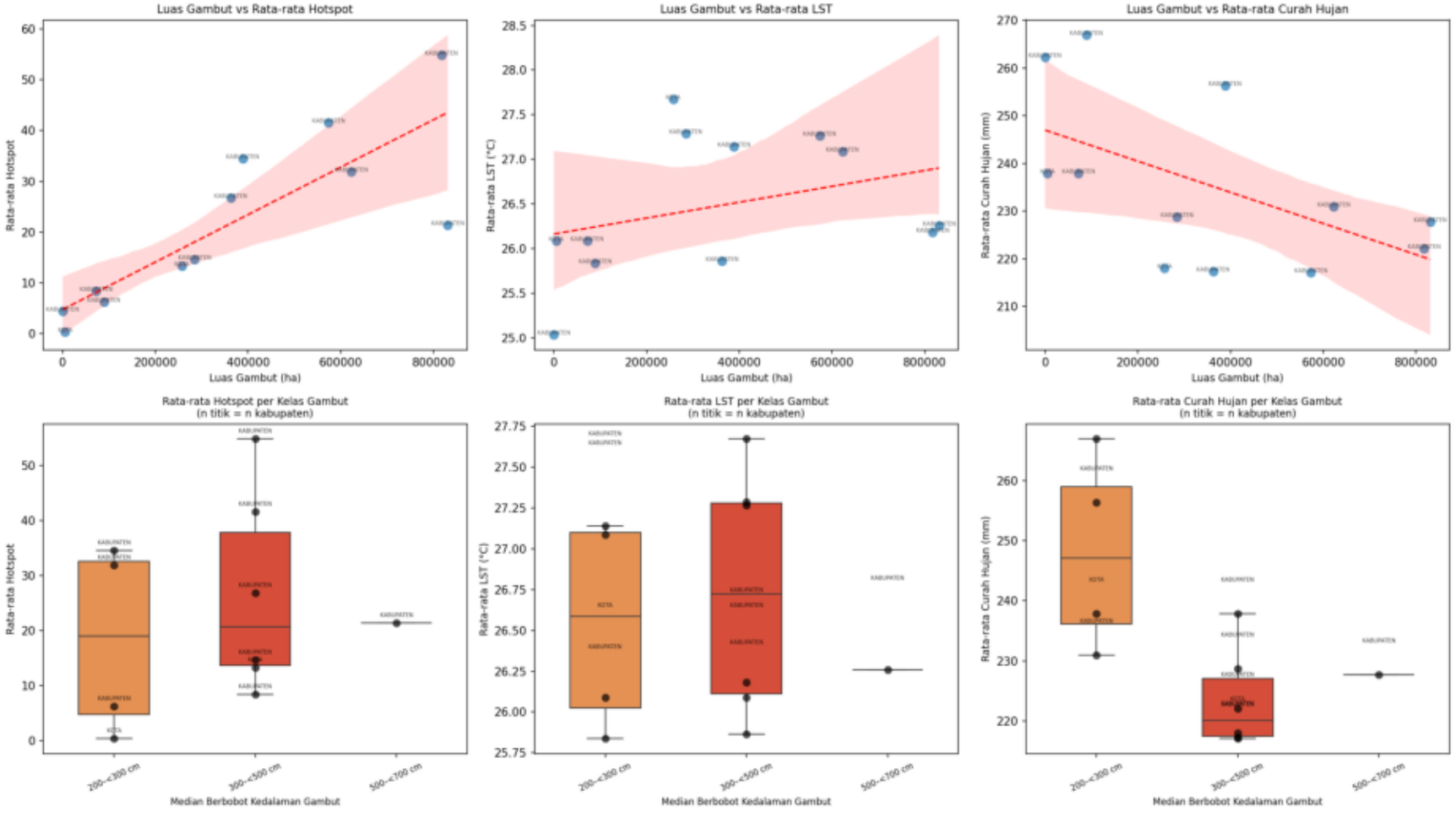
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



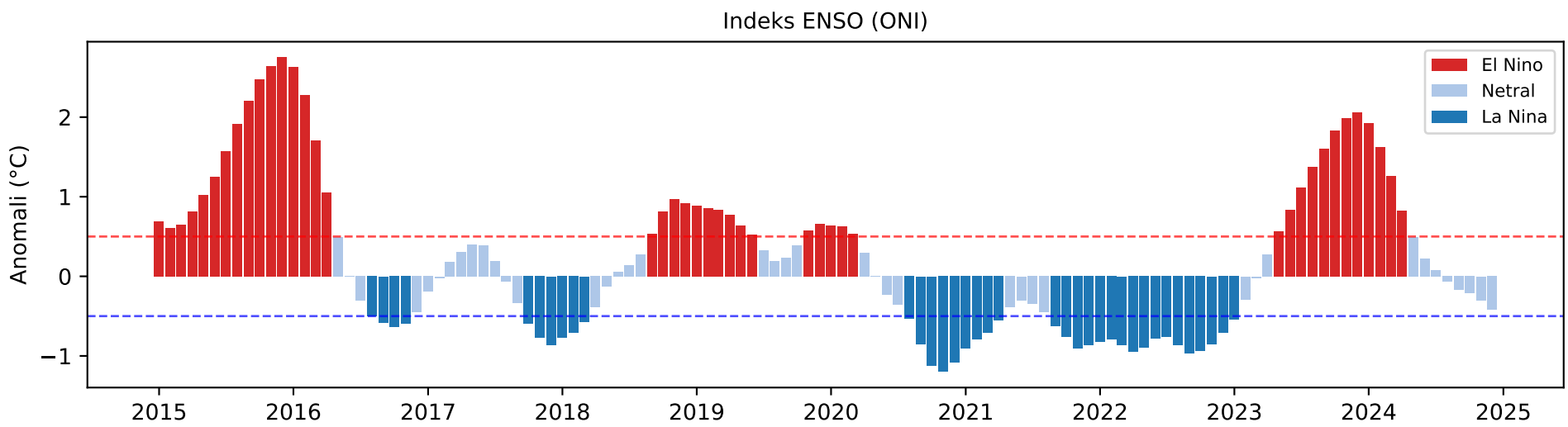
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN INDRAGIRI HILIR

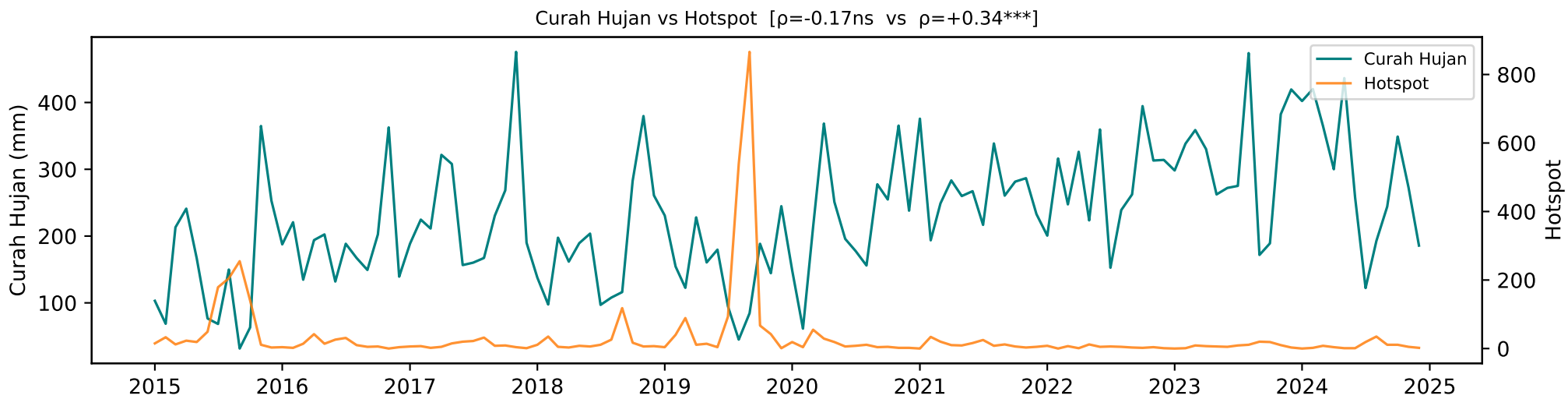
```
═══════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.494 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.173 ns    (p=0.059, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.173 ns    (p=0.059, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.345 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

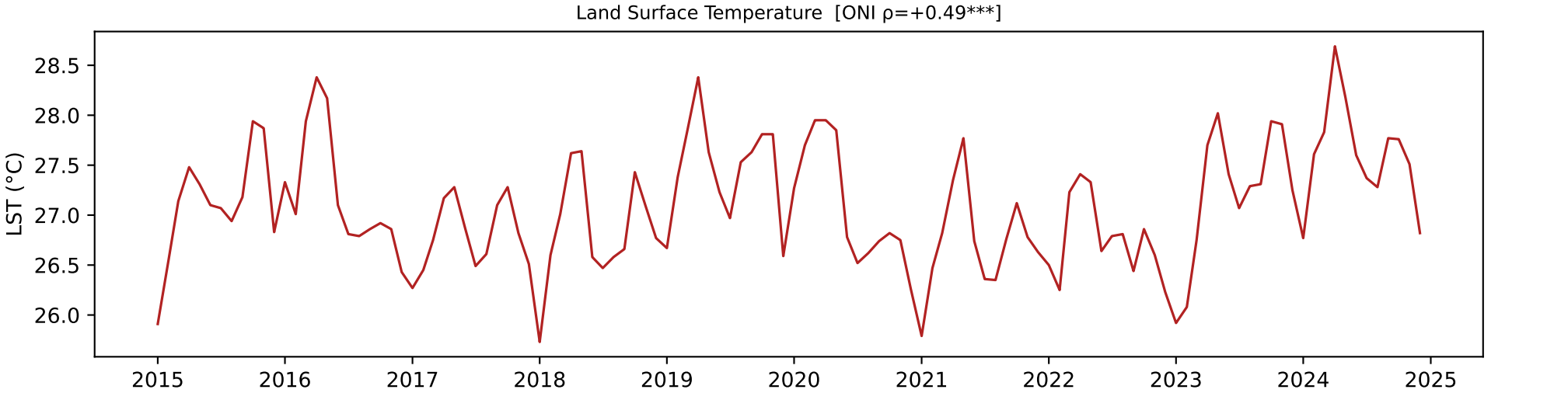
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)
  Luas Gambut (ha) vs LST                       : r=+0.000 ns    (p=1.000, constant, efek=diabaikan, n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────
  Hotspot     : H=15.82, p=0.000***, η²=0.118
  LST         : H=25.29, p=0.000, η²=0.199
  Curah Hujan : H=5.89, p=0.053, η²=0.033

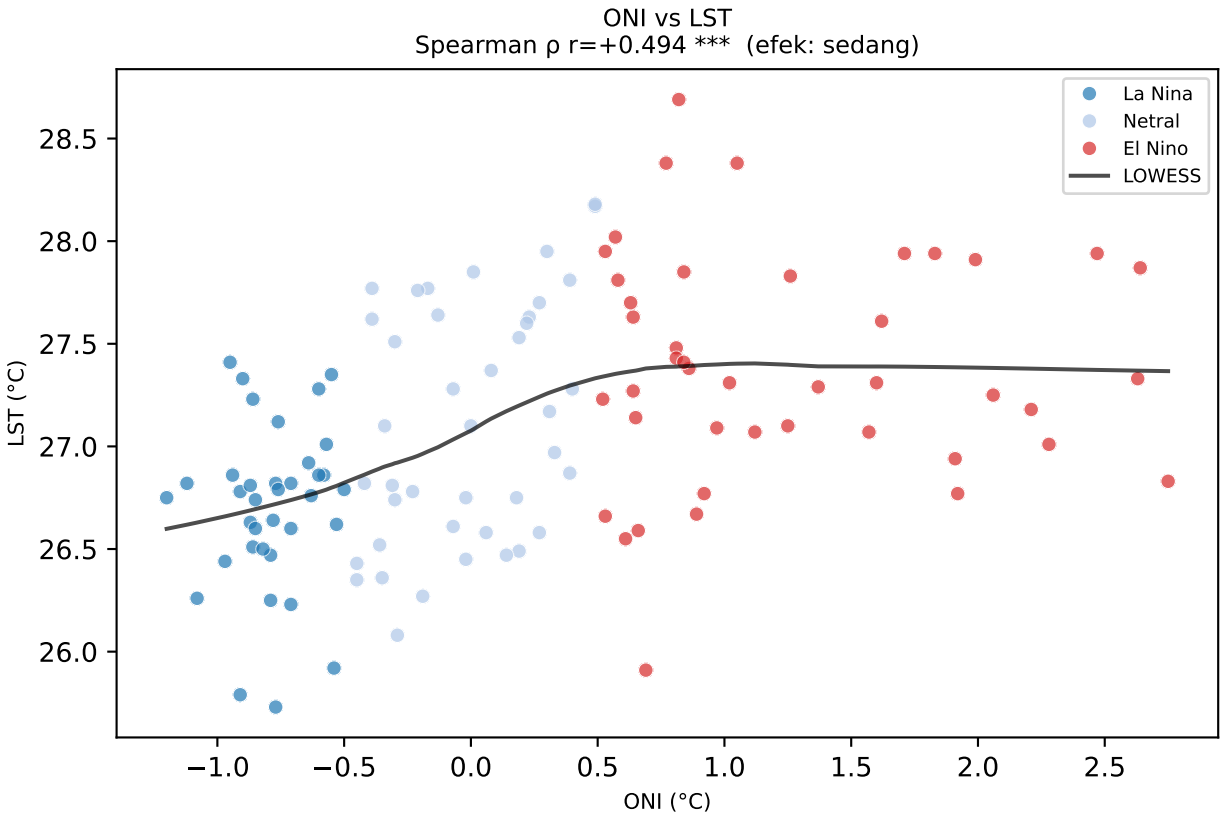
  INFO GAMBUT : total=622,921 ha | kelas dominan=100–<200 cm | median berbobot=200–<300 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

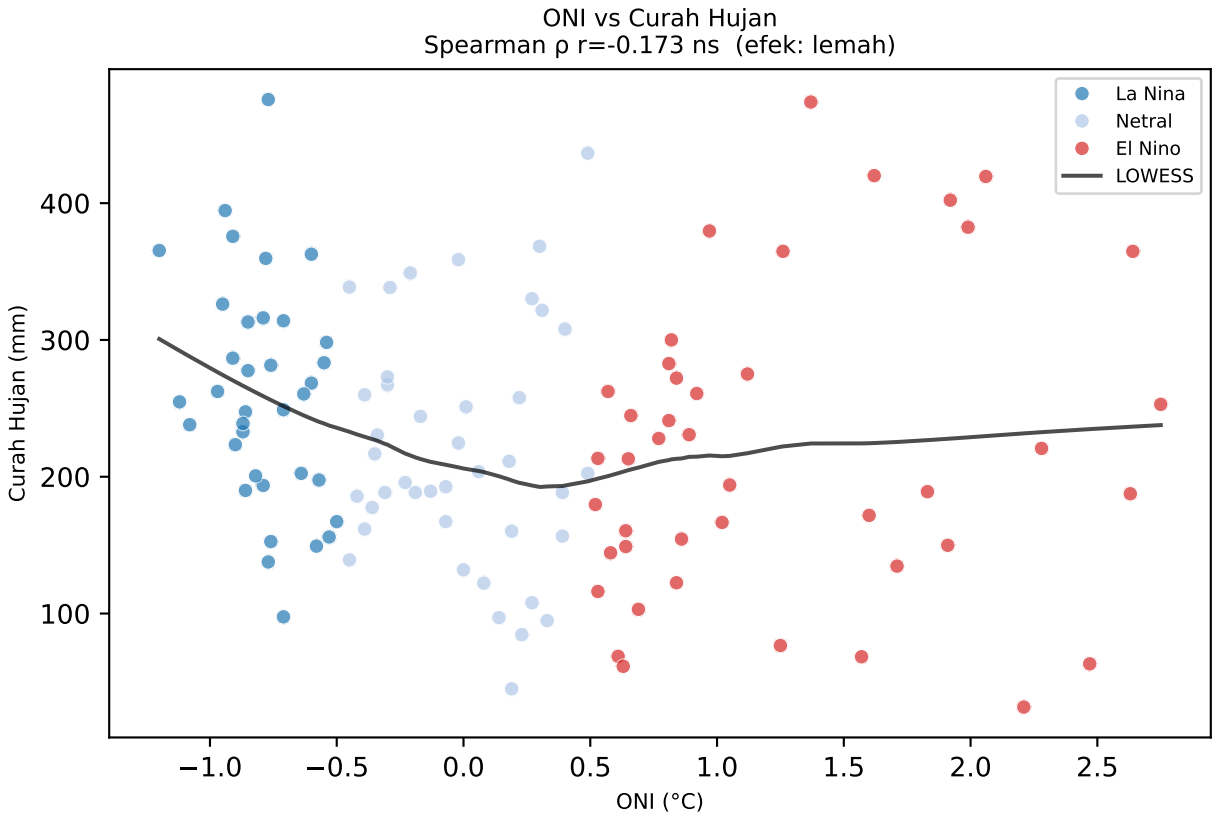


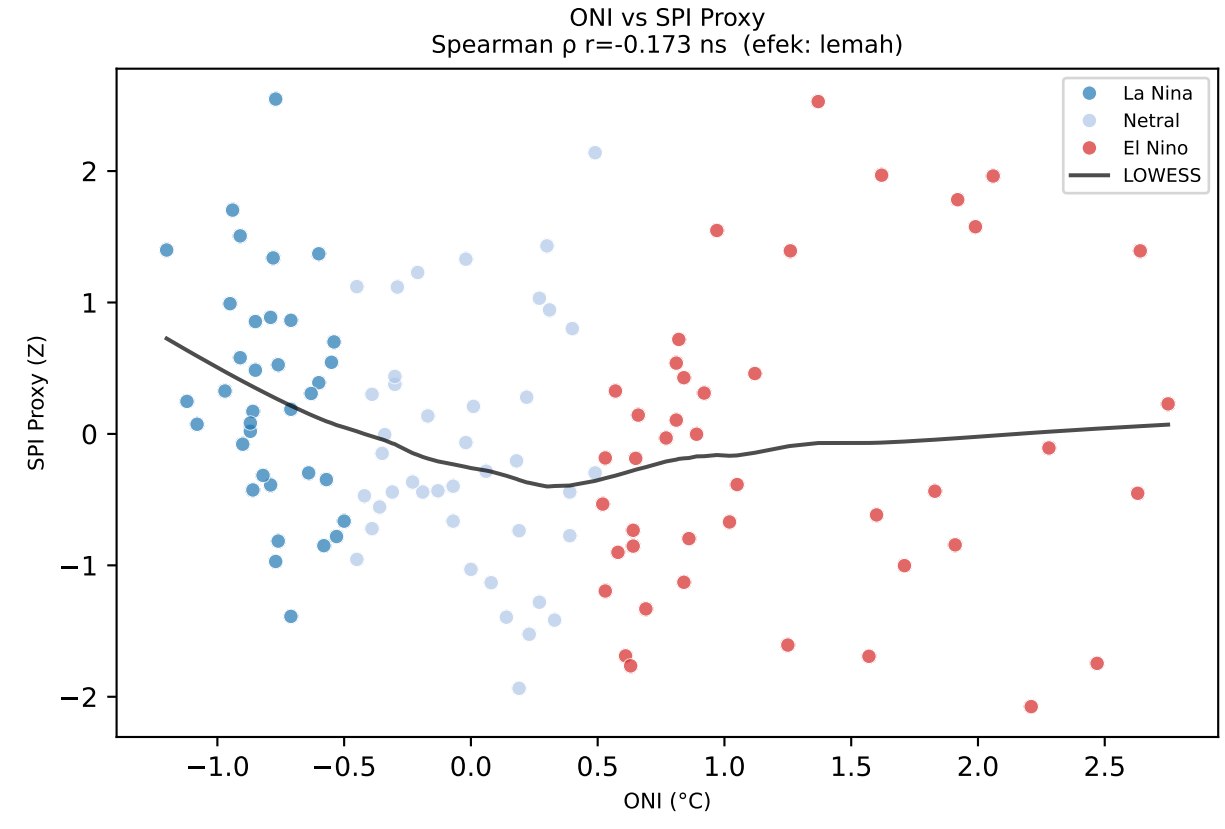


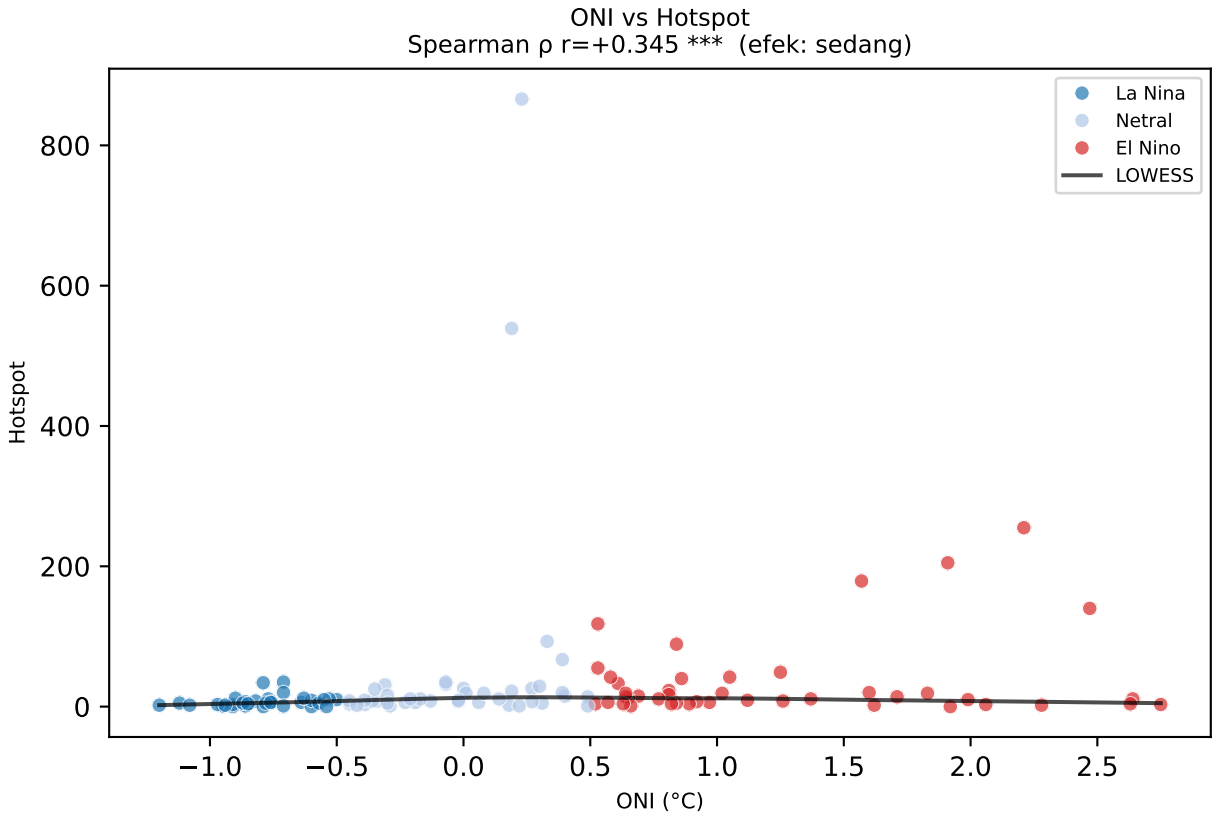


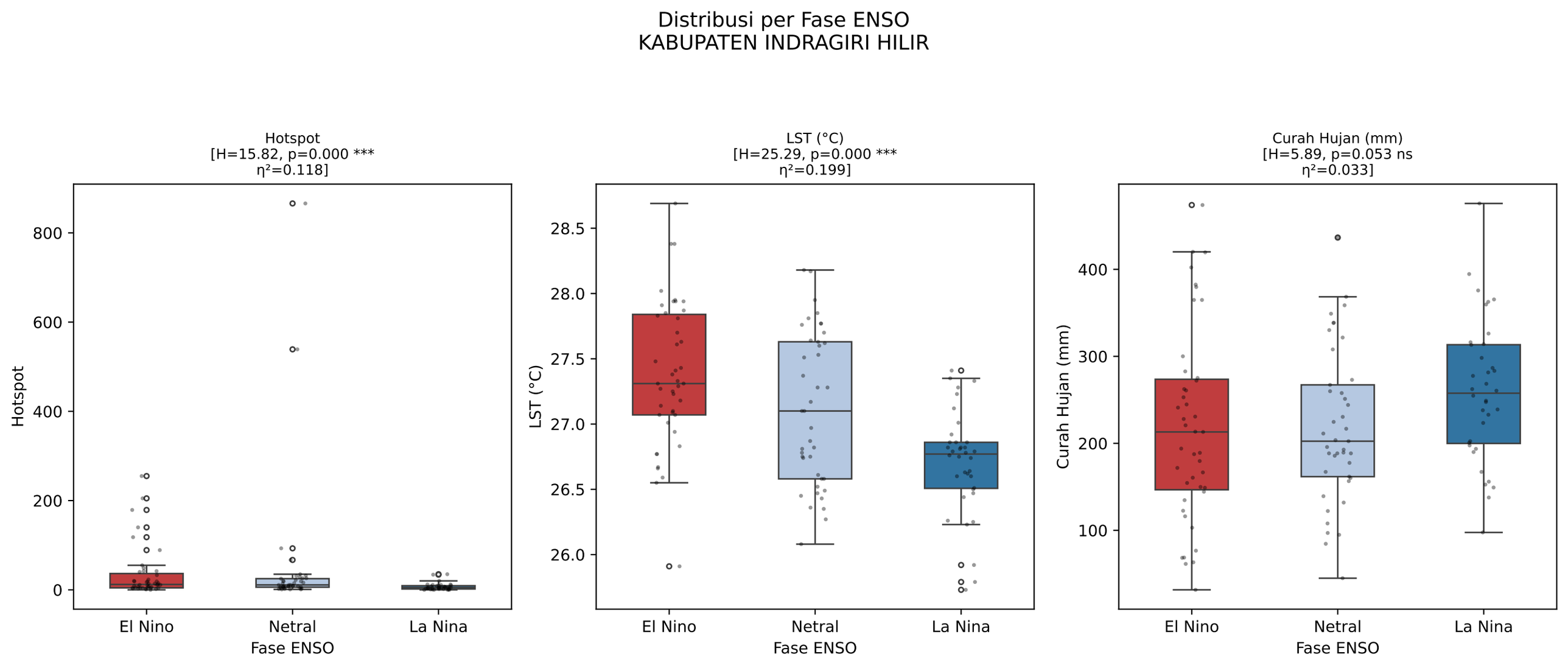
Scatter – ONI vs Variabel Iklim
KABUPATEN INDRAGIRI HILIR





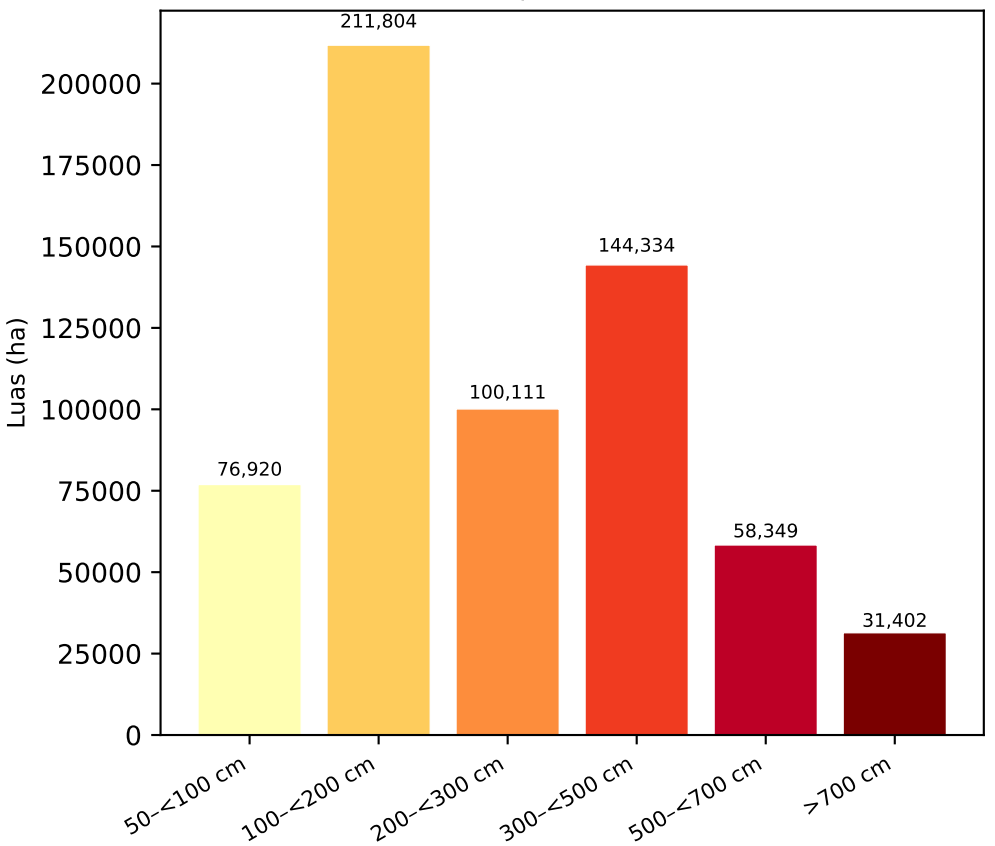




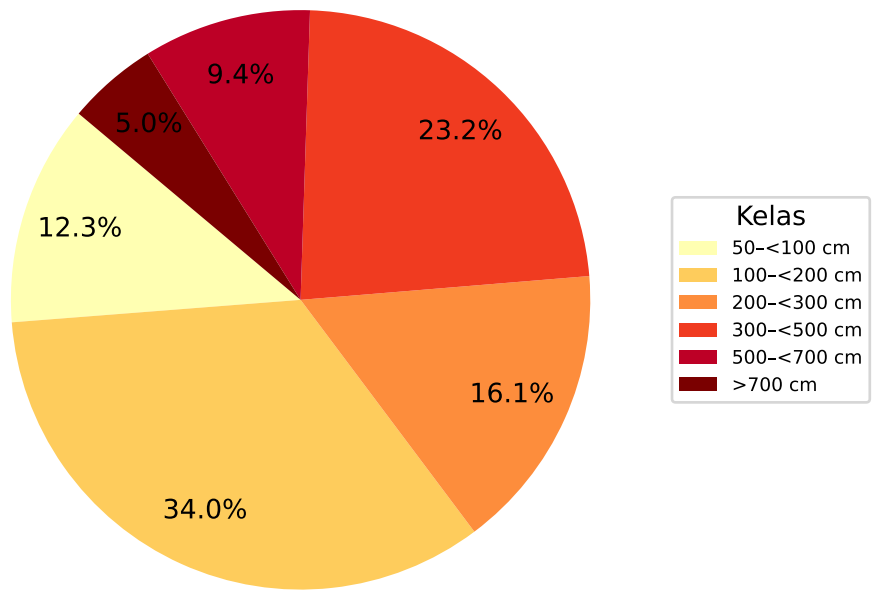

Distribusi per Fase ENSO
KABUPATEN INDRAGIRI HILIR
Hotspot
[H=15.82, p=0.000 ***
η²=0.118]
Hotspot
0
200
400
600
800
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=25.29, p=0.000 ***
η²=0.199]
LST (°C)
26.0
26.5
27.0
27.5
28.0
28.5
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=5.89, p=0.053 ns
η²=0.033]
Curah Hujan (mm)
100
200
300
400
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN INDRAGIRI HILIR

A. Luas Area per Kelas Kedalaman



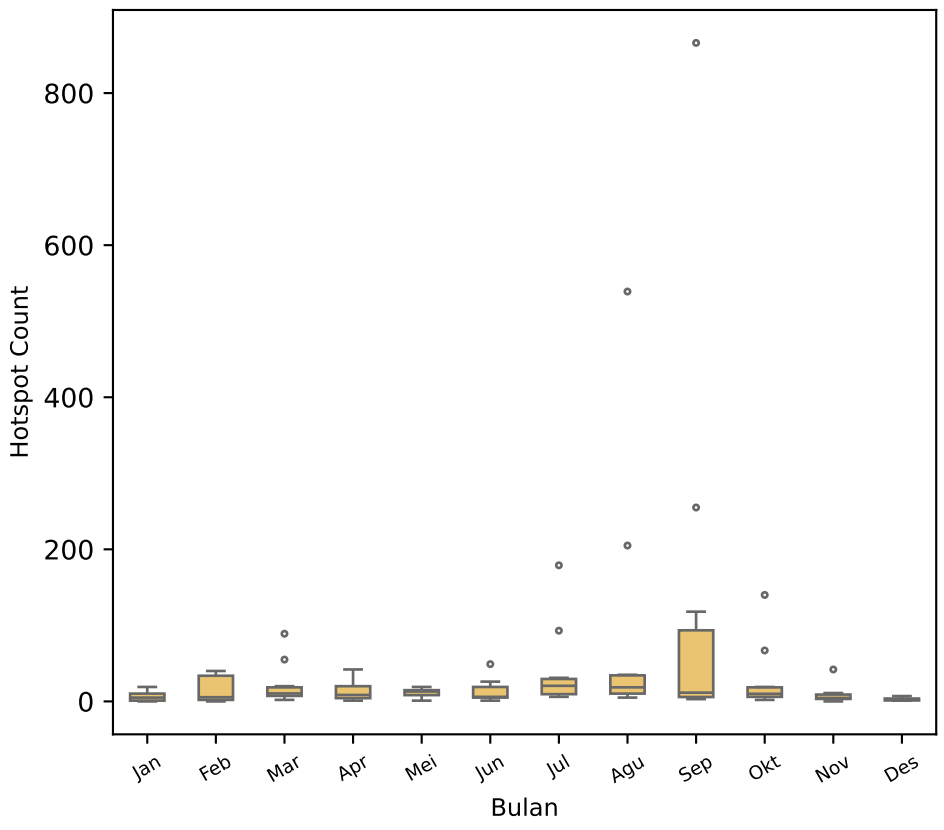
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════
  Statistik Gambut — Kabupaten Indragiri
───────────────────────────────────────────────
  Total Luas          :      622,921 ha
───────────────────────────────────────────────
  Kelas 1    50–<100 cm:     76,920 ha  ( 12.3%)
  Kelas 2   100–<200 cm:    211,804 ha  ( 34.0%)
  Kelas 3   200–<300 cm:    100,111 ha  ( 16.1%)
  Kelas 4   300–<500 cm:    144,334 ha  ( 23.2%)
  Kelas 5   500–<700 cm:     58,349 ha  (  9.4%)
  Kelas 6       >700 cm:     31,402 ha  (  5.0%)
───────────────────────────────────────────────
  Kelas Dominan       : 100–<200 cm
  Median Berbobot (σ) : 200–<300 cm
───────────────────────────────────────────────
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 100–<200 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

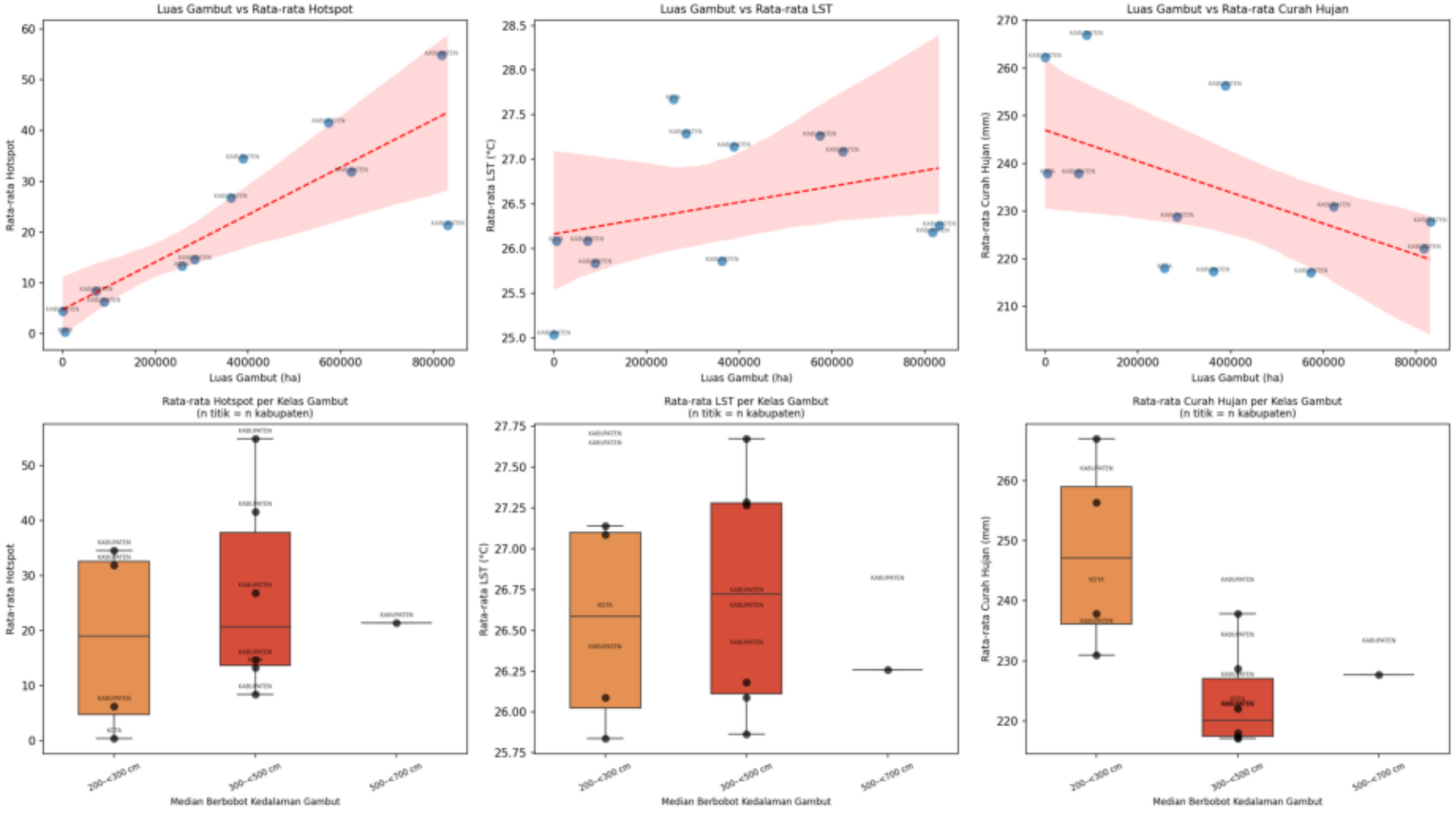
```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten



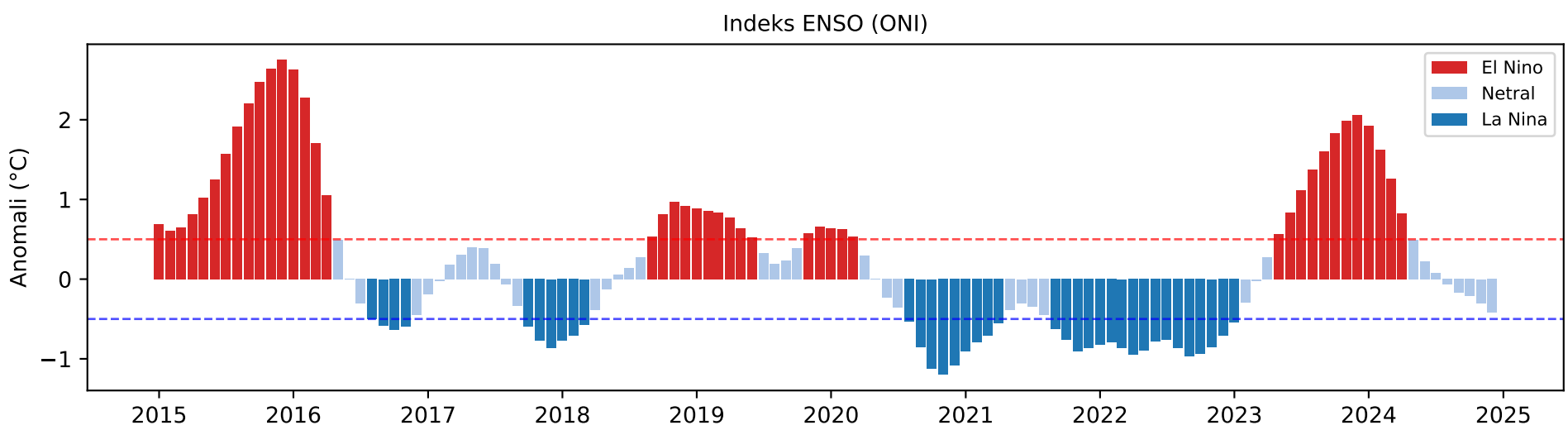
Laporan Korelasi Iklim, Gambut & Hotspot
KABUPATEN BENGKALIS

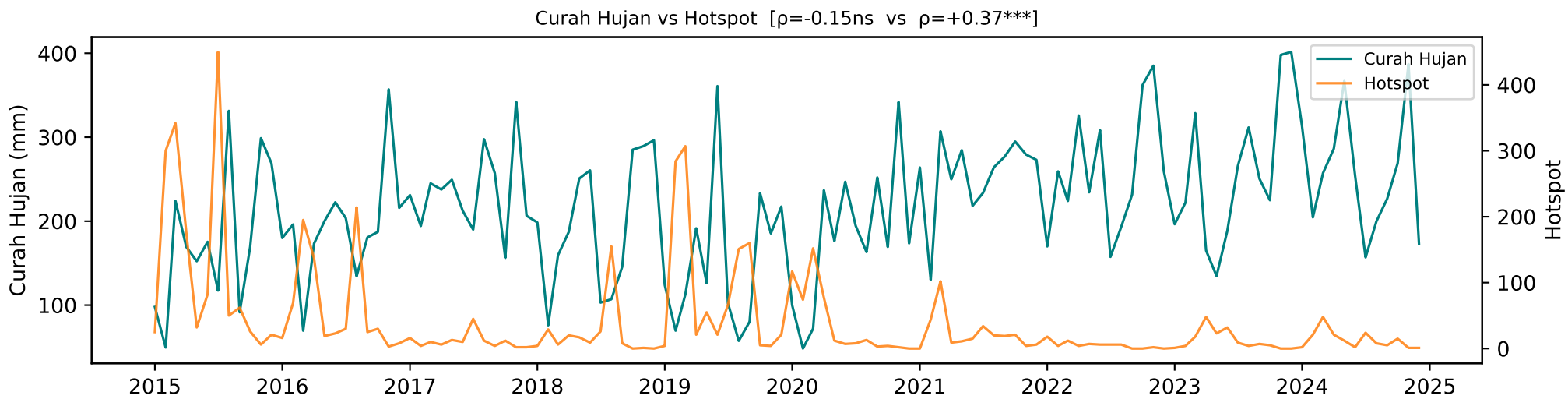
```
═══════════════════════════════════════════════════════════════════════
  ENSO (ONI) — Spearman ρ
──────────────────────────────────────────────────────────────────────
  ONI vs LST                                    : r=+0.454 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)
  ONI vs Curah Hujan                            : r=-0.150 ns    (p=0.103, Spearman ρ, efek=lemah, n=120)
  ONI vs SPI Proxy (kekeringan)                 : r=-0.150 ns    (p=0.103, Spearman ρ, efek=lemah, n=120)
  ONI vs Hotspot                                : r=+0.372 ***   (p=0.000, Spearman ρ, efek=sedang, n=120)

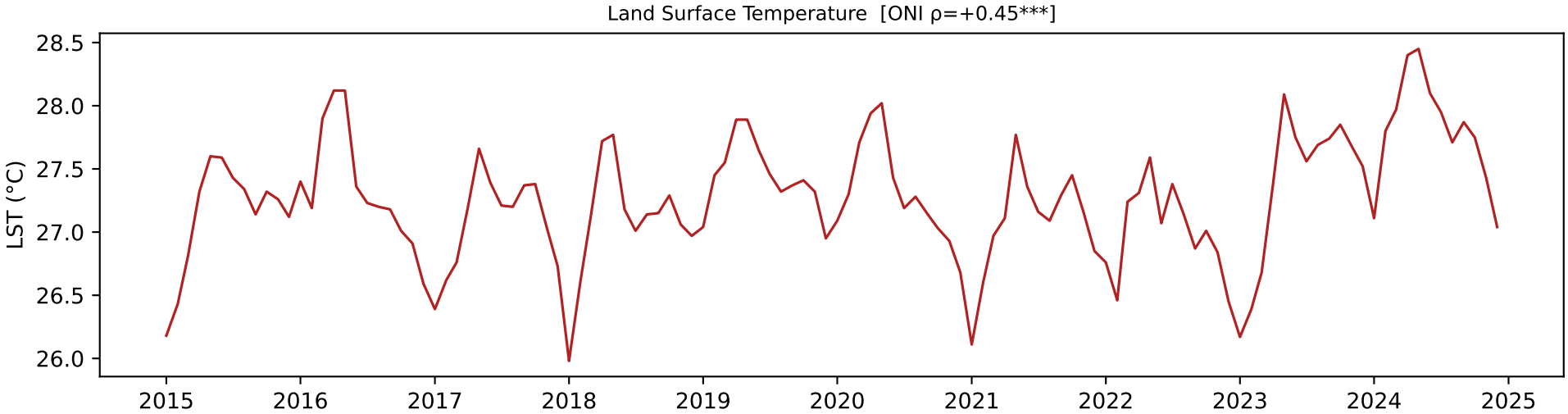
  GAMBUT — Spearman ρ (luas ha, kontinu)
──────────────────────────────────────────────────────────────────────
  Luas Gambut (ha) vs Hotspot                   : n/a (n=120)
  Luas Gambut (ha) vs LST                       : n/a (n=120)

  KRUSKAL-WALLIS antar fase ENSO
──────────────────────────────────────────────────────────────────────
  Hotspot     : H=15.40, p=0.000***, η²=0.115
  LST         : H=26.28, p=0.000, η²=0.207
  Curah Hujan : H=3.29, p=0.193, η²=0.011

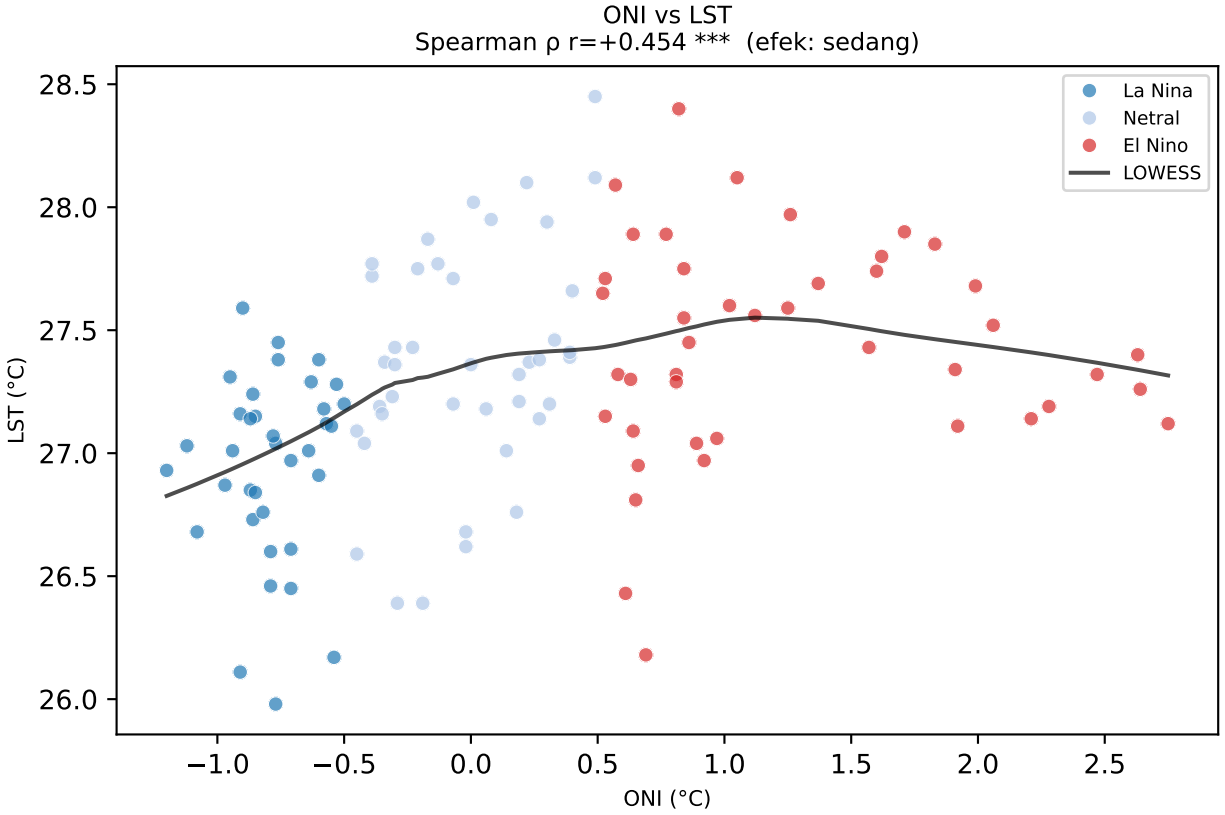
  INFO GAMBUT : total=573,294 ha | kelas dominan=500–<700 cm | median berbobot=300–<500 cm
  → Distribusi kelas gambut & musiman hotspot: Halaman 4
  → Korelasi kedalaman vs hotspot/LST lintas kabupaten: Halaman 5
  Sig.: *** p<0.001  ** p<0.01  * p<0.05  ns p≥0.05
```

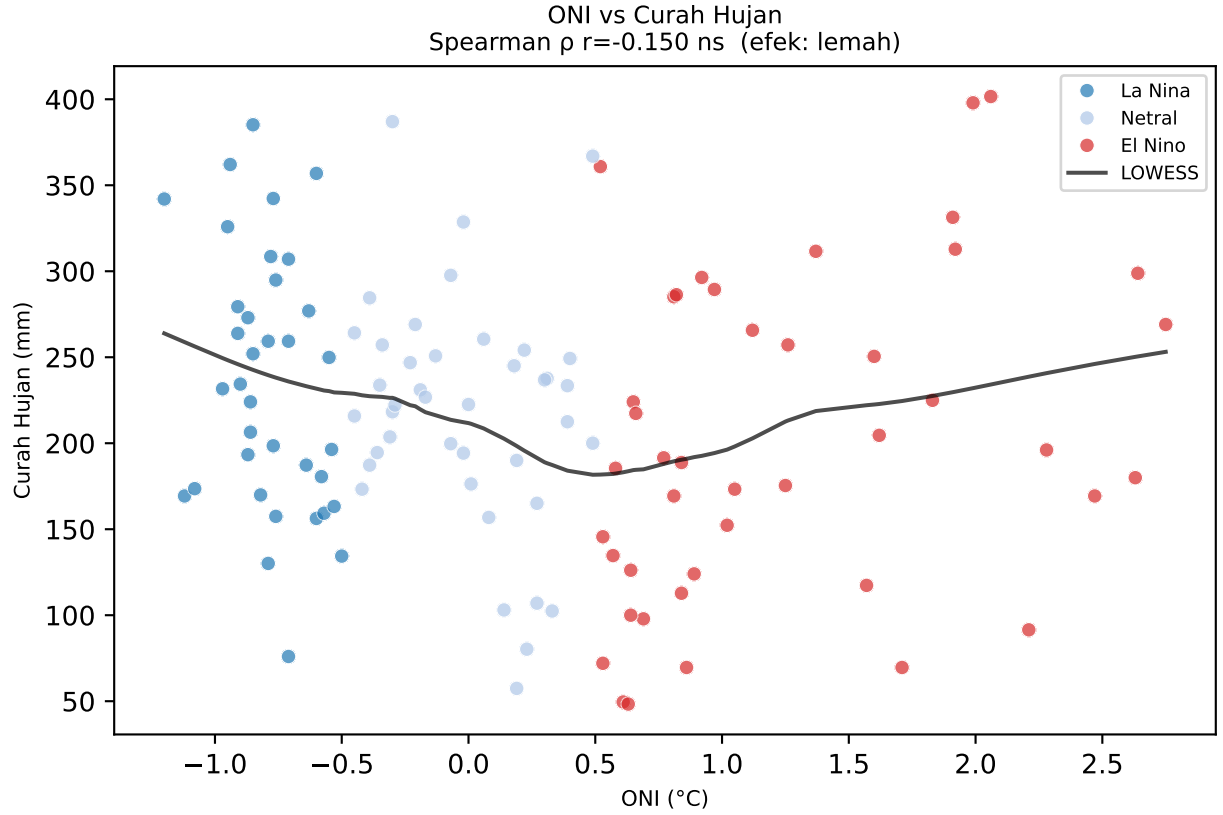


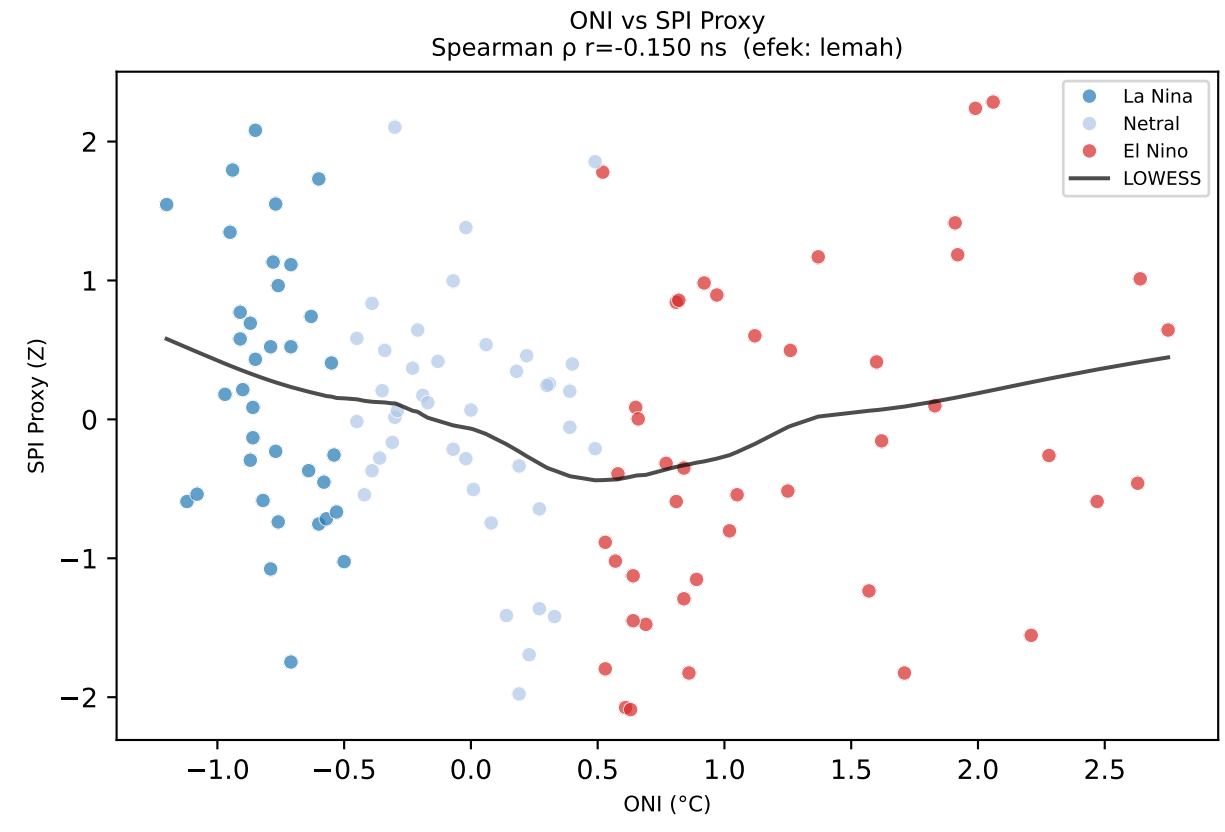


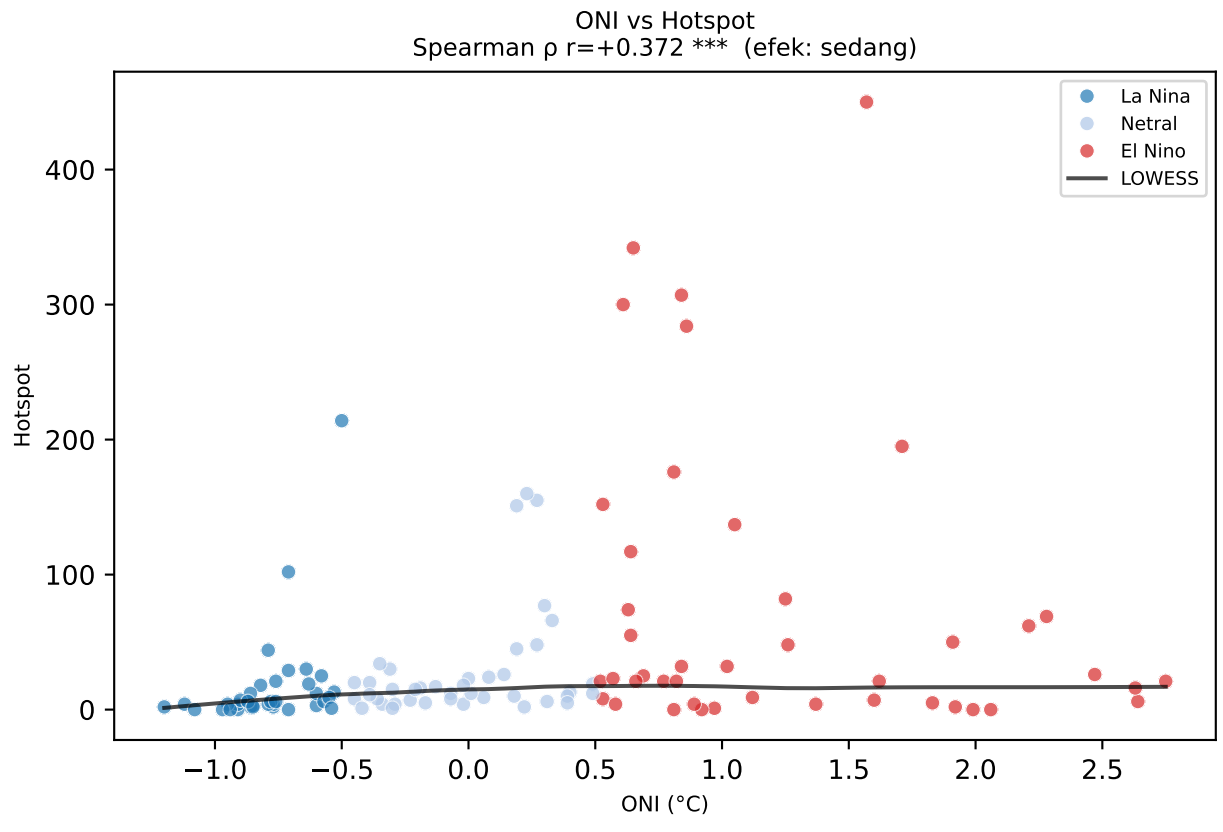


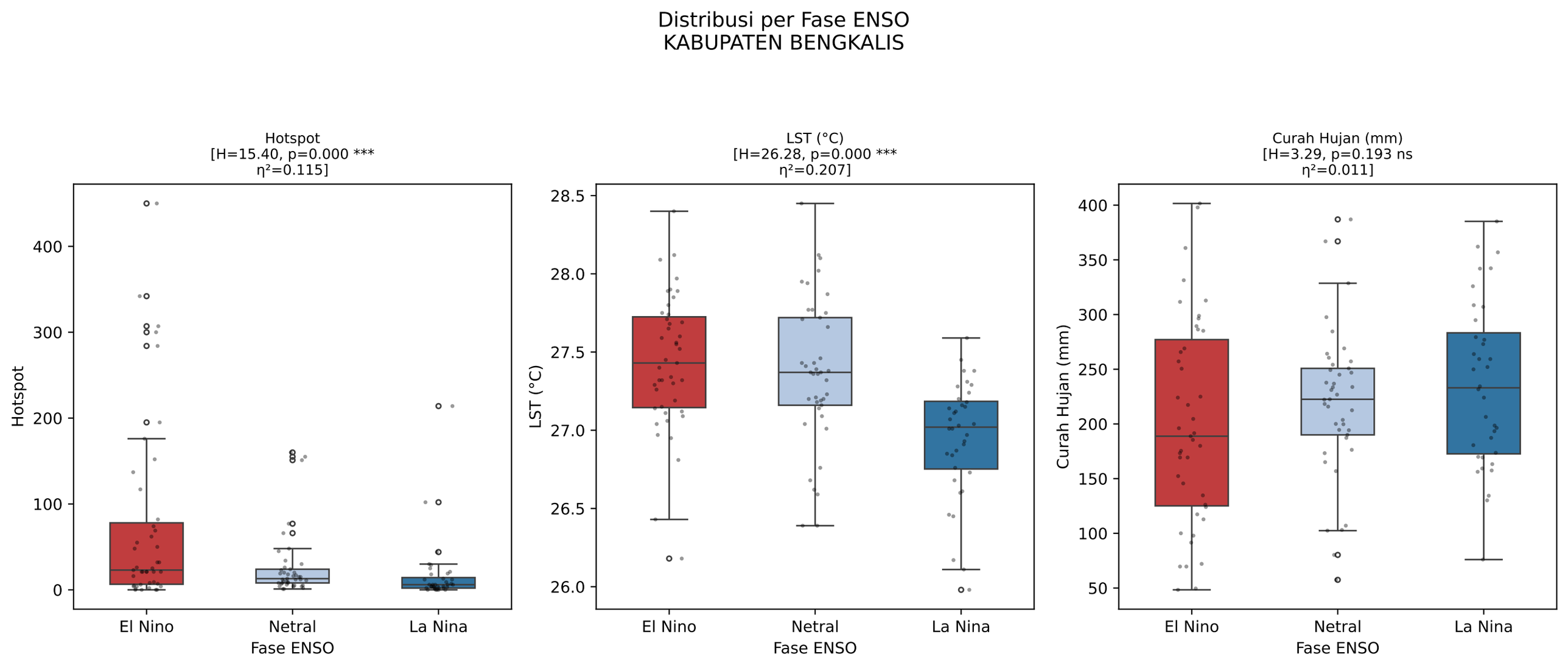
Scatter – ONI vs Variabel Iklim
KABUPATEN BENGKALIS





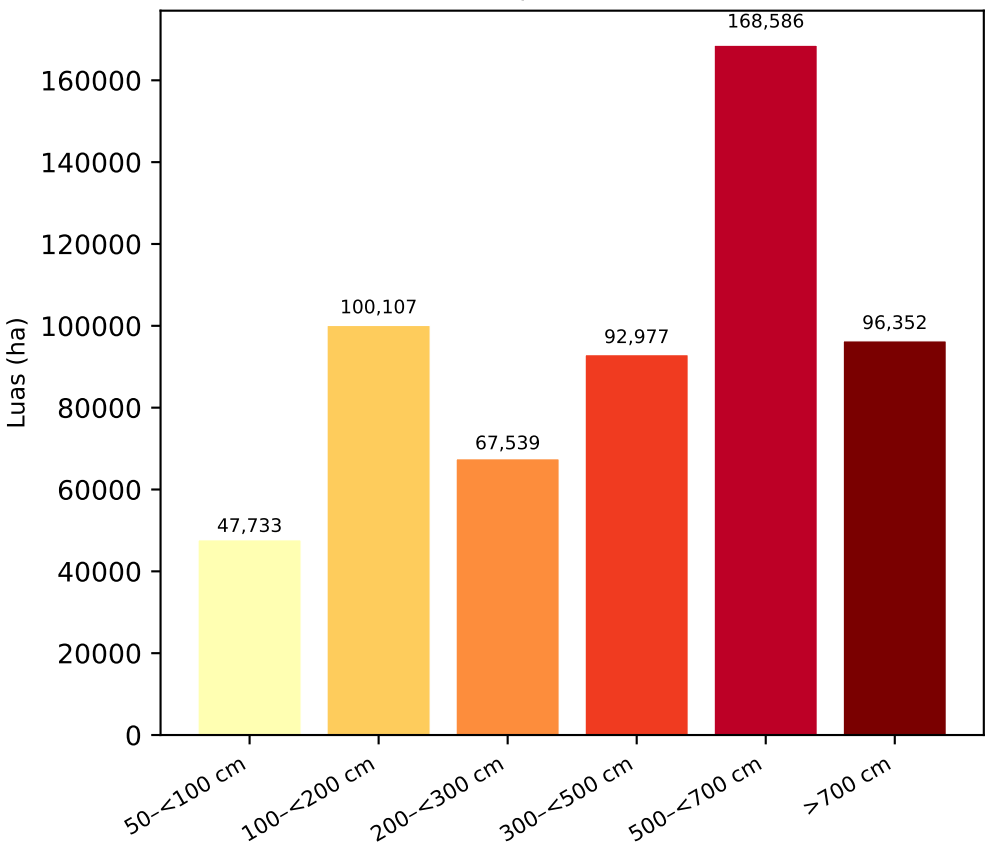




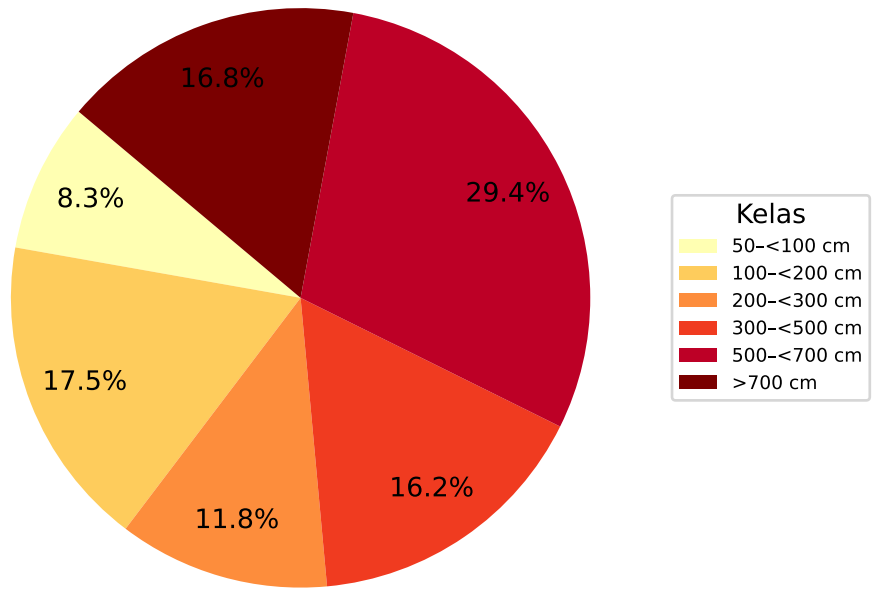

Distribusi per Fase ENSO
KABUPATEN BENGKALIS
Hotspot
[H=15.40, p=0.000 ***
η²=0.115]
Hotspot
0
100
200
300
400
El Nino
Netral
La Nina
Fase ENSO
LST (°C)
[H=26.28, p=0.000 ***
η²=0.207]
LST (°C)
26.0
26.5
27.0
27.5
28.0
28.5
El Nino
Netral
La Nina
Fase ENSO
Curah Hujan (mm)
[H=3.29, p=0.193 ns
η²=0.011]
Curah Hujan (mm)
50
100
150
200
250
300
350
400
El Nino
Netral
La Nina
Fase ENSO

Distribusi Kelas Kedalaman Gambut & Pola Musiman
KABUPATEN BENGKALIS

A. Luas Area per Kelas Kedalaman



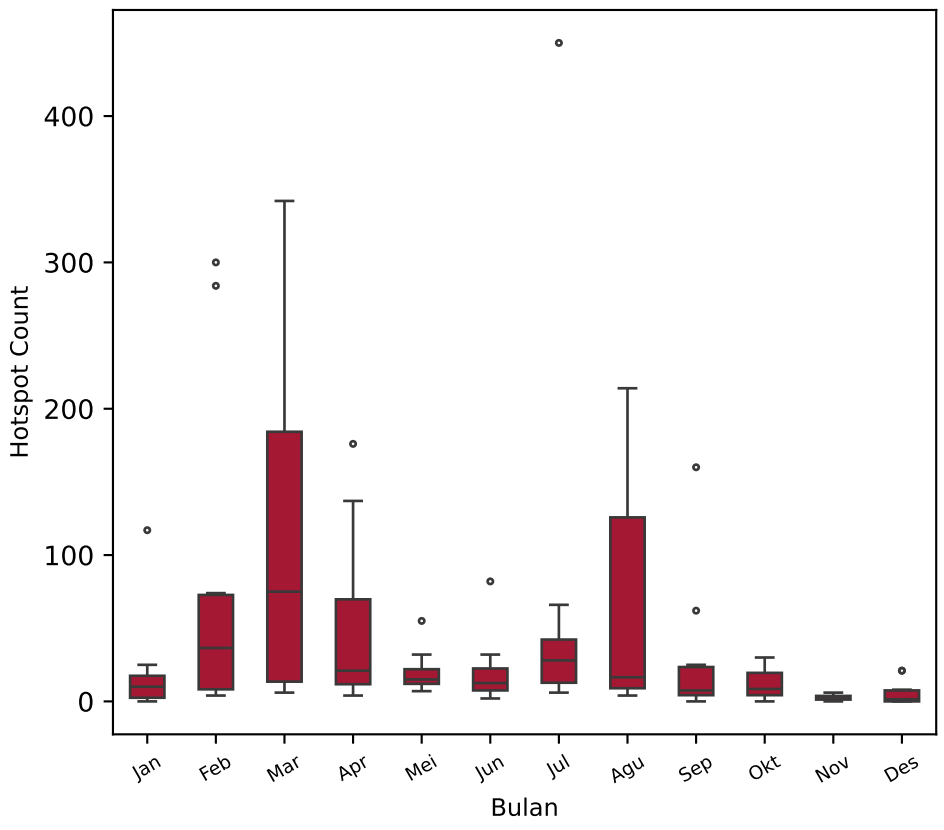
B. Proporsi Luas (%)



C. Ringkasan Statistik Gambut

```
═══════════════════════════════════════════════════
  Statistik Gambut — Kabupaten Bengkalis
═══════════════════════════════════════════════════
  Total Luas          :      573,294 ha
═══════════════════════════════════════════════════
  Kelas 1    50–<100 cm:     47,733 ha  (  8.3%)
  Kelas 2   100–<200 cm:    100,107 ha  ( 17.5%)
  Kelas 3   200–<300 cm:     67,539 ha  ( 11.8%)
  Kelas 4   300–<500 cm:     92,977 ha  ( 16.2%)
  Kelas 5   500–<700 cm:    168,586 ha  ( 29.4%)
  Kelas 6       >700 cm:     96,352 ha  ( 16.8%)
═══════════════════════════════════════════════════
  Kelas Dominan       : 500–<700 cm
  Median Berbobot (σ) : 300–<500 cm
═══════════════════════════════════════════════════
  ⚠  Kedalaman gambut = atribut statis
     Korelasi lintas kab. → lihat Hal. 5
```

D. Sebaran Hotspot per Bulan
(gambut dominan: 500–<700 cm)



Korelasi Peatland – Lintas Kabupaten (Global)

```
Korelasi Gambut vs Hotspot/LST (n = jumlah kabupaten):

  Kedalaman → weighted_median_depth (median berbobot area) + Kendall τ-b

  Luas      → total_peatland_area_ha (kontinu) + Spearman ρ

  Median Kedalaman (ord.) vs Avg Hotspot              : r=+0.312 ns    (p=0.200, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg LST                  : r=+0.370 ns    (p=0.131, Kendall τ-b, efek=lemah, n=12)
  Median Kedalaman (ord.) vs Avg ONI                  : n/a (n=12)
  Median Kedalaman (ord.) vs Avg Curah Hujan          : r=-0.592 *     (p=0.016, Kendall τ-b, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg Hotspot                     : r=+0.860 ***   (p=0.000, Spearman ρ, efek=kuat, n=12)
  Luas Gambut (ha) vs Avg LST                         : r=+0.445 ns    (p=0.147, Spearman ρ, efek=sedang, n=12)
  Luas Gambut (ha) vs Avg ONI                         : n/a (n=12)
```

Visualisasi Global: Gambut, Hotspot & LST per Kabupaten

